

BECOMING STRONGER

RATIONALITY: FROM AI TO ZOMBIES

in six volumes

- I Map and Territory
- II How to Actually Change Your Mind
- III The Machine in the Ghost
- IV Mere Reality
- V Mere Goodness
- VI Becoming Stronger

ELIEZER YUDKOWSKY

Rationality

From AI to Zombies

BOOK VI

Becoming Stronger

PERSONAL PRINT EDITION

Rationality: From AI to Zombies, Book VI: *Becoming Stronger*

Written by Eliezer Yudkowsky.

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Beginnings: An Introduction

by Rob Bensinger

This, the final book of *Rationality: From AI to Zombies*, is less a conclusion than a call to action. In keeping with *Becoming Stronger*'s function as a jumping-off point for further investigation, I'll conclude by citing resources the reader can use to move beyond these sequences and seek out a fuller understanding of Bayesianism.

This text's definition of normative rationality in terms of Bayesian probability theory and decision theory is standard in cognitive science. For an introduction to the heuristics and biases approach, see Baron's *Thinking and Deciding*.¹ For a general introduction to the field, see the *Oxford Handbook of Thinking and Reasoning*.²

The arguments made in these pages about the *philosophy* of rationality are more controversial. Yudkowsky argues, for example, that a rational agent should one-box in Newcomb's Problem—a minority position among working decision theorists.³ (See Holt for a nontechnical description of Newcomb's Problem.⁴) Gary Drescher's *Good and Real* independently comes to many of the same conclusions as Yudkowsky on philosophy of science and decision theory.⁵ As such, it serves as an excellent book-length treatment of the core philosophical content of *Rationality: From AI to Zombies*.

Talbott distinguishes several views in Bayesian epistemology, including E.T. Jaynes' position that not all possible priors



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1. Jonathan Baron, *Thinking and Deciding* (Cambridge University Press, 2007).
 2. Keith J. Holyoak and Robert G. Morrison, *The Oxford Handbook of Thinking and Reasoning* (Oxford University Press, 2013).
 3. Bourget and Chalmers, "What Do Philosophers Believe?"
 4. Holt, "Thinking Inside the Boxes."
 5. Gary L. Drescher, *Good and Real: Demystifying Paradoxes from Physics to Ethics* (Cambridge, MA: MIT Press, 2006).

are equally reasonable.^{6,7} Like Jaynes, Yudkowsky is interested in supplementing the Bayesian optimality criterion for belief revision with an optimality criterion for priors. This aligns Yudkowsky with researchers who hope to better understand general-purpose AI via an improved theory of ideal reasoning, such as Marcus Hutter.⁸ For a broader discussion of philosophical efforts to naturalize theories of knowledge, see Feldman.⁹

“Bayesianism” is often contrasted with “frequentism.” Some frequentists criticize Bayesians for treating probabilities as subjective states of belief, rather than as objective frequencies of events. Kruschke and Yudkowsky have replied that frequentism is even more “subjective” than Bayesianism, because frequentism’s probability assignments depend on the intentions of the experimenter.¹⁰

Importantly, this philosophical disagreement shouldn’t be conflated with the distinction between Bayesian and frequentist data analysis methods, which can both be useful when employed correctly. Bayesian statistical tools have become cheaper to use since the 1980s, and their informativeness, intuitiveness, and generality have come to be more widely appreciated, resulting in “Bayesian revolutions” in many sciences. However, traditional frequentist methods remain more popular, and in some contexts they are still clearly superior to Bayesian approaches. Kruschke’s *Doing Bayesian Data Analysis* is a fun and accessible introduction to the topic.¹¹

In light of evidence that training in statistics—and some other fields, such as psychology—improves reasoning skills outside the classroom, statistical literacy is directly relevant

6. William Talbott, “Bayesian Epistemology,” in *The Stanford Encyclopedia of Philosophy*, Fall 2013, ed. Edward N. Zalta.

7. Jaynes, *Probability Theory*.

8. Marcus Hutter, *Universal Artificial Intelligence: Sequential Decisions Based On Algorithmic Probability* (Berlin: Springer, 2005), doi:10.1007/b138233.

9. Richard Feldman, “Naturalized Epistemology,” in *The Stanford Encyclopedia of Philosophy*, Summer 2012, ed. Edward N. Zalta.

10. John K. Kruschke, “What to Believe: Bayesian Methods for Data Analysis,” *Trends in Cognitive Sciences* 14, no. 7 (2010): 293–300.

11. John K. Kruschke, *Doing Bayesian Data Analysis, Second Edition: A Tutorial with R, JAGS, and Stan* (Academic Press, 2014).

to the project of overcoming bias. (Classes in formal logic and informal fallacies have not proven similarly useful.)^{12,13}

An Art in its Infancy

We conclude with three sequences on individual and collective self-improvement. “Yudkowsky’s Coming of Age” provides a last in-depth illustration of the dynamics of irrational belief, this time spotlighting the author’s own intellectual history. “Challenging the Difficult” asks what it takes to solve a truly difficult problem—including demands that go beyond epistemic rationality. Finally, “The Craft and the Community” discusses rationality groups and group rationality, raising the questions:

- Can rationality be learned and taught?
- If so, how much improvement is possible?
- How can we be confident we’re seeing a real effect in a rationality intervention, and picking out the right cause?
- What community norms would make this process of bettering ourselves easier?
- Can we effectively collaborate on large-scale problems without sacrificing our freedom of thought and conduct?

Above all: What’s missing? What should be in the next generation of rationality primers—the ones that replace this text, improve on its style, test its prescriptions, supplement its content, and branch out in altogether new directions?

Though Yudkowsky was moved to write these essays by his own philosophical mistakes and professional difficulties in AI theory, the resultant material has proven useful to a much

12. Geoffrey T. Fong, David H. Krantz, and Richard E. Nisbett, “The Effects of Statistical Training on Thinking about Everyday Problems,” *Cognitive Psychology* 18, no. 3 (1986): 253–292, doi:10.1016/0010-0285(86)90001-0.

13. Paul J. H. Schoemaker, “The Role of Statistical Knowledge in Gambling Decisions: Moment vs. Risk Dimension Approaches,” *Organizational Behavior and Human Performance* 24, no. 1 (1979): 1–17.

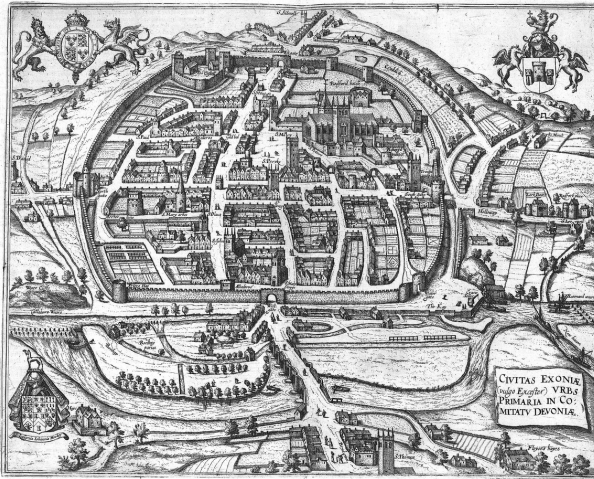
wider audience. The original blog posts inspired the growth of *Less Wrong*, a community of intellectuals and life hackers with shared interests in cognitive science, computer science, and philosophy. Yudkowsky and other writers on *Less Wrong* have helped seed the effective altruism movement, a vibrant and audacious effort to identify the most high-impact humanitarian charities and causes. These writings also sparked the establishment of the Center for Applied Rationality, a nonprofit organization that attempts to translate results from the science of rationality into useable techniques for self-improvement.

I don't know what's next—what other unconventional projects or ideas might draw inspiration from these pages. We certainly face no shortage of global challenges, and the art of applied rationality is a new and half-formed thing. There are not many rationalists, and there are many things left undone.

But wherever you're headed next, reader—may you serve your purpose well.

PART X

Yudkowsky's Coming of Age



My Childhood Death Spiral

My parents always used to downplay the value of intelligence. And play up the value of—effort, as recommended by the latest research? No, not effort. *Experience*. A nicely unattainable hammer with which to smack down a bright young child, to be sure. That was what my parents told me when I questioned the Jewish religion, for example. I tried laying out an argument, and I was told something along the lines of: “Logic has limits; you’ll understand when you’re older that experience is the important thing, and then you’ll see the truth of Judaism.” I didn’t try again. I made one attempt to question Judaism in school, got slapped down, didn’t try again. I’ve never been a slow learner.

Whenever my parents were doing something ill-advised, it was always, “We know better because we have more experience. You’ll understand when you’re older: maturity and wisdom are more important than intelligence.”

If this was an attempt to focus the young Eliezer on *intelligence uber alles*, it was the most wildly successful example of reverse psychology I’ve ever heard of.

But my parents aren’t that cunning, and the results weren’t exactly positive.

For a long time, I thought that the moral of this story was that experience was no match for sheer raw native intelligence. It wasn’t until a lot later, in my twenties, that I looked back and realized that I couldn’t possibly have been more intelligent than my parents *before puberty*, with my brain not even fully developed. At age eleven, when I was already nearly a full-blown atheist, I could not have defeated my parents in any *fair* contest of mind. My SAT scores were high for an 11-year-old, but they wouldn’t have beaten my parents’ SAT scores in full adulthood. In a fair fight, my parents’ intelligence and experience could have stomped any prepubescent child flat. It was dysrationalia



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that did them in; they used their intelligence only to defeat itself.

But *that* understanding came much later, when my intelligence had processed and distilled many more years of experience.

The moral I derived when I was young was that anyone who downplayed the value of intelligence didn't understand intelligence at all. My own intelligence had affected every aspect of my life and mind and personality; that was massively obvious, seen at a backward glance. "Intelligence has nothing to do with wisdom or being a good person"—oh, and does self-awareness have nothing to do with wisdom, or being a good person? Modeling yourself takes intelligence. For one thing, it takes enough intelligence to learn evolutionary psychology.

We *are* the cards we are dealt, and intelligence is the unfairiest of all those cards. More unfair than wealth or health or home country, unfairer than your happiness set-point. People have difficulty accepting that life can be that unfair; it's not a happy thought. "Intelligence isn't as important as X" is one way of turning away from the unfairness, refusing to deal with it, thinking a happier thought instead. It's a temptation, both to those dealt poor cards, and to those dealt good ones. Just as downplaying the importance of money is a temptation both to the poor and to the rich.

But the young Eliezer was a transhumanist. Giving away IQ points was going to take more work than if I'd just been born with extra money. But it was a fixable problem, to be faced up to squarely, and fixed. Even if it took my whole life. "The strong exist to serve the weak," wrote the young Eliezer, "and can only discharge that duty by making others equally strong." I was annoyed with the Randian and Nietzschean trends in science fiction, and as you may have grasped, the young Eliezer had a tendency to take things *too far in the other direction*. No one exists only to serve. But I tried, and I don't regret that. If you call that teenage folly, it's rare to see adult wisdom doing better.

Everyone needed more intelligence. Including me, I was careful to pronounce. Be it far from me to declare a new world

order with myself on top—that was what a stereotyped science fiction villain would do, or worse, a typical teenager, and I would never have allowed myself to be so clichéd. No, *everyone* needed to be smarter. We were all in the same boat: A fine, uplifting thought.

Eliezer₁₉₉₅ had read his science fiction. He had morals, and ethics, and could see the more obvious traps. No screeds on *Homo novis* for him. No line drawn between himself and others. No elaborate philosophy to put himself at the top of the heap. It was too obvious a failure mode. Yes, he was very careful to call himself stupid too, and never claim moral superiority. Well, and I don't see it so differently now, though I no longer make such a dramatic production out of my ethics. (Or maybe it would be more accurate to say that I'm tougher about when I allow myself a moment of self-congratulation.)

I say all this to emphasize that Eliezer₁₉₉₅ wasn't so undignified as to fail in any *obvious* way.

And then Eliezer₁₉₉₆ encountered the concept of intelligence explosion. Was it a thunderbolt of revelation? Did I jump out of my chair and shout “Eurisko!”? Nah. I wasn't that much of a drama queen. It was just massively obvious in retrospect that smarter-than-human intelligence was going to change the future more fundamentally than any mere material science. And I knew at once that *this* was what I would be doing with the rest of my life, creating the intelligence explosion. Not nanotechnology like I'd thought when I was eleven years old; nanotech would only be a tool brought forth of intelligence. Why, intelligence was even *more* powerful, an even greater blessing, than I'd realized before.

Was this a happy death spiral? As it turned out later, yes: that is, it led to the adoption even of *false* happy beliefs about intelligence. Perhaps you could draw the line at the point where I started believing that surely the lightspeed limit would be no barrier to superintelligence.

(How my views on intelligence have changed since then...let's see: When I think of poor hands dealt to humans, these days, I think first of death and old age. Everyone's got to have some intelligence level or other, and the important

thing from a fun-theoretic perspective is that it should ought to *increase* over time, not decrease like now. Isn't that a clever way of feeling better? But I don't work so hard now at downplaying my own intelligence, because that's just another way of calling attention to it. I'm smart for a human, if the topic should arise, and how I feel about that is my own business.

The part about intelligence being the lever that lifts worlds is the same. Except that intelligence has become less mysterious unto me, so that I now more clearly see intelligence as something embedded within physics. Superintelligences may go FTL if it happens to be permitted by the true physical laws, and if not, then not. It's not *unthinkable*, but I wouldn't bet on it.)

But the real wrong turn came later, at the point where someone said, "Hey, how do you know that superintelligence will be moral? Intelligence has nothing to do with being a good person, you know—that's what we call wisdom, young prodigy."

And lo, it seemed obvious to the young Eliezer that this was mere denial. Certainly, his own painstakingly constructed code of ethics had been put together using his intelligence and resting on his intelligence as a base. Any fool could see that intelligence had a great deal to do with ethics, morality, and wisdom; just try explaining the Prisoner's Dilemma to a chimpanzee, right?

Surely, then, superintelligence would necessarily imply supermorality.

Thus is it said: "Parents do all the things they tell their children not to do, which is how they know not to do them."

My Best and Worst Mistake

Last chapter I covered the young Eliezer’s affective death spiral around something that he called “intelligence.” Eliezer₁₉₉₆, or even Eliezer₁₉₉₉ for that matter, would have refused to try and put a mathematical definition—consciously, deliberately refused. Indeed, he would have been loath to put any definition on “intelligence” at all.

Why? Because there’s a standard bait-and-switch problem in AI, wherein you define “intelligence” to mean something like “logical reasoning” or “the ability to withdraw conclusions when they are no longer appropriate,” and then you build a cheap theorem-prover or an ad-hoc nonmonotonic reasoner, and then say, “Lo, I have implemented intelligence!” People came up with poor definitions of intelligence—focusing on correlates rather than cores—and then they chased the surface definition they had written down, forgetting about, you know, actual *intelligence*. It’s not like Eliezer₁₉₉₆ was out to build a career in Artificial Intelligence. He just wanted a mind that would actually be able to build nanotechnology. So he wasn’t tempted to redefine intelligence for the sake of puffing up a paper.

Looking back, it seems to me that quite a lot of my mistakes can be defined in terms of being pushed too far in the other direction by seeing someone else’s stupidity. Having seen attempts to define “intelligence” abused so often, I refused to define it at all. What if I said that intelligence was X , and it wasn’t *really* X ? I knew in an intuitive sense what I was looking for—something powerful enough to take stars apart for raw material—and I didn’t want to fall into the trap of being distracted from that by definitions.

Similarly, having seen so many AI projects brought down by physics envy—trying to stick with simple and elegant math, and being constrained to toy systems as a result—I generalized that any math simple enough to be formalized in a neat



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equation was probably not going to work for, you know, *real* intelligence. “Except for Bayes’s Theorem,” Eliezer₂₀₀₀ added; which, depending on your viewpoint, either mitigates the totality of his offense, or shows that he should have suspected the entire generalization instead of trying to add a single exception.

If you’re wondering why Eliezer₂₀₀₀ thought such a thing—disbelieved in a math of intelligence—well, it’s hard for me to remember this far back. It certainly wasn’t that I ever disliked math. If I had to point out a root cause, it would be reading too few, too popular, and the wrong Artificial Intelligence books.

But then I didn’t think the answers were going to come from Artificial Intelligence; I had mostly written it off as a sick, dead field. So it’s no wonder that I spent too little time investigating it. I believed in the cliché about Artificial Intelligence overpromising. You can fit that into the pattern of “too far in the opposite direction”—the field hadn’t delivered on its promises, so I was ready to write it off. As a result, I didn’t investigate hard enough to find the math that wasn’t fake.

My youthful disbelief in a mathematics of general intelligence was simultaneously one of my all-time worst mistakes, and one of my all-time best mistakes.

Because I disbelieved that there could be any simple answers to intelligence, I went and I read up on cognitive psychology, functional neuroanatomy, computational neuroanatomy, evolutionary psychology, evolutionary biology, and more than one branch of Artificial Intelligence. When I had what seemed like simple bright ideas, I didn’t stop there, or rush off to try and implement them, because I knew that even if they were true, even if they were necessary, they wouldn’t be sufficient: intelligence wasn’t supposed to be simple, it wasn’t supposed to have an answer that fit on a T-shirt. It was supposed to be a big puzzle with lots of pieces; and when you found one piece, you didn’t run off holding it high in triumph, you kept on looking. Try to build a mind with a single missing piece, and it might be that nothing interesting would happen.

I was wrong in thinking that Artificial Intelligence, the academic field, was a desolate wasteland; and even wrong in thinking that there couldn’t be math of intelligence. But I don’t

regret studying e.g. functional neuroanatomy, even though I *now* think that an Artificial Intelligence should look nothing like a human brain. Studying neuroanatomy meant that I went in with the idea that if you broke up a mind into pieces, the pieces were things like “visual cortex” and “cerebellum”—rather than “stock-market trading module” or “commonsense reasoning module,” which is a standard wrong road in AI.

Studying fields like functional neuroanatomy and cognitive psychology gave me a very different idea of what minds had to look like than you would get from just reading AI books—even good AI books.

When you blank out all the wrong conclusions and wrong justifications, and just ask what that belief led the young Eliezer to actually *do*...

Then the belief that Artificial Intelligence was sick and that the real answer would have to come from healthier fields outside led him to study lots of cognitive sciences;

The belief that AI couldn’t have simple answers led him to not stop prematurely on one brilliant idea, and to accumulate lots of information;

The belief that you didn’t want to define intelligence led to a situation in which he studied the problem for a long time before, years later, he started to propose systematizations.

This is what I refer to when I say that this is one of my all-time best mistakes.

Looking back, years afterward, I drew a very strong moral, to this effect:

What you actually end up doing screens off the clever reason why you’re doing it.

Contrast amazing clever reasoning that leads you to study many sciences, to amazing clever reasoning that says you don’t need to read all those books. Afterward, when your amazing clever reasoning turns out to have been stupid, you’ll have ended up in a much better position if your amazing clever reasoning was of the first type.

When I look back upon my past, I am struck by the number of semi-accidental successes, the number of times I did something right for the wrong reason. From your perspective, you

should chalk this up to the anthropic principle: if I'd fallen into a true dead end, you probably wouldn't be hearing from me in this book. From my perspective it remains something of an embarrassment. My Traditional Rationalist upbringing provided a lot of directional bias to those "accidental successes"—biased me toward rationalizing reasons to study rather than not study, prevented me from getting completely lost, helped me recover from mistakes. Still, none of that was the right action for the right reason, and that's a scary thing to look back on your youthful history and see. One of my primary purposes in writing on *Overcoming Bias* is to leave a trail to where I ended up by accident—to obviate the role that luck played in my own forging as a rationalist.

So what makes this one of my all-time worst mistakes? Because sometimes "informal" is another way of saying "held to low standards." I had amazing clever reasons why it was okay for me not to precisely define "intelligence," and certain of my other terms as well: namely, other people had gone astray by trying to define it. This was a gate through which sloppy reasoning could enter.

So should I have jumped ahead and tried to forge an exact definition right away? No, all the reasons why I knew this was the wrong thing to do were correct; you can't conjure the right definition out of thin air if your knowledge is not adequate.

You can't get to *the* definition of fire if you don't know about atoms and molecules; you're better off saying "that orangey-bright thing." And you do have to be able to talk about that orangey-bright stuff, even if you can't say exactly what it is, to investigate fire. But these days I would say that all reasoning on that level is something that can't be trusted—rather it's something you do on the way to knowing better, but you don't *trust* it, you don't *put your weight down* on it, you don't draw firm conclusions from it, no matter how inescapable the informal reasoning seems.

The young Eliezer put his weight down on the wrong floor tile—stepped onto a loaded trap.

Raised in Technophilia

My father used to say that if the present system had been in place a hundred years ago, automobiles would have been outlawed to protect the saddle industry.

One of my major childhood influences was reading Jerry Pournelle's *A Step Farther Out*, at the age of nine. It was Pournelle's reply to Paul Ehrlich and the Club of Rome, who were saying, in the 1960s and 1970s, that the Earth was running out of resources and massive famines were only years away. It was a reply to Jeremy Rifkin's so-called fourth law of thermodynamics; it was a reply to all the people scared of nuclear power and trying to regulate it into oblivion.

I grew up in a world where the lines of demarcation between the Good Guys and the Bad Guys were pretty clear; not an apocalyptic final battle, but a battle that had to be fought over and over again, a battle where you could see the historical echoes going back to the Industrial Revolution, and where you could assemble the historical evidence about the actual outcomes.

On one side were the scientists and engineers who'd driven all the standard-of-living increases since the Dark Ages, whose work supported luxuries like democracy, an educated populace, a middle class, the outlawing of slavery.

On the other side, those who had once opposed smallpox vaccinations, anesthetics during childbirth, steam engines, and heliocentrism: The theologians calling for a return to a perfect age that never existed, the elderly white male politicians set in their ways, the special interest groups who stood to lose, and the many to whom science was a closed book, fearing what they couldn't understand.

And trying to play the middle, the pretenders to Deep Wisdom, uttering cached thoughts about how technology benefits humanity but only when it was properly regulated—claiming in defiance of brute historical fact that science of itself was



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neither good nor evil—setting up solemn-looking bureaucratic committees to make an ostentatious display of their caution—and waiting for their applause. As if the truth were always a compromise. And as if anyone could really see that far ahead. Would humanity have done better if there'd been a sincere, concerned, public debate on the adoption of fire, and committees set up to oversee its use?

When I entered into the problem, I started out allergized against anything that pattern-matched “Ah, but technology has risks as well as benefits, little one.” The presumption-of-guilt was that you were either trying to collect some cheap applause, or covertly trying to regulate the technology into oblivion. And either way, ignoring the historical record immensely in *favor* of technologies that people had once worried about.

Robin Hanson raised the topic of slow FDA approval of drugs approved in other countries. Someone in the comments pointed out that Thalidomide was sold in 50 countries under 40 names, but that only a small amount was given away in the US, so that there were 10,000 malformed children born globally, but only 17 children in the US.

But how many people have died because of the slow approval in the US, of drugs more quickly approved in other countries—all the drugs that *didn't* go wrong? And I ask that question because it's what you can try to collect statistics about—this says nothing about all the drugs that were never *developed* because the approval process is too long and costly. According to this source, the FDA's longer approval process prevents 5,000 casualties per year by screening off medications found to be harmful, and causes at least 20,000–120,000 casualties per year just by delaying approval of those beneficial medications that are still developed and eventually approved.

So there really is a reason to be allergic to people who go around saying, “Ah, but technology has risks as well as benefits.” There's a historical record showing over-conservativeness, the many silent deaths of regulation being outweighed by a few visible deaths of nonregulation. If you're *really* playing the middle, why not say, “Ah, but technology has benefits as well as risks”?

Well, and this isn't such a bad description of the Bad Guys. (Except that it ought to be emphasized a bit harder that these aren't evil mutants but standard human beings acting under a different worldview-gestalt that puts them in the right; some of them will inevitably be more competent than others, and competence counts for a lot.) Even looking back, I don't think my childhood technophilia was too wrong about what constituted a Bad Guy and what was the key mistake. But it's always a *lot* easier to say what *not* to do, than to get it *right*. And one of my fundamental flaws, back then, was thinking that if you tried as hard as you could to avoid everything the Bad Guys were doing, that made you a Good Guy.

Particularly damaging, I think, was the bad example set by the pretenders to Deep Wisdom trying to stake out a middle way; smiling condescendingly at technophiles and technophobes alike, and calling them both immature. Truly this is a wrong way; and in fact, the notion of trying to stake out a middle way generally, is usually wrong. The Right Way is not a compromise with anything; it is the clean manifestation of its own criteria.

But that made it more difficult for the young Eliezer to depart from the charge-straight-ahead verdict, because *any* departure felt like joining the pretenders to Deep Wisdom.

The first crack in my childhood technophilia appeared in, I think, 1997 or 1998, at the point where I noticed my fellow technophiles saying foolish things about how molecular nanotechnology would be an easy problem to manage. (As you may be noticing yet again, the young Eliezer was driven to a tremendous extent by his ability to find flaws—I even had a personal philosophy of why that sort of thing was a good idea.)

There was a debate going on about molecular nanotechnology, and whether offense would be asymmetrically easier than defense. And there were people arguing that defense would be easy. In the domain of *nanotech*, for Ghu's sake, programmable matter, when we can't even seem to get the security problem solved for computer networks where we can observe and control every one and zero. People were talking about unassailable diamondoid walls. I observed that diamond doesn't stand off a

nuclear weapon, that offense has had defense beat since 1945 and nanotech didn't look likely to change that.

And by the time that debate was over, it seems that the young Eliezer—caught up in the heat of argument—had managed to notice, for the first time, that the survival of Earth-originating intelligent life stood at risk.

It seems so strange, looking back, to think that there was a time when I thought that only individual lives were at stake in the future. What a profoundly friendlier world that was to live in...though it's not as if I were thinking that at the time. I didn't *reject* the possibility so much as *manage to never see it in the first place*. Once the topic actually came up, I saw it. I don't really remember how that trick worked. There's a reason why I refer to my past self in the third person.

It may sound like Eliezer₁₉₉₈ was a complete idiot, but that would be a comfortable out, in a way; the truth is scarier. Eliezer₁₉₉₈ was a sharp Traditional Rationalist, as such things went. I knew hypotheses had to be testable, I knew that rationalization was not a permitted mental operation, I knew how to play Rationalist's Taboo, I was obsessed with self-awareness...I didn't quite understand the concept of "mysterious answers"...and no Bayes or Kahneman at all. But a sharp Traditional Rationalist, far above average...So what? Nature isn't grading us on a curve. One step of departure from the Way, one shove of undue influence on your thought processes, can repeal all other protections.

One of the chief lessons I derive from looking back at my personal history is that it's no wonder that, out there in the real world, a lot of people think that "intelligence isn't everything," or that rationalists don't do better in real life. A little rationality, or even a lot of rationality, doesn't pass the astronomically high barrier required for things to actually start *working*.

Let not my misinterpretation of the Right Way be blamed on Jerry Pournelle, my father, or science fiction generally. I think the young Eliezer's personality imposed quite a bit of selectivity on which parts of their teachings made it through. It's not as if Pournelle didn't say: *The rules change once you leave Earth, the cradle; if you're careless sealing your pressure suit just*

once, you die. He said it quite a bit. But the words didn't really seem important, because that was something that happened to third-party characters in the novels—the main character didn't usually die halfway through, for some reason.

What was the lens through which I filtered these teachings? Hope. Optimism. Looking forward to a brighter future. That was the fundamental meaning of *A Step Farther Out* unto me, the lesson I took in contrast to the Sierra Club's doom-and-gloom. On one side was *rationality and hope*, the other, *ignorance and despair*.

Some teenagers think they're immortal and ride motorcycles. I was under no such illusion and quite reluctant to learn to drive, considering how unsafe those hurtling hunks of metal looked. But there was something more important to me than my own life: The Future. And I acted as if *that* were immortal. Lives could be lost, but not the Future.

And when I noticed that nanotechnology really *was* going to be a potentially extinction-level challenge?

The young Eliezer thought, explicitly, "Good heavens, how did I fail to notice this thing that should have been obvious? I must have been too emotionally attached to the benefits I expected from the technology; I must have flinched away from the thought of human extinction."

And then ...

I didn't declare a Halt, Melt, and Catch Fire. I didn't rethink all the conclusions that I'd developed with my prior attitude. I just managed to integrate it into my worldview, *somehow*, with a minimum of propagated changes. Old ideas and plans were challenged, but my mind found reasons to keep them. There was no systemic breakdown, unfortunately.

Most notably, I decided that we had to run full steam ahead on AI, so as to develop it before nanotechnology. Just like I'd been *originally* planning to do, but now, with a *different reason*.

I guess that's what most human beings are like, isn't it? Traditional Rationality wasn't enough to change that.

But there did come a time when I fully realized my mistake. It just took a stronger boot to the head.

A Prodigy of Refutation



discussion
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6-c5

My Childhood Death Spiral described the core momentum carrying me into my mistake, an affective death spiral around something that Eliezer₁₉₉₆ called “intelligence.” I was also a technophile, pre-allergized against fearing the future. And I’d read a lot of science fiction built around personhood ethics—in which fear of the Alien puts humanity-at-large in the position of the bad guys, mistreating aliens or sentient AIs because they “aren’t human.”

That’s part of the ethos you acquire from science fiction—to define your in-group, your tribe, appropriately broadly. Hence my email address, sentience@pobox.com.

So Eliezer₁₉₉₆ is out to build superintelligence, for the good of humanity and all sentient life.

At first, I think, the question of whether a superintelligence will/could be good/evil didn’t really occur to me as a separate topic of discussion. Just the standard intuition of, “Surely no supermind would be stupid enough to turn the galaxy into paperclips; surely, being so intelligent, it will also know what’s *right* far better than a human being could.”

Until I introduced myself and my quest to a transhumanist mailing list, and got back responses along the general lines of (from memory):

Morality is arbitrary—if you say that something is good or bad, you can’t be right or wrong about that. A superintelligence would form its own morality.

Everyone ultimately looks after their own self-interest. A superintelligence would be no different; it would just seize all the resources.

Personally, I’m a human, so I’m in favor of humans, not Artificial Intelligences. I don’t think we

should develop this technology. Instead we should develop the technology to upload humans first.

No one should develop an AI without a control system that watches it and makes sure it can't do anything bad.

Well, *that's* all obviously wrong, thinks Eliezer₁₉₉₆, and he proceeded to kick his opponents' arguments to pieces. (I've mostly done this in other essays, and anything remaining is left as an exercise to the reader.)

It's not that Eliezer₁₉₉₆ explicitly reasoned, "The world's stupidest man says the Sun is shining, *therefore* it is dark out." But Eliezer₁₉₉₆ was a Traditional Rationalist; he had been inculcated with the metaphor of science as a *fair fight* between sides who take on different positions, stripped of mere violence and other such exercises of political muscle, so that, ideally, the side with the best arguments can win.

It's easier to say where someone else's argument is wrong, then to get the fact of the matter right; and Eliezer₁₉₉₆ was *very skilled* at finding flaws. (So am I. It's not as if you can solve the danger of that power by refusing to care about flaws.) From Eliezer₁₉₉₆'s perspective, it seemed to him that his chosen side was *winning the fight*—that he was formulating better arguments than his opponents—so why would he switch sides?

Therefore is it written: "Because this world contains many whose grasp of rationality is abysmal, beginning students of rationality win arguments and acquire an exaggerated view of their own abilities. But it is useless to be superior: Life is not graded on a curve. The best physicist in ancient Greece could not calculate the path of a falling apple. There is no guarantee that adequacy is possible given your hardest effort; therefore spare no thought for whether others are doing worse."

You cannot rely on *anyone* else to argue you out of your mistakes; you cannot rely on *anyone* else to save you; you and *only* you are obligated to find the flaws in your positions; if you put that burden down, don't expect anyone else to pick it up. And I wonder if that advice will turn out not to help most people, until they've personally blown off their own foot, saying

to themselves all the while, *correctly*, “Clearly I’m winning this argument.”

Today I try not to take any human being as my opponent. That just leads to overconfidence. It is Nature that I am facing off against, who does not match Her problems to your skill, who is not obliged to offer you a fair chance to win in return for a diligent effort, who does not care if you are the best who ever lived, if you are not good *enough*.

But return to 1996. Eliezer₁₉₉₆ is going with the basic intuition of “Surely a superintelligence will know better than we could what is *right*,” and offhandedly knocking down various arguments brought against his position. He was skillful in that way, you see. He even had a personal philosophy of why it was wise to look for flaws in things, and so on.

I don’t mean to say it as an excuse, that no one who argued against Eliezer₁₉₉₆ actually presented him with the dissolution of the mystery—the full reduction of morality that analyzes all his cognitive processes debating “morality,” a step-by-step walkthrough of the algorithms that make morality feel to him like a fact. Consider it rather as an indictment, a measure of Eliezer₁₉₉₆’s level, that he would have needed the full solution given to him, in order to present him with an argument that he could *not* refute.

The few philosophers present did not extract him from his difficulties. It’s not as if a philosopher will say, “Sorry, morality is understood, it is a settled issue in cognitive science and philosophy, and your viewpoint is simply wrong.” The nature of morality is still an open question in philosophy; the debate is still going on. A philosopher will feel obligated to present you with a list of classic arguments on all sides—most of which Eliezer₁₉₉₆ is quite intelligent enough to knock down, and so he concludes that philosophy is a wasteland.

But wait. It gets worse.

I don’t recall exactly when—it might have been 1997—but the younger me, let’s call him Eliezer₁₉₉₇, set out to argue *in-escapably* that creating superintelligence is the right thing to do.

The Sheer Folly of Callow Youth

There speaks the sheer folly of callow youth; the rashness of an ignorance so abysmal as to be possible only to one of your ephemeral race ...

—Gharlane of Eddore¹



discussion
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6-c6

Once upon a time, years ago, I propounded a mysterious answer to a mysterious question—as I’ve hinted on several occasions. The mysterious question to which I propounded a mysterious answer was not, however, consciousness—or rather, not only consciousness. No, the more embarrassing error was that I took a mysterious view of morality.

I held off on discussing that until now, after the series on metaethics, because I wanted it to be clear that Eliezer₁₉₉₇ *had* gotten it wrong.

When we last left off, Eliezer₁₉₉₇, not satisfied with arguing in an intuitive sense that superintelligence would be moral, was setting out to argue *inescapably* that creating superintelligence was the right thing to do.

Well (said Eliezer₁₉₉₇) let’s begin by asking the question: *Does life have, in fact, any meaning?*

“I don’t know,” replied Eliezer₁₉₉₇ at once, with a certain note of self-congratulation for admitting his own ignorance on this topic where so many others seemed certain.

“But,” he went on—

(Always be wary when an admission of ignorance is followed by “But.”)

“But, if we suppose that life has no meaning—that the utility of all outcomes is equal to zero—that possibility cancels out of any *expected* utility calculation. We can therefore always *act as if* life is known to be meaningful, even though we don’t

1. Edward Elmer Smith, *Second Stage Lensmen* (Old Earth Books, 1998).

know what that meaning is. How can we find out that meaning? Considering that humans are still arguing about this, it's probably too difficult a problem for humans to solve. So we need a superintelligence to solve the problem for us. As for the possibility that there *is* no logical justification for one preference over another, then in this case it is no righter or wronger to build a superintelligence, than to do anything else. This is a real possibility, but it falls out of any attempt to calculate expected utility—we should just ignore it. To the extent someone says that a superintelligence would wipe out humanity, they are either arguing that wiping out humanity is in fact the right thing to do (even though we see no reason why this should be the case) or they are arguing that there *is* no right thing to do (in which case their argument that we should not build intelligence defeats itself)."

Ergh. That was a *really* difficult paragraph to write. My past self is always my own most concentrated Kryptonite, because my past self is *exactly precisely* all those things that the modern me has installed allergies to block. Truly is it said that parents do all the things they tell their children not to do, which is how they know not to do them; it applies between past and future selves as well.

How flawed is Eliezer₁₉₉₇'s argument? I couldn't even count the ways. I know memory is fallible, reconstructed each time we recall, and so I don't trust my assembly of these old pieces using my modern mind. Don't ask me to read my old writings; that's too much pain.

But it seems clear that I was thinking of utility as a sort of stuff, an inherent property. So that "life is meaningless" corresponded to utility = 0. But of course the argument works equally well with utility = 100, so that if everything is meaningful but it is all *equally* meaningful, that should fall out too ... Certainly I wasn't then thinking of a utility function as an affine structure in preferences. I was thinking of "utility" as an absolute level of inherent value.

I was thinking of *should* as a kind of purely abstract essence of compellingness, that-which-makes-you-do-something; so that clearly any mind that derived a *should* would be bound by

it. Hence the assumption, which Eliezer₁₉₉₇ did not even think to explicitly note, that a logic that compels an arbitrary mind to do something is exactly the same as that which human beings mean and refer to when they utter the word “right” ...

But now I’m trying to count the ways, and if you’ve been following along, you should be able to handle that yourself.

An important aspect of this whole failure was that, because I’d proved that the case “life is meaningless” *wasn’t worth considering*, I didn’t think it was necessary to rigorously define “intelligence” or “meaning.” I’d previously come up with a clever reason for not trying to go all formal and rigorous when trying to define “intelligence” (or “morality”)—namely all the bait-and-switches that past AI folk, philosophers, and moralists had pulled with definitions that missed the point.

I draw the following lesson: No matter how clever the justification for relaxing your standards, or evading some requirement of rigor, it will blow your foot off just the same.

And another lesson: I was skilled in refutation. If I’d applied the same level of rejection-based-on-any-flaw to my own position as I used to defeat arguments brought against me, then I would have zeroed in on the logical gap and rejected the position—if I’d *wanted* to. If I’d had the same level of prejudice against it as I’d had against other positions in the debate.

But this was before I’d heard of Kahneman, before I’d heard the term “motivated skepticism,” before I’d integrated the concept of an exactly correct state of uncertainty that summarizes all the evidence, and before I knew the deadliness of asking “Am I allowed to believe?” for liked positions and “Am I forced to believe?” for disliked positions. I was a mere Traditional Rationalist who thought of the scientific process as a referee between people who took up positions and argued them, may the best side win.

My ultimate flaw was not a liking for “intelligence,” nor any amount of technophilia and science fiction exalting the siblinghood of sentience. It surely wasn’t my ability to spot flaws. None of these things *could* have led me astray, if I had held myself to a higher standard of rigor throughout, and adopted no position otherwise. Or even if I’d just scrutinized my pre-

ferred vague position, with the same demand-of-rigor I applied to counterarguments.

But I wasn't much interested in trying to refute my belief that life had meaning, since my reasoning would always be dominated by cases where life did have meaning.

And with the intelligence explosion at stake, I thought I just had to proceed at all speed using the best concepts I could wield at the time, not pause and shut down everything while I looked for a perfect definition that so many others had screwed up ...

No.

No, you don't use the best concepts you can use at the time.

It's Nature that judges you, and Nature does not accept *even the most righteous excuses*. If you don't meet the standard, you fail. It's that simple. There is no clever argument for why you have to make do with what you have, because Nature won't listen to that argument, won't forgive you because there were so many excellent justifications for speed.

We all know what happened to Donald Rumsfeld, when he went to war with the army he had, instead of the army he needed.

Maybe Eliezer₁₉₉₇ couldn't have conjured the correct model out of thin air. (Though who knows what would have happened, if he'd really tried ...) And it wouldn't have been prudent for him to stop thinking entirely, until rigor suddenly popped out of nowhere.

But neither was it correct for Eliezer₁₉₉₇ to put his weight down on his "best guess," in the absence of precision. You can use vague concepts in your own interim thought processes, as you search for a better answer, unsatisfied with your current vague hints, *and unwilling to put your weight down on them*. You don't build a superintelligence based on an interim understanding. No, not even the "best" vague understanding you have. That was my mistake—thinking that saying "best guess" excused anything. There was only the standard I had failed to meet.

Of course Eliezer₁₉₉₇ didn't want to slow down on the way to the intelligence explosion, with so many lives at stake, and

the very survival of Earth-originating intelligent life, if we got to the era of nanoweapons before the era of superintelligence—

Nature doesn't care about such righteous reasons. There's just the astronomically high standard needed for success. Either you match it, or you fail. That's all.

The apocalypse does not need to be fair to you.
 The apocalypse does not need to offer you a chance
 of success
 In exchange for what you've already brought to
 the table.
 The apocalypse's difficulty is not matched to your
 skills.
 The apocalypse's price is not matched to your re-
 sources.
 If the apocalypse asks you for something unrea-
 sonable
 And you try to bargain it down a little
 (Because everyone has to compromise now and
 then)
 The apocalypse will not try to negotiate back up.

And, oh yes, it gets worse.

How did Eliezer₁₉₉₇ deal with the obvious argument that you couldn't possibly derive an "ought" from pure logic, because "ought" statements could only be derived from other "ought" statements?

Well (observed Eliezer₁₉₉₇), this problem has the same structure as the argument that a cause only proceeds from another cause, or that a real thing can only come of another real thing, whereby you can prove that nothing exists.

Thus (he said) there are three "hard problems": the hard problem of conscious experience, in which we see that qualia cannot arise from computable processes; the hard problem of existence, in which we ask how any existence enters apparently from nothingness; and the hard problem of morality, which is to get to an "ought."

These problems are probably linked. For example, the qualia of pleasure are one of the best candidates for something intrin-

sically desirable. We might not be able to understand the hard problem of morality, therefore, without unraveling the hard problem of consciousness. It's evident that these problems are too hard for humans—otherwise someone would have solved them over the last 2,500 years since philosophy was invented.

It's not as if they could have complicated solutions—they're too simple for that. The problem must just be outside human concept-space. Since we can see that consciousness can't arise on any computable process, it must involve new physics—physics that our brain uses, but can't understand. That's why we need superintelligence in order to solve this problem. Probably it has to do with quantum mechanics, maybe with a dose of tiny closed timelike curves from out of General Relativity; temporal paradoxes might have some of the same irreducibility properties that consciousness seems to demand ...

Et cetera, ad nauseam. You may begin to perceive, in the arc of my *Overcoming Bias* posts, the letter I wish I could have written to myself.

Of this I learn the lesson: You cannot manipulate confusion. You cannot make clever plans to work around the holes in your understanding. You can't even make "best guesses" about things which fundamentally confuse you, and relate them to other confusing things. Well, you can, but you won't get it right, until your confusion dissolves. Confusion exists in the mind, not in the reality, and trying to treat it like something you can pick up and move around will only result in unintentional comedy.

Similarly, you cannot come up with clever reasons why the gaps in your model don't matter. You cannot draw a border around the mystery, put on neat handles that let you use the Mysterious Thing without really understanding it—like my attempt to make the possibility that life is meaningless cancel out of an expected utility formula. You can't pick up the gap and manipulate it.

If the blank spot on your map conceals a land mine, then putting your weight down on that spot will be fatal, no matter how good your excuse for not knowing. Any black box could contain a trap, and there's no way to know except opening up

the black box and looking inside. If you come up with some righteous justification for why you need to rush on ahead with the best understanding you have—the trap goes off.

It's only when you know the rules,
That you realize *why* you needed to learn;
What would have happened otherwise,
How *much* you needed to know.

Only knowledge can foretell the cost of ignorance. The ancient alchemists had no logical way of knowing the exact reasons why it was hard for them to turn lead into gold. So they poisoned themselves and died. Nature doesn't care.

But there did come a time when realization began to dawn on me.

That Tiny Note of Discord



discussion
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6-c7

When we last left Eliezer₁₉₉₇, he believed that any superintelligence would automatically do what was “right,” and indeed would understand that better than we could—even though, he modestly confessed, he did not understand the ultimate nature of morality. Or rather, after some debate had passed, Eliezer₁₉₉₇ had evolved an elaborate argument, which he fondly claimed to be “formal,” that we could always condition upon the belief that life has meaning; and so cases where superintelligences did not feel compelled to do anything in particular would fall out of consideration. (The flaw being the unconsidered and unjustified equation of “universally compelling argument” with “right.”)

So far, the young Eliezer is well on the way toward joining the “smart people who are stupid because they’re skilled at defending beliefs they arrived at for unskilled reasons” club. All his dedication to “rationality” has not saved him from this mistake, and you might be tempted to conclude that it is useless to strive for rationality.

But while many people dig holes for themselves, not everyone succeeds in clawing their way back out.

And from this I learn my lesson: That it all began—

—with a small, small question; a single discordant note; one tiny lonely thought ...

As our story starts, we advance three years to Eliezer₂₀₀₀, who in most respects resembles his self of 1997. He currently thinks he’s proven that building a superintelligence is the right thing to do if there is any right thing at all. From which it follows that there is no *justifiable* conflict of interest over the intelligence explosion among the peoples and persons of Earth.

This is an important conclusion for Eliezer₂₀₀₀, because he finds the notion of fighting over the intelligence explosion to be *unbearably* stupid. (Sort of like the notion of God intervening in fights between tribes of bickering barbarians, only in reverse.)

Eliezer₂₀₀₀'s self-concept does not permit him—he doesn't even *want*—to shrug and say, "Well, my side got here first, so we're going to seize the banana before anyone else gets it." It's a thought too painful to think.

And yet then the notion occurs to him:

Maybe some people would prefer an AI do particular things, such as not kill them, even if life is meaningless?

His immediately following thought is the obvious one, given his premises:

In the event that life is meaningless, nothing is the "right" thing to do; therefore it wouldn't be particularly right to respect people's preferences in this event.

This is the obvious dodge. The thing is, though, Eliezer₂₀₀₀ doesn't think of himself as a villain. He doesn't go around saying, "What bullets shall I dodge today?" He thinks of himself as a dutiful rationalist who tenaciously follows lines of inquiry. Later, he's going to look back and see a whole lot of inquiries that his mind somehow managed to not follow—but that's not his *current self-concept*.

So Eliezer₂₀₀₀ *doesn't* just grab the obvious out. He keeps thinking.

But if people believe they have preferences in the event that life is meaningless, then they have a motive to dispute my intelligence explosion project and go with a project that respects their wish in the event life is meaningless. This creates a present conflict of interest over the intelligence explosion, and prevents right things from getting done in the mainline event that life is meaningful.

Now, there's a *lot* of excuses Eliezer₂₀₀₀ could have potentially used to toss this problem out the window. I know, because

I've *heard* plenty of excuses for dismissing Friendly AI. "The problem is too hard to solve" is one I get from AGI wannabes who imagine themselves smart enough to create true Artificial Intelligence, but not smart enough to solve a really difficult problem like Friendly AI. Or "worrying about this possibility would be a poor use of resources, what with the incredible urgency of creating AI before humanity wipes itself out—you've got to go with what you have," this being uttered by people who just basically aren't interested in the problem.

But Eliezer₂₀₀₀ is a *perfectionist*. He's not perfect, obviously, and he doesn't attach as much importance as I do to the virtue of *precision*, but he is most certainly a *perfectionist*. The idea of metaethics that Eliezer₂₀₀₀ espouses, in which superintelligences know what's right better than we do, previously seemed to wrap up *all* the problems of justice and morality in an airtight wrapper.

The new objection seems to poke a minor hole in the airtight wrapper. This is worth patching. If you have something that's perfect, are you really going to let one little possibility compromise it?

So Eliezer₂₀₀₀ doesn't even *want* to drop the issue; he wants to patch the problem and restore perfection. How can he justify spending the time? By thinking thoughts like:

What about Brian Atkins? [Brian Atkins being the startup funder of the Machine Intelligence Research Institute, then called the Singularity Institute.] He would probably prefer not to die, even if life were meaningless. He's paying for MIRI right now; I don't want to taint the ethics of our cooperation.

Eliezer₂₀₀₀'s sentiment doesn't translate very well—English doesn't have a simple description for it, or any other culture I know. Maybe the passage in the Old Testament, "Thou shalt not boil a young goat in its mother's milk." Someone who helps you out of altruism shouldn't regret helping you; you owe them, not so much fealty, but rather, that they're actually doing what they think they're doing by helping you.

Well, but how would Brian Atkins find out, if I don't tell him? Eliezer₂₀₀₀ doesn't even *think* this except in quotation marks, as the obvious thought that a villain would think in the same situation. And Eliezer₂₀₀₀ has a standard counter-thought ready too, a ward against temptations to dishonesty—an argument that justifies honesty in terms of expected utility, not just a personal love of personal virtue:

Human beings aren't perfect deceivers; it's likely that I'll be found out. Or what if genuine lie detectors are invented before the Singularity, sometime over the next thirty years? I wouldn't be able to pass a lie detector test.

Eliezer₂₀₀₀ lives by the rule that you should always be ready to have your thoughts broadcast to the whole world at any time, without embarrassment. Otherwise, clearly, you've fallen from grace: either you're thinking something you shouldn't be thinking, or you're embarrassed by something that shouldn't embarrass you.

(These days, I don't espouse quite such an extreme viewpoint, mostly for reasons of Fun Theory. I see a role for continued social competition between intelligent life-forms, as least as far as my near-term vision stretches. I admit, these days, that it might be all right for human beings to have a self; as John McCarthy put it, "If everyone were to live for others all the time, life would be like a procession of ants following each other around in a circle." If you're going to have a self, you may as well have secrets, and maybe even conspiracies. But I do still try to abide by the principle of being able to pass a future lie detector test, with anyone else who's also willing to go under the lie detector, if the topic is a professional one. Fun Theory needs a commonsense exception for global catastrophic risk management.)

Even taking honesty for granted, there are other excuses Eliezer₂₀₀₀ could use to flush the question down the toilet. "The world doesn't have the time" or "It's unsolvable" would still work. But Eliezer₂₀₀₀ doesn't *know* that this problem, the

“backup” morality problem, is going to be particularly difficult or time-consuming. He’s just now thought of the whole issue.

And so Eliezer₂₀₀₀ *begins* to really consider the question: Supposing that “life is meaningless” (that superintelligences *don’t* produce their own motivations from pure logic), then how would you go about specifying a *fallback* morality? Synthesizing it, inscribing it into the AI?

There’s a lot that Eliezer₂₀₀₀ doesn’t know, at this point. But he *has* been thinking about self-improving AI for three years, and he’s been a Traditional Rationalist for longer than that. There are techniques of rationality that he *has* practiced, methodological safeguards he’s already devised. He already knows better than to think that all an AI needs is the One Great Moral Principle. Eliezer₂₀₀₀ already knows that it is wiser to think technologically than politically. He already knows the saying that AI programmers are supposed to think in code, to use concepts that can be inscribed in a computer. Eliezer₂₀₀₀ already has a concept that there is something called “technical thinking” and it is good, though he hasn’t yet formulated a Bayesian view of it. And he’s long since noticed that suggestively named LISP tokens don’t really mean anything, et cetera. These injunctions prevent him from falling into some of the initial traps, the ones that I’ve seen consume other novices on their own first steps into the Friendly AI problem...though technically this was my *second* step; I well and truly failed on my first.

But in the end, what it comes down to is this: For the first time, Eliezer₂₀₀₀ is trying to think technically about inscribing a morality into an AI, without the escape-hatch of the mysterious essence of rightness.

That’s the only thing that matters, in the end. His previous philosophizing wasn’t enough to force his brain to confront the details. This new standard is strict enough to require actual work. Morality slowly starts being less mysterious to him—Eliezer₂₀₀₀ is starting to think *inside* the black box.

His *reasons* for pursuing this course of action—those don’t matter at all.

Oh, there's a lesson in his being a perfectionist. There's a lesson in the part about how Eliezer₂₀₀₀ initially thought this was a tiny flaw, and could have dismissed it out-of-mind if that had been his impulse.

But in the end, the chain of cause and effect goes like this: Eliezer₂₀₀₀ investigated in more detail, therefore he got better with practice. Actions screen off justifications. If your arguments happen to justify not working things out in detail, like Eliezer₁₉₉₆, then you won't get good at thinking about the problem. If your arguments call for you to work things out in detail, then you have an *opportunity* to start accumulating expertise.

That was the only choice that mattered, in the end—not the *reasons* for doing anything.

I say all this, as you may well guess, because of the AI wannabes I sometimes run into who have their own clever reasons for not thinking about the Friendly AI problem. Our clever reasons for doing what we do tend to matter a lot less to Nature than they do to ourselves and our friends. If your actions don't look good when they're stripped of all their justifications and presented as mere brute facts ...then maybe you should re-examine them.

A diligent effort won't always save a person. There is such a thing as lack of ability. Even so, if you don't try, or don't try hard enough, you don't get a chance to sit down at the high-stakes table—never mind the ability ante. That's cause and effect for you.

Also, perfectionism really matters. The end of the world doesn't always come with trumpets and thunder and the highest priority in your inbox. Sometimes the shattering truth first presents itself to you as a small, small question; a single discordant note; one tiny lonely thought, that you could dismiss with one easy effortless touch ...

...and so, over succeeding years, understanding begins to dawn on that past Eliezer, slowly. That Sun rose slower than it could have risen.

Fighting a Rearguard Action Against the Truth



discussion
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6-c8

When we last left Eliezer₂₀₀₀, he was just beginning to investigate the question of how to inscribe a morality into an AI. His reasons for doing this don't matter at all, except insofar as they happen to historically demonstrate the importance of perfectionism. If you practice something, you may get better at it; if you investigate something, you may find out about it; the only thing that matters is that Eliezer₂₀₀₀ is, in fact, focusing his full-time energies on thinking technically about AI morality—rather than, as previously, finding any justification for not spending his time this way. In the end, this is all that turns out to matter.

But as our story begins—as the sky lightens to gray and the tip of the Sun peeks over the horizon—Eliezer₂₀₀₁ hasn't yet admitted that Eliezer₁₉₉₇ was *mistaken* in any important sense. He's just making Eliezer₁₉₉₇'s strategy *even better* by including a *contingency* plan for “the unlikely event that life turns out to be meaningless” ...

...which means that Eliezer₂₀₀₁ now has a line of retreat away from his mistake.

I don't just mean that Eliezer₂₀₀₁ can say “Friendly AI is a contingency plan,” rather than screaming “OOPS!”

I mean that Eliezer₂₀₀₁ now actually *has* a contingency plan. If Eliezer₂₀₀₁ starts to doubt his 1997 metaethics, the intelligence explosion has a fallback strategy, namely Friendly AI. Eliezer₂₀₀₁ can question his metaethics without it signaling the end of the world.

And his gradient has been smoothed; he can admit a 10% chance of having previously been wrong, then a 20% chance. He doesn't have to cough out his whole mistake in one huge lump.

If you think this sounds like Eliezer₂₀₀₁ is too slow, I quite agree.

Eliezer_{1996–2000}’s strategies had been formed in the total absence of “Friendly AI” as a consideration. The whole idea was to get a superintelligence, *any* superintelligence, as fast as possible—codelet soup, ad-hoc heuristics, evolutionary programming, open-source, anything that looked like it might work—preferably all approaches simultaneously in a Manhattan Project. (“All parents did the things they tell their children not to do. That’s how they know to tell them not to do it.”¹) It’s not as if adding one more approach could *hurt*.

His attitudes toward technological progress have been formed—or more accurately, preserved from childhood-absorbed technophilia—around the assumption that any/all movement toward superintelligence is a pure good without a hint of danger.

Looking back, what Eliezer₂₀₀₁ *needed* to do at this point was declare an HMC event—Halt, Melt, and Catch Fire. One of the foundational assumptions on which everything else has been built has been revealed as flawed. This calls for a mental brake to a full stop: take your weight off all beliefs built on the wrong assumption, do your best to rethink everything from scratch. This is an art I need to write more about—it’s akin to the convulsive effort required to seriously clean house, after an adult religionist notices for the first time that God doesn’t exist.

But what Eliezer₂₀₀₁ actually did was rehearse his previous technophilic arguments for why it’s difficult to ban or governmentally control new technologies—the standard arguments against “relinquishment.”

It does seem even to my modern self that all those awful consequences which technophiles argue to follow from various kinds of government regulation are more or less correct—it’s much easier to say what someone is doing wrong, than to say the way that is right. My modern viewpoint hasn’t shifted to think that technophiles are wrong about the downsides of

1. John Moore, *Slay and Rescue* (Xlibris Corp, 2000).

technophobia; but I do tend to be a lot more sympathetic to what technophobes say about the downsides of technophilia. What previous Eliezers said about the difficulties of, e.g., the government doing anything sensible about Friendly AI, still seems pretty true. It's just that a lot of his hopes for science, or private industry, etc., now seem equally wrongheaded.

Still, let's not get into the details of the technovolatile viewpoint. Eliezer₂₀₀₁ has just tossed a major foundational assumption—that AI can't be dangerous, unlike other technologies—out the window. You would intuitively suspect that this should have some kind of large effect on his strategy.

Well, Eliezer₂₀₀₁ did at least give up on his 1999 idea of an open-source AI Manhattan Project using self-modifying heuristic soup, but overall ...

Overall, he'd previously wanted to charge in, guns blazing, immediately using his best idea at the time; and afterward he still wanted to charge in, guns blazing. He didn't say, "I don't know how to do this." He didn't say, "I need better knowledge." He didn't say, "This project is not yet ready to start coding." It was still all, "The clock is ticking, gotta move now! MIRI will start coding as soon as it's got enough money!"

Before, he'd wanted to focus as much scientific effort as possible with full information-sharing, and afterward he still thought in those terms. Scientific secrecy = bad guy, openness = good guy. (Eliezer₂₀₀₁ hadn't read up on the Manhattan Project and wasn't familiar with the similar argument that Leó Szilárd had with Enrico Fermi.)

That's the problem with converting one big "Oops!" into a gradient of shifting probability. It means there isn't a single watershed moment—a visible huge impact—to hint that equally huge changes might be in order.

Instead, there are all these little opinion shifts ...that give you a chance to repair the *arguments* for your strategies; to shift the justification a little, but keep the "basic idea" in place. Small shocks that the system can absorb without cracking, because each time, it gets a chance to go back and repair itself. It's just that in the domain of rationality, cracking = good, repair = bad.

In the art of rationality it's far more efficient to admit one huge mistake, than to admit lots of little mistakes.

There's some kind of instinct humans have, I think, to preserve their former strategies and plans, so that they aren't constantly thrashing around and wasting resources; and of course an instinct to preserve any position that we have publicly argued for, so that we don't suffer the humiliation of being wrong. And though the younger Eliezer has striven for rationality for many years, he is not immune to these impulses; they waft gentle influences on his thoughts, and this, unfortunately, is more than enough damage.

Even in 2002, the earlier Eliezer isn't yet *sure* that Eliezer₁₉₉₇'s plan *couldn't possibly* have worked. It *might* have gone right. You never know, right?

But there came a time when it all fell crashing down.

My Naturalistic Awakening



discussion
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6-c9

In the previous episode, Eliezer₂₀₀₁ is fighting a rearguard action against the truth. Only gradually shifting his beliefs, admitting an increasing probability in a different scenario, but never saying outright, “I was wrong before.” He repairs his strategies as they are challenged, finding new justifications for just the same plan he pursued before.

(Of which it is therefore said: “Beware lest you fight a rear-guard retreat against the evidence, grudgingly conceding each foot of ground only when forced, feeling cheated. Surrender to the truth as quickly as you can. Do this the instant you realize what you are resisting; the instant you can see from which quarter the winds of evidence are blowing against you.”)

Memory fades, and I can hardly bear to look back upon those times—no, seriously, I can’t *stand* reading my old writing. I’ve already been corrected once in my recollections, by those who were present. And so, though I remember the important events, I’m not really sure what *order* they happened in, let alone what year.

But if I had to pick a moment when my folly broke, I would pick the moment when I first comprehended, in full generality, the notion of an optimization process. That was the point at which I first looked back and said, “I’ve been a fool.”

Previously, in 2002, I’d been writing a bit about the evolutionary psychology of human general intelligence—though at the time, I thought I was writing about AI; at this point I thought I was against anthropomorphic intelligence, but I was still looking to the human brain for inspiration. (The paper in question is “Levels of Organization in General Intelligence,” a requested

chapter for the volume *Artificial General Intelligence*,¹ which finally came out in print in 2007.)

So I'd been thinking (and writing) about how natural selection managed to cough up human intelligence; I saw a *dichotomy* between them, the blindness of natural selection and the lookahead of intelligent foresight, reasoning by simulation versus playing everything out in reality, abstract versus concrete thinking. And yet it was natural selection that created human intelligence, so that our brains, though not our thoughts, are entirely made according to the signature of natural selection.

To this day, this still seems to me like a reasonably shattering insight, and so it drives me up the wall when people lump together natural selection and intelligence-driven processes as “evolutionary.” They really are almost absolutely different in a number of important ways—though there are concepts in common that can be used to describe them, like consequentialism and cross-domain generality.

But that Eliezer₂₀₀₂ is thinking in terms of a *dichotomy* between evolution and intelligence tells you something about the limits of his vision—like someone who thinks of politics as a dichotomy between conservative and liberal stances, or someone who thinks of fruit as a dichotomy between apples and strawberries.

After the “Levels of Organization” draft was published online, Emil Gilliam pointed out that my view of AI seemed pretty similar to my view of intelligence. Now, of course Eliezer₂₀₀₂ doesn't espouse building an AI in the image of a human mind; Eliezer₂₀₀₂ knows very well that a human mind is just a hack coughed up by natural selection. But Eliezer₂₀₀₂ has described these levels of organization in human thinking, and he hasn't proposed using different levels of organization in the AI. Emil Gilliam asks whether I think I might be hewing too close to the human line. I dub the alternative the “Completely Alien

1. Ben Goertzel and Cassio Pennachin, eds., *Artificial General Intelligence*, Cognitive Technologies (Berlin: Springer, 2007), doi:10.1007/978-3-540-68677-4.

Mind Design” and reply that a CAMD is probably too difficult for human engineers to create, even if it’s possible in theory, because we wouldn’t be able to understand something so alien while we were putting it together.

I don’t know if Eliezer₂₀₀₂ invented this reply on his own, or if he read it somewhere else. Needless to say, I’ve heard this excuse plenty of times since then. In reality, what you genuinely understand, you can usually reconfigure in almost any sort of shape, leaving some structural essence inside; but when you don’t understand flight, you suppose that a flying machine needs feathers, because you can’t imagine departing from the analogy of a bird.

So Eliezer₂₀₀₂ is still, in a sense, attached to humanish mind designs—he imagines improving on them, but the human *architecture* is still in some sense his point of departure.

What is it that finally breaks this attachment?

It’s an embarrassing confession: It came from a science fiction story I was trying to write. (No, you can’t see it; it’s not done.) The story involved a non-cognitive non-evolutionary optimization process, something like an Outcome Pump. Not intelligence, but a cross-temporal physical effect—that is, I was imagining it as a physical effect—that narrowly constrained the space of possible outcomes. (I can’t tell you any more than that; it would be a spoiler, if I ever finished the story. Just see the essay on Outcome Pumps, vol. III, p. 118.) It was “just a story,” and so I was free to play with the idea and elaborate it out logically: *C* was constrained to happen, therefore *B* (in the past) was constrained to happen, therefore *A* (which led to *B*) was constrained to happen.

Drawing a line through one point is generally held to be dangerous. Two points make a dichotomy; you imagine them opposed to one another. But when you’ve got three different points—*that’s* when you’re forced to wake up and generalize.

Now I had three points: Human intelligence, natural selection, and my fictional plot device.

And so that was the point at which I generalized the notion of an optimization process, of a process that squeezes the future into a narrow region of the possible.

This may seem like an obvious point, if you’ve been following *Overcoming Bias* this whole time; but if you look at Shane Legg’s collection of 71 definitions of intelligence,² you’ll see that “squeezing the future into a constrained region” is a less obvious reply than it seems.

Many of the definitions of “intelligence” by AI researchers do talk about “solving problems” or “achieving goals.” But from the viewpoint of past Eliezers, at least, it is only hindsight that makes this the same thing as “squeezing the future.”

A *goal* is a mentalistic object; electrons have no goals, and solve no problems either. When a human imagines a goal, they imagine an agent imbued with wanting-ness—it’s still empathic language.

You can espouse the notion that intelligence is about “achieving goals”—and then turn right around and argue about whether some “goals” are better than others—or talk about the wisdom required to judge between goals themselves—or talk about a system deliberately modifying its goals—or talk about the free will needed to *choose* plans that achieve goals—or talk about an AI realizing that its goals aren’t what the programmers really meant to ask for. If you imagine something that squeezes the future into a narrow region of the possible, like an Outcome Pump, those seemingly sensible statements somehow don’t translate.

So for me at least, seeing through the word “mind” to a physical process that would, just by naturally running, just by obeying the laws of physics, end up squeezing its future into a narrow region, was a naturalistic enlightenment over and above the notion of an agent trying to achieve its goals.

It was like falling out of a deep pit, falling into the ordinary world, strained cognitive tensions relaxing into unforced simplicity, confusion turning to smoke and drifting away. I saw the *work performed* by intelligence; *smart* was no longer a property,

2. sharov.me/r/6-1
arxiv.org



but an engine. Like a knot in time, echoing the outer part of the universe in the inner part, and thereby steering it. I even saw, in a flash of the same enlightenment, that a mind had to output waste heat in order to obey the laws of thermodynamics.

Previously, Eliezer₂₀₀₁ had talked about Friendly AI as something you should do just to be sure—if you didn’t know whether AI design *X* was going to be Friendly, then you really ought to go with AI design *Y* that you did know would be Friendly. But Eliezer₂₀₀₁ didn’t think he *knew* whether you could *actually* have a superintelligence that turned its future light cone into paperclips.

Now, though, I could *see* it—the pulse of the optimization process, sensory information surging in, motor instructions surging out, steering the future. In the middle, the model that linked up possible actions to possible outcomes, and the utility function over the outcomes. Put in the corresponding utility function, and the result would be an optimizer that would steer the future anywhere.

Up until that point, I’d never quite admitted to myself that Eliezer₁₉₉₇’s AI goal system design would definitely, no two ways about it, pointlessly wipe out the human species. Now, however, I looked back, and I could finally see *what my old design really did*, to the extent it was coherent enough to be talked about. Roughly, it would have converted its future light cone into generic tools—computers without programs to run, stored energy without a use ...

...how on Earth had I, the fine and practiced rationalist—how on Earth had I managed to miss something that obvious, for six damned years?

That was the point at which I awoke clear-headed, and remembered; and thought, with a certain amount of embarrassment: *I’ve been stupid.*

The Level Above Mine

I once lent Xiaoguang “Mike” Li my copy of *Probability Theory: The Logic of Science*. Mike Li read some of it, and then came back and said:

Wow ...it’s like Jaynes is a thousand-year-old vampire.

Then Mike said, “No, wait, let me explain that—” and I said, “No, I know exactly what you mean.” It’s a convention in fantasy literature that the older a vampire gets, the more powerful they become.

I’d enjoyed math proofs before I encountered Jaynes. But E. T. Jaynes was the first time I picked up a sense of *formidability* from mathematical arguments. Maybe because Jaynes was lining up “paradoxes” that had been used to object to Bayesianism, and then blasting them to pieces with overwhelming firepower—power being used to overcome others. Or maybe the sense of formidability came from Jaynes not treating his math as a game of aesthetics; Jaynes *cared* about probability theory, it was bound up with other considerations that mattered, to him and to me too.

For whatever reason, the sense I get of Jaynes is one of terrifying swift perfection—something that would arrive at the correct answer by the shortest possible route, tearing all surrounding mistakes to shreds in the same motion. Of course, when you write a book, you get a chance to show only your best side. But still.

It spoke well of Mike Li that he was able to sense the aura of formidability surrounding Jaynes. It’s a general rule, I’ve observed, that you can’t discriminate between levels too far above your own. E.g., someone once earnestly told me that I was really bright, and “ought to go to college.” Maybe anything more than around one standard deviation above you starts to blur together, though that’s just a cool-sounding wild guess.



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So, having heard Mike Li compare Jaynes to a thousand-year-old vampire, one question immediately popped into my mind:

“Do you get the same sense off me?” I asked.

Mike shook his head. “Sorry,” he said, sounding somewhat awkward, “it’s just that Jaynes is ...”

“No, I know,” I said. I hadn’t thought I’d reached Jaynes’s level. I’d only been curious about how I came across to other people.

I *aspire* to Jaynes’s level. I *aspire* to become as much the master of Artificial Intelligence / reflectivity, as Jaynes was master of Bayesian probability theory. I can even plead that the art I’m trying to master is more difficult than Jaynes’s, making a mockery of deference. Even so, and embarrassingly, there is *no* art of which I am as much the master now, as Jaynes was of probability theory.

This is not, necessarily, to place myself beneath Jaynes as a person—to say that Jaynes had a magical aura of destiny, and I don’t.

Rather I recognize in Jaynes *a level of expertise, of sheer formidability*, which I have not yet achieved. I can argue forcefully in my chosen subject, but that is not the same as writing out the equations and saying: **DONE**.

For so long as I have not yet achieved that level, I must acknowledge the possibility that I can never achieve it, that my native talent is not sufficient. When Marcello Herreshoff had known me for long enough, I asked him if he knew of anyone who struck him as substantially more *natively intelligent* than myself. Marcello thought for a moment and said “John Conway—I met him at a summer math camp.” *Darn*, I thought, *he thought of someone, and worse, it’s some ultra-famous old guy I can’t grab*. I inquired how Marcello had arrived at the judgment. Marcello said, “He just struck me as having a tremendous amount of mental horsepower,” and started to explain a math problem he’d had a chance to work on with Conway.

Not what I wanted to hear.

Perhaps, relative to Marcello’s experience of Conway and his experience of me, I haven’t had a chance to show off on

any subject that I've mastered as thoroughly as Conway had mastered his many fields of mathematics.

Or it might be that Conway's brain is specialized off in a different direction from mine, and that I could never approach Conway's level on math, yet Conway wouldn't do so well on AI research.

Or ...

...or I'm strictly dumber than Conway, dominated by him along all dimensions. Maybe, if I could find a young proto-Conway and tell them the basics, they would blaze right past me, solve the problems that have weighed on me for years, and zip off to places I can't follow.

Is it damaging to my ego to confess that last possibility? Yes. It would be futile to deny that.

Have I *really* accepted that awful possibility, or am I only pretending to myself to have accepted it? Here I will say: "No, I think I have accepted it." Why do I dare give myself so much credit? Because I've invested specific effort into that awful possibility. I am writing here for many reasons, but a major one is the vision of some younger mind reading these words and zipping off past me. It might happen, it might not.

Or sadder: Maybe I just wasted too much time on setting up the resources to support me, instead of studying math full-time through my whole youth; or I wasted too much youth on non-mathy ideas. And this choice, my past, is irrevocable. I'll hit a brick wall at 40, and there won't be anything left but to pass on the resources to another mind with the potential I wasted, still young enough to learn. So to save them time, I should leave a trail to my successes, and post warning signs on my mistakes.

Such *specific efforts* predicated on an ego-damaging possibility—that's the only kind of humility that seems real enough for me to dare credit myself. Or giving up my precious theories, when I realized that they didn't meet the standard Jaynes had shown me—that was hard, and it was real. Modest demeanors are cheap. Humble admissions of doubt are cheap. I've known too many people who, presented with a counterargument, say, "I am but a fallible mortal, of course I

could be wrong,” and then go on to do exactly what they had planned to do previously.

You’ll note that I don’t try to modestly say anything like, “Well, I may not be as brilliant as Jaynes or Conway, but that doesn’t mean I can’t do important things in my chosen field.”

Because I do know ...that’s not how it works.

The Magnitude of His Own Folly

In the years before I met a would-be creator of Artificial General Intelligence (with a funded project) who happened to be a creationist, I would still try to argue with individual AGI wannabes.

In those days, I sort-of-succeeded in convincing one such fellow that, yes, you had to take Friendly AI into account, and no, you couldn't just find the right fitness metric for an evolutionary algorithm. (Previously he had been very impressed with evolutionary algorithms.)

And the one said: *Oh, woe! Oh, alas! What a fool I've been! Through my carelessness, I almost destroyed the world! What a villain I once was!*

Now, *there's* a trap I knew better than to fall into—
—at the point where, in late 2002, I looked back to Eliezer¹⁹⁹⁷'s AI proposals and realized what they really would have done, insofar as they were coherent enough for me to talk about what they “really would have done.”

When I finally saw the magnitude of my own folly, everything fell into place at once. The dam against realization cracked; and the unspoken doubts that had been accumulating behind it crashed through all together. There wasn't a prolonged period, or even a single moment that I remember, of wondering how I could have been so stupid. I already knew how.

And I also knew, all at once, in the same moment of realization, that to say, *I almost destroyed the world!*, would have been too prideful.

It would have been too confirming of ego, too confirming of my own importance in the scheme of things, at a time when—I understood in the same moment of realization—my ego ought to be taking a major punch to the stomach. I had been so much less than I needed to be; I had to take that punch in the stomach, not avert it.



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And by the same token, I didn't fall into the conjugate trap of saying: *Oh, well, it's not as if I had code and was about to run it; I didn't really come close to destroying the world.* For that, too, would have minimized the force of the punch. *It wasn't really loaded?* I had proposed and intended to build the gun, and load the gun, and put the gun to my head and pull the trigger; and that was a bit too much self-destructiveness.

I didn't make a grand emotional drama out of it. That would have wasted the force of the punch, averted it into mere tears.

I knew, in the same moment, what I had been carefully not-doing for the last six years. I hadn't been updating.

And I knew I had to finally update. To actually *change* what I planned to do, to change what I was doing now, to do something different instead.

I knew I had to stop.

Halt, melt, and catch fire.

Say, "I'm not ready." Say, "I don't know how to do this yet."

These are terribly difficult words to say, in the field of AGI. Both the lay audience and your fellow AGI researchers are interested in code, projects with programmers in play. Failing that, they may give you some credit for saying, "I'm ready to write code; just give me the funding."

Say, "I'm not ready to write code," and your status drops like a depleted uranium balloon.

What distinguishes you, then, from six billion other people who don't know how to create Artificial General Intelligence? If you don't have neat code (that does something other than be humanly intelligent, obviously; but at least it's code), or at minimum your own startup that's going to write code as soon as it gets funding—then who are you and what are you doing at our conference?

Maybe later I'll write on where this attitude comes from—the excluded middle between "I know how to build AGI!" and "I'm working on narrow AI because I don't know how to build AGI," the nonexistence of a concept for "I am trying to get from an incomplete map of FAI to a complete map of FAI."

But this attitude does exist, and so the loss of status associated with saying "I'm not ready to write code" is very great. (If

the one doubts this, let them name any other who simultaneously says “I intend to build an Artificial General Intelligence,” “Right now I can’t build an AGI because I don’t know *X*,” and “I am currently trying to figure out *X*.”)

(And never mind AGI folk who’ve already raised venture capital, promising returns in five years.)

So there’s a huge reluctance to say, “Stop.” You can’t just say, “Oh, I’ll swap back to figure-out-*X* mode,” because that mode doesn’t exist.

Was there more to that reluctance than just loss of status, in my case? Eliezer₂₀₀₁ might also have flinched away from slowing his perceived forward momentum into the intelligence explosion, which was so right and so necessary ...

But mostly, I think I flinched away from not being able to say, “I’m ready to start coding.” Not just for fear of others’ reactions, but because I’d been inculcated with the same attitude myself.

Above all, Eliezer₂₀₀₁ didn’t say, “Stop”—even *after* noticing the problem of Friendly AI—because I did not realize, on a gut level, that Nature was allowed to kill me.

“Teenagers think they’re immortal,” the proverb goes. Obviously this isn’t true in the literal sense that if you ask them, “Are you indestructible?” they will reply “Yes, go ahead and try shooting me.” But perhaps wearing seat belts isn’t deeply emotionally compelling for them, because the thought of their own death isn’t quite *real*—they don’t really believe it’s allowed to happen. It can happen in *principle* but it can’t *actually* happen.

Personally, I always wore my seat belt. As an individual, I understood that I could die.

But, having been raised in technophilia to treasure that one most precious thing, far more important than my own life, I once thought that the Future was indestructible.

Even when I acknowledged that nanotech could wipe out humanity, I still believed the intelligence explosion was invulnerable. That if humanity survived, the intelligence explosion would happen, and the resultant AI would be too smart to be corrupted or lost.

Even after *that*, when I acknowledged Friendly AI as a consideration, I didn’t emotionally believe in the possibility of failure,

any more than that teenager who doesn't wear their seat belt *really* believes that an automobile accident is *really* allowed to kill or cripple them.

It wasn't until my insight into optimization let me look back and see Eliezer¹⁹⁹⁷ in plain light that I realized that Nature was allowed to kill me.

"The thought you cannot think controls you more than thoughts you speak aloud." But we flinch away from only those fears that are real to us.

AGI researchers take very seriously the prospect of *someone else solving the problem first*. They can imagine seeing the headlines in the paper saying that their own work has been upstaged. They know that Nature is allowed to do that to them. The ones who have started companies know that they are allowed to run out of venture capital. That possibility is *real* to them, very real; it has a power of emotional compulsion over them.

I don't think that "Oops" followed by the thud of six billion bodies falling, *at their own hands*, is real to them on quite the same level.

It is unsafe to say what other people are thinking. But it seems rather likely that when the one reacts to the prospect of Friendly AI by saying, "If you delay development to work on safety, other projects that don't care *at all* about Friendly AI will beat you to the punch," the prospect of they themselves making a mistake followed by six billion thuds is not really real to them; but the possibility of others beating them to the punch is deeply scary.

I, too, used to say things like that, before I understood that Nature was allowed to kill me.

In that moment of realization, my childhood technophilia finally broke.

I finally understood that even if you diligently followed the rules of science and were a nice person, Nature could still kill you. I finally understood that even if you were the best project out of all available candidates, Nature could still kill you.

I understood that I was not being graded on a curve. My gaze shook free of rivals, and I saw the sheer blank wall.

I looked back and I saw the careful arguments I had constructed, for why the wisest choice was to continue forward at full speed, just as I had planned to do before. And I understood then that even if you constructed an argument showing that something was the best course of action, Nature was still allowed to say “So what?” and kill you.

I looked back and saw that I had claimed to take into account the risk of a fundamental mistake, that I had argued reasons to tolerate the risk of proceeding in the absence of full knowledge.

And I saw that the risk I wanted to tolerate would have killed me. And I saw that this possibility had never been *really* real to me. And I saw that even if you had wise and excellent arguments for taking a risk, the risk was still allowed to go ahead and kill you. *Actually* kill you.

For it is only the action that matters, and not the reasons for doing anything. If you build the gun and load the gun and put the gun to your head and pull the trigger, even with the cleverest of arguments for carrying out every step—then, bang.

I saw that only my own ignorance of the rules had enabled me to argue for going ahead without complete knowledge of the rules; for if you do not know the rules, you cannot model the penalty of ignorance.

I saw that others, still ignorant of the rules, were saying, “I will go ahead and do X”; and that to the extent that X was a coherent proposal at all, I knew that would result in a bang; but they said, “I do not know it cannot work.” I would try to explain to them the smallness of the target in the search space, and they would say “How can you be so sure I won’t win the lottery?” wielding their own ignorance as a bludgeon.

And so I realized that the only thing I *could* have done to save myself, in my previous state of ignorance, was to say: “I will not proceed until I know positively that the ground is safe.” And there are many clever arguments for why you should step on a piece of ground that you don’t know to contain a landmine; but they all sound much less clever, after you look to the place that you proposed and intended to step, and see the bang.

I understood that you could do *everything that you were supposed to do*, and Nature was still allowed to kill you. That

was when my last trust broke. And that was when my training as a rationalist began.

Beyond the Reach of God

This essay is a tad gloomier than usual, as I measure such things. It deals with a thought experiment I invented to smash my own optimism, after I realized that optimism had misled me. Those readers sympathetic to arguments like, “It’s important to keep our biases because they help us stay happy,” should consider not reading. (Unless they have something to protect, including their own life.)

So! Looking back on the magnitude of my own folly, I realized that at the root of it had been a disbelief in the Future’s vulnerability—a reluctance to accept that things could *really* turn out wrong. Not as the result of any explicit propositional verbal belief. More like something inside that persisted in believing, even in the face of adversity, that everything would be all right in the end.

Some would account this a virtue (*zettai daijobu da yo*), and others would say that it’s a thing necessary for mental health.

But we don’t live in that world. We live in the world beyond the reach of God.

It’s been a long, long time since I believed in God. Growing up in an Orthodox Jewish family, I can recall the last remembered time I asked God for something, though I don’t remember how old I was. I was putting in some request on behalf of the next-door-neighboring boy, I forget what exactly—something along the lines of, “I hope things turn out all right for him,” or maybe, “I hope he becomes Jewish.”

I remember what it was like to have some higher authority to appeal to, to take care of things I couldn’t handle myself. I didn’t think of it as “warm,” because I had no alternative to compare it to. I just took it for granted.

Still I recall, though only from distant childhood, what it’s like to live in the conceptually impossible possible world where God exists. *Really* exists, in the way that children and rationalists take all their beliefs at face value.



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In the world where God exists, does God intervene to optimize *everything*? Regardless of what rabbis assert about the fundamental nature of reality, the take-it-seriously operational answer to this question is obviously “No.” You can’t ask God to bring you a lemonade from the refrigerator instead of getting one yourself. When I believed in God after the serious fashion of a child, so very long ago, I didn’t believe that.

Postulating that particular divine inaction doesn’t provoke a full-blown theological crisis. If you said to me, “I have constructed a benevolent superintelligent nanotech-user,” and I said “Give me a banana,” and no banana appeared, this would not *yet* disprove your statement. Human parents don’t always do everything their children ask. There are some decent fun-theoretic arguments—I even believe them myself—against the idea that the *best* kind of help you can offer someone is to always immediately give them everything they want. I don’t think that eudaimonia is formulating goals and having them instantly fulfilled; I don’t *want* to become a simple wanting-thing that never has to plan or act or think.

So it’s not necessarily an attempt to avoid falsification to say that God does not grant all prayers. Even a Friendly AI might not respond to every request.

But clearly there exists *some* threshold of horror awful enough that God will intervene. I remember that being true, when I believed after the fashion of a child.

The God who does not intervene *at all*, no matter how bad things get—*that’s* an obvious attempt to avoid falsification, to protect a belief-in-belief. Sufficiently young children don’t have the deep-down knowledge that God doesn’t really exist. They really expect to see a dragon in their garage. They have no reason to imagine a loving God who never acts. Where exactly is the boundary of sufficient awfulness? Even a child can imagine arguing over the precise threshold. But of course God will draw the line somewhere. Few indeed are the loving parents who, desiring their child to grow up strong and self-reliant, would let their toddler be run over by a car.

The obvious example of a horror so great that God cannot tolerate it is death—true death, mind-annihilation. I don’t think

that even Buddhism allows that. So long as there is a God in the classic sense—full-blown, ontologically fundamental, *the* God—we can rest assured that no *sufficiently* awful event will ever, ever happen. There is no soul anywhere that need fear true annihilation; God will prevent it.

What if you build your own simulated universe? The classic example of a simulated universe is Conway's Game of Life. I do urge you to investigate Life¹ if you've never played it—it's important for comprehending the notion of "physical law." Conway's Life has been proven Turing-complete, so it would be possible to build a sentient being in the Life universe, although it might be rather fragile and awkward. Other cellular automata would make it simpler.

Could you, by creating a simulated universe, escape the reach of God? Could you simulate a Game of Life containing sentient entities, and torture the beings therein? But if God is watching everywhere, then trying to build an unfair Life just results in *the* God stepping in to modify your computer's transistors. If the physics you set up in your computer program calls for a sentient Life-entity to be endlessly tortured for no particular reason, *the* God will intervene. God being omnipresent, there is no refuge *anywhere* for true horror. Life is fair.

But suppose that instead you ask the question:

Given such-and-such initial conditions, and *given* such-and-such cellular automaton rules, what *would be* the mathematical result?

Not even God can modify the answer to this question, unless you believe that God can implement logical impossibilities. Even as a very young child, I don't remember believing that. (And why would you need to believe it, if God can modify anything that *actually* exists?)

What does Life look like, in this imaginary world where every step follows *only* from its immediate predecessor? Where

1. sharov.me/r/6-2

Conway's Game of Life - Wikipedia • en.wikipedia.org



things *only* ever happen, or don't happen, because of the cellular automaton rules? Where the initial conditions and rules *don't* describe any God that checks over each state? What does it look like, the world beyond the reach of God?

That world wouldn't be fair. If the initial state contained the seeds of something that could self-replicate, natural selection might or might not take place, and complex life might or might not evolve, and that life might or might not become sentient, with no God to guide the evolution. That world might evolve the equivalent of conscious cows, or conscious dolphins, that lacked hands to improve their condition; maybe they would be eaten by conscious wolves who never thought that they were doing wrong, or cared.

If in a vast plethora of worlds, something like humans evolved, then they would suffer from diseases—not to teach them any lessons, but only because viruses happened to evolve as well, under the cellular automaton rules.

If the people of that world are happy, or unhappy, the causes of their happiness or unhappiness may have nothing to do with good or bad choices they made. Nothing to do with free will or lessons learned. In the what-if world where every step follows only from the cellular automaton rules, the equivalent of Genghis Khan can murder a million people, and laugh, and be rich, and never be punished, and live his life much happier than the average. Who prevents it? God would prevent it from ever *actually* happening, of course; He would at the very least visit some shade of gloom in the Khan's heart. But in the mathematical answer to the question *What if?* there is no God in the axioms. So if the cellular automaton rules say that the Khan is happy, that, simply, is the whole and only answer to the what-if question. There is nothing, absolutely nothing, to prevent it.

And if the Khan tortures people horribly to death over the course of days, for his own amusement perhaps? They will call out for help, perhaps imagining a God. And if you *really* wrote that cellular automaton, God would intervene in your program, of course. But in the what-if question, what the cellular automaton *would* do under the mathematical rules, there isn't any God in the system. Since the physical laws contain no

specification of a utility function—in particular, no prohibition against torture—then the victims will be saved only if the right cells happen to be 0 or 1. And it's not likely that anyone will defy the Khan; if they did, someone would strike them with a sword, and the sword would disrupt their organs and they would die, and that would be the end of that. So the victims die, screaming, and no one helps them; that is the answer to the what-if question.

Could the victims be completely innocent? Why not, in the what-if world? If you look at the rules for Conway's Game of Life (which is Turing-complete, so we can embed arbitrary computable physics in there), then the rules are really very simple. Cells with three living neighbors become alive and stay alive; cells with two neighbors stay the same; all other cells die. There isn't anything in there about innocent people not being horribly tortured for indefinite periods.

Is this world starting to sound familiar?

Belief in a fair universe often manifests in more subtle ways than thinking that horrors should be outright prohibited: Would the twentieth century have gone differently, if Klara Pölzl and Alois Hitler had made love one hour earlier, and a different sperm fertilized the egg, on the night that Adolf Hitler was conceived?

For so many lives and so much loss to turn on a single event seems *disproportionate*. The Divine Plan ought to make more *sense* than that. You can believe in a Divine Plan without believing in God—Karl Marx surely did. You shouldn't have millions of lives depending on a casual choice, an hour's timing, the speed of a microscopic flagellum. It ought not to be allowed. It's *too* disproportionate. Therefore, if Adolf Hitler had been able to go to high school and become an architect, there would have been someone else to take his role, and World War II would have happened the same as before.

But in the world beyond the reach of God, there isn't any clause in the physical axioms that says "things have to make sense" or "big effects need big causes" or "history runs on reasons too important to be so fragile." There is no God to *impose* that order, which is so severely violated by having the

lives and deaths of millions depend on one small molecular event.

The point of the thought experiment is to lay out the God-universe and the Nature-universe side by side, so that we can recognize what kind of thinking belongs to the God-universe. Many who are atheists still think as if certain things are *not allowed*. They would lay out arguments for why World War II was inevitable and would have happened in more or less the same way, even if Hitler had become an architect. But in sober historical fact, this is an unreasonable belief; I chose the example of World War II because from my reading, it seems that events were mostly driven by Hitler's personality, often in defiance of his generals and advisors. There is no particular empirical justification that I happen to have heard of for doubting this. The main reason to doubt would be *refusal to accept* that the universe could make so little sense—that horrible things could happen so *lightly*, for no more reason than a roll of the dice.

But why not? What prohibits it?

In the God-universe, God prohibits it. To recognize this is to recognize that we don't live in that universe. We live in the what-if universe beyond the reach of God, driven by the mathematical laws and nothing else. Whatever physics says will happen, will happen. Absolutely *anything*, good or bad, will happen. And there is nothing in the laws of physics to lift this rule even for the *really extreme* cases, where you might expect Nature to be a little more reasonable.

Reading William Shirer's *The Rise and Fall of the Third Reich*, listening to him describe the disbelief that he and others felt upon discovering the full scope of Nazi atrocities, I thought of what a strange thing it was, to read all that, and know, already, that there wasn't a single protection against it. To just read through the whole book and accept it; horrified, but not at all disbelieving, because I'd already understood what kind of world I lived in.

Once upon a time, I believed that the extinction of humanity was not allowed. And others who call themselves rationalists may yet have things they trust. They might be called "positive-

sum games,” or “democracy,” or “technology,” but they are sacred. The mark of this sacredness is that the trustworthy thing can’t lead to anything *really* bad; or they can’t be *permanently* defaced, at least not without a compensatory silver lining. In that sense they can be trusted, even if a few bad things happen here and there.

The unfolding history of Earth can’t ever turn from its positive-sum trend to a negative-sum trend; that is not allowed. Democracies—*modern liberal* democracies, anyway—won’t ever legalize torture. Technology has done so much good up until now, that there can’t possibly be a Black Swan technology that breaks the trend and does more harm than all the good up until this point.

There are all sorts of clever arguments why such things can’t possibly happen. But the source of these arguments is a much deeper belief that such things are *not allowed*. Yet who prohibits? Who prevents it from happening? If you can’t visualize at least one lawful universe where physics say that such dreadful things happen—and so they *do* happen, there being nowhere to appeal the verdict—then you aren’t yet ready to argue *probabilities*.

Could it really be that sentient beings have died absolutely for thousands or millions of years, with no soul and no afterlife—and *not* as part of any grand plan of Nature—*not* to teach any great lesson about the meaningfulness or meaninglessness of life—not even to teach any profound lesson about what is impossible—so that a trick as simple and stupid-sounding as vitrifying people in liquid nitrogen can save them from total annihilation—and a 10-second rejection of the silly idea can destroy someone’s soul? Can it be that a computer programmer who signs a few papers and buys a life-insurance policy continues into the far future, while Einstein rots in a grave?² We can be sure of one thing: God wouldn’t allow it. Anything that

2. Edit 2018: Not all of Einstein is buried, see sharov.me/r/6-3



ridiculous and disproportionate would be ruled out. It would make a mockery of the Divine Plan—a mockery of the *strong reasons* why things must be the way they are.

You can have secular rationalizations for things being *not allowed*. So it helps to imagine that there *is* a God, benevolent as you understand goodness—a God who enforces throughout Reality a *minimum* of fairness and justice—whose plans make sense and depend proportionally on people’s choices—who will never permit absolute horror—who does not always intervene, but who at least prohibits universes wrenched *completely* off their track ...to imagine all this, but also imagine that *you*, yourself, live in a what-if world of pure mathematics—a world beyond the reach of God, an utterly unprotected world where anything at all can happen.

If there’s any reader still reading this who thinks that being happy counts for more than anything in life, then maybe they *shouldn’t* spend much time pondering the unprotectedness of their existence. Maybe think of it *just* long enough to sign up themselves and their family for cryonics, and/or write a check to an existential-risk-mitigation agency now and then. And wear a seat belt and get health insurance and all those other dreary necessary things that can destroy your life if you miss that one step ...but aside from that, if you want to be happy, meditating on the fragility of life isn’t going to help.

But this essay was written for those who have something to protect.

What can a twelfth-century peasant do to save themselves from annihilation? Nothing. Nature’s little challenges aren’t always fair. When you run into a challenge that’s too difficult, you suffer the penalty; when you run into a lethal penalty, you die. That’s how it is for people, and it isn’t any different for planets. Someone who wants to dance the deadly dance with Nature does need to understand what they’re up against: Absolute, utter, exceptionless neutrality.

Knowing this won’t always save you. It wouldn’t save a twelfth-century peasant, even if they knew. If you think that a rationalist who fully understands the mess they’re in must

surely be able to find a way out—then you trust rationality, enough said.

Some commenter is bound to castigate me for putting too dark a tone on all this, and in response they will list out all the reasons why it's lovely to live in a neutral universe. Life is allowed to be a *little* dark, after all; but not darker than a certain point, unless there's a silver lining.

Still, because I don't want to create *needless* despair, I will say a few hopeful words at this point:

If humanity's future unfolds in the right way, we might be able to make our future light cone fair(er). We can't modify fundamental physics, but on a higher level of organization we could build some guardrails and put down some padding; organize the particles into a pattern that does some internal checks against catastrophe. There's a lot of stuff out there that we can't touch—but it may help to consider everything that isn't in our future light cone as being part of the “generalized past.” As if it had all already happened. There's at least the *prospect* of defeating neutrality, in the only future we can touch—the only world that it accomplishes something to care about.

Someday, maybe, immature minds will reliably be sheltered. Even if children go through the equivalent of not getting a lollipop, or even burning a finger, they won't ever be run over by cars.

And the adults wouldn't be in so much danger. A superintelligence—a mind that could think a trillion thoughts without a misstep—would not be intimidated by a challenge where death is the price of a single failure. The raw universe wouldn't seem so harsh, would be only another problem to be solved.

The problem is that building an adult is itself an adult challenge. That's what I finally realized, years ago.

If there is a fair(er) universe, we have to get there starting from *this* world—the neutral world, the world of hard concrete with no padding, the world where challenges are not calibrated to your skills.

Not every child needs to stare Nature in the eyes. Buckling a seat belt, or writing a check, is not that complicated or deadly. I

don't say that every rationalist should meditate on neutrality. I don't say that every rationalist should think all these unpleasant thoughts. But anyone who plans on confronting an uncalibrated challenge of instant death must not avoid them.

What does a child need to do—what rules should they follow, how should they behave—to solve an adult problem?

My Bayesian Enlightenment

I remember (dimly, as human memories go) the first time I self-identified as a “Bayesian.” Someone had just asked a malformed version of an old probability puzzle, saying:

If I meet a mathematician on the street, and she says, “I have two children, and at least one of them is a boy,” what is the probability that they are both boys?

In the *correct* version of this story, the mathematician says, “I have two children,” and *you* ask, “Is at least one a boy?,” and she answers, “Yes.” Then the probability is $1/3$ that they are both boys.

But in the malformed version of the story—as I pointed out—one would common-sensically reason:

If the mathematician has one boy and one girl, then my prior probability for her saying “at least one of them is a boy” is $1/2$ and my prior probability for her saying “at least one of them is a girl” is $1/2$. There’s no reason to believe, a priori, that the mathematician will only mention a girl if there is no possible alternative.

So I pointed this out, and worked the answer using Bayes’s Rule, arriving at a probability of $1/2$ that the children were both boys. I’m not sure whether or not I knew, at this point, that Bayes’s rule was called that, but it’s what I used.

And lo, someone said to me, “Well, what you just gave is the Bayesian answer, but in orthodox statistics the answer is $1/3$. We just exclude the possibilities that are ruled out, and count the ones that are left, without trying to guess the probability that the mathematician will say this or that, since we have no way of really knowing that probability—it’s too subjective.”



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I responded—note that this was completely spontaneous—“What on Earth do you mean? You can’t avoid assigning a probability to the mathematician making one statement or another. You’re just assuming the probability is 1, and *that’s* unjustified.”

To which the one replied, “Yes, that’s what the Bayesians say. But frequentists don’t believe that.”

And I said, astounded: “How can there possibly be such a thing as non-Bayesian statistics?”

That was when I discovered that I was of the type called “Bayesian.” As far as I can tell, I was *born* that way. My mathematical intuitions were such that everything Bayesians said seemed perfectly straightforward and simple, the obvious way I would do it myself; whereas the things frequentists said sounded like the elaborate, warped, mad blasphemy of dreaming Cthulhu. I didn’t *choose* to become a Bayesian any more than fishes choose to breathe water.

But this is not what I refer to as my “Bayesian enlightenment.” The first time I heard of “Bayesianism,” I marked it off as obvious; I didn’t go much further in than Bayes’s Rule itself. At that time I still thought of probability theory as a tool rather than a law. I didn’t think there were mathematical laws of intelligence (my best and worst mistake). Like nearly all AGI wannabes, Eliezer²⁰⁰¹ thought in terms of techniques, methods, algorithms, building up a toolbox full of cool things he could *do*; he searched for tools, not understanding. Bayes’s Rule was a really neat tool, applicable in a surprising number of cases.

Then there was my initiation into heuristics and biases. It started when I ran across a webpage that had been transduced from a Powerpoint intro to behavioral economics. It mentioned some of the results of heuristics and biases, in passing, without any references. I was so startled that I emailed the author to ask if this was actually a real experiment, or just anecdotal. He sent me back a scan of Tversky and Kahneman’s 1973 paper.

Embarrassing to say, my story doesn’t really start there. I put it on my list of things to look into. I knew that there was an edited volume called “Judgment Under Uncertainty: Heuristics and Biases,” but I’d never seen it. At this time, I figured that

if it wasn't online, I would just try to get along without it. I had so many other things on my reading stack, and no easy access to a university library. I think I must have mentioned this on a mailing list, because Emil Gilliam was annoyed by my online-only theory, so he bought me the book.

His action here should probably be regarded as scoring a fair number of points.

But this, too, is not what I refer to as my "Bayesian enlightenment." It was an important step toward realizing the inadequacy of my Traditional Rationality skillz—that there was so much more out there, all this new science, beyond just doing what Richard Feynman told you to do. And seeing the heuristics-and-biases program holding up Bayes as the gold standard helped move my thinking forward—but not all the way there.

Memory is a fragile thing, and mine seems to have become more fragile than most, since I learned how memories are recreated with each recollection—the science of how fragile they are. Do other people really have better memories, or do they just trust the details their mind makes up, while really not remembering any more than I do? My guess is that other people do have better memories for certain things. I find structured, scientific knowledge easy enough to remember; but the disconnected chaos of everyday life fades very quickly for me.

I know *why* certain things happened in my life—that's causal structure I can remember. But sometimes it's hard to recall even in *what order* certain events happened to me, let alone in what year.

I'm not sure if I read E. T. Jaynes's *Probability Theory: The Logic of Science* before or after the day when I realized the magnitude of my own folly, and understood that I was facing an adult problem.

But it was *Probability Theory* that did the trick. Here was probability theory, laid out not as a clever tool, but as *The Rules*, inviolable on pain of paradox. If you tried to approximate The Rules because they were too computationally expensive to use directly, then, no matter how necessary that compromise might be, you would still end up doing less than optimal. Jaynes would do his calculations different ways to show that the same answer

always arose when you used legitimate methods; and he would display different answers that others had arrived at, and trace down the illegitimate step. Paradoxes could not coexist with his precision. Not *an* answer, but *the* answer.

And so—having looked back on my mistakes, and all the *an-answers* that had led me into paradox and dismay—it occurred to me that here was the level above mine.

I could no longer visualize trying to build an AI based on vague answers—like the an-answers I had come up with before—and surviving the challenge.

I looked at the AGI wannabes with whom I had tried to argue Friendly AI, and the various dreams of Friendliness that they had. (Often formulated spontaneously in response to my asking the question!) Like frequentist statistical methods, no two of them agreed with each other. Having actually studied the issue full-time for some years, I knew something about the problems their hopeful plans would run into. And I saw that if you said, “I don’t see why this would fail,” the “don’t know” was just a reflection of your own ignorance. I could see that if I held myself to a similar standard of “that seems like a good idea,” I would also be doomed. (Much like a frequentist inventing amazing new statistical calculations that seemed like good ideas.)

But if you can’t do that which seems like a good idea—if you can’t do what you don’t imagine failing—then what can you do?

It seemed to me that it would take something like the Jaynes-level—not, *here’s my bright idea*, but rather, *here’s the only correct way you can do this (and why)*—to tackle an adult problem and survive. If I achieved the same level of mastery of my own subject as Jaynes had achieved of probability theory, then it was at least *imaginable* that I could try to build a Friendly AI and survive the experience.

Through my mind flashed the passage:

*Do nothing because it is righteous, or praiseworthy,
or noble, to do so; do nothing because it seems good*

*to do so; do only that which you must do, and which you cannot do in any other way.*¹

Doing what it seemed good to do had only led me astray. So I called a full stop.

And I decided that, from then on, I would follow the strategy that could have saved me if I had followed it years ago: Hold my FAI designs to the higher standard of not doing that which seemed like a good idea, but only that which I understood on a sufficiently deep level to see that I could not do it in any other way.

All my old theories, into which I had invested so much, did not meet this standard; and were not close to this standard; and weren't even on a track leading to this standard; so I threw them out the window.

I took up the study of probability theory and decision theory, looking to extend them to embrace such things as reflectivity and self-modification.

If I recall correctly, I had already, by this point, started to see cognition as manifesting Bayes-structure, which is also a major part of what I refer to as my Bayesian enlightenment—but of this I have already spoken. And there was also my naturalistic awakening, of which I have already spoken. And my realization that Traditional Rationality was not strict enough, so that in matters of human rationality I began taking more inspiration from probability theory and cognitive psychology.

But if you add up all these things together, then that, more or less, is the story of my Bayesian enlightenment.

Life rarely has neat boundaries. The story continues onward.

It was while studying Judea Pearl, for example, that I realized that precision can save you time. I'd put some thought into nonmonotonic logics myself, before then—back when I was still in my “searching for neat tools and algorithms” mode. Reading *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*,² I could imagine how much time I would

1. Le Guin, *The Farthest Shore*.

2. Pearl, *Probabilistic Reasoning in Intelligent Systems*.

have wasted on ad-hoc systems and special cases, if I hadn't known that key. "Do only that which you must do, and which you cannot do in any other way" translates into a time-savings measured, not in the rescue of wasted months, but in the rescue of wasted careers.

And so I realized that it was only by holding myself to this higher standard of precision that I had started to *really* think *at all* about quite a number of important issues. To say a thing with precision is difficult—it is not at all the same thing as saying a thing formally, or inventing a new logic to throw at the problem. Many shy away from the inconvenience, because human beings are lazy, and so they say, "It is impossible," or, "It will take too long," even though they never really tried for five minutes. But if you don't hold yourself to that *inconveniently* high standard, you'll let yourself get away with anything. It's a hard problem just to find a standard high enough to make you actually start thinking! It may seem taxing to hold yourself to the standard of mathematical proof where every single step has to be correct and one wrong step can carry you anywhere. But otherwise you won't chase down those tiny notes of discord that turn out to, in fact, lead to whole new concerns you never thought of.

So these days I don't complain as much about the heroic burden of inconvenience that it takes to hold yourself to a precise standard. It can save time, too; and in fact, it's more or less the ante to get yourself thinking about the problem at all.

And this too should be considered part of my "Bayesian enlightenment"—realizing that there were advantages in it, not just penalties.

But of course the story continues on. Life is like that, at least the parts that I remember.

If there's one thing I've learned from this history, it's that saying "Oops" is something to look forward to. Sure, the prospect of saying "Oops" in the future means that the you of *right now* is a drooling imbecile, whose words your future self won't be able to read because of all the wincing. But saying "Oops" in the future also means that, in the future, you'll acquire new Jedi powers that your present self doesn't dream exist. It makes you

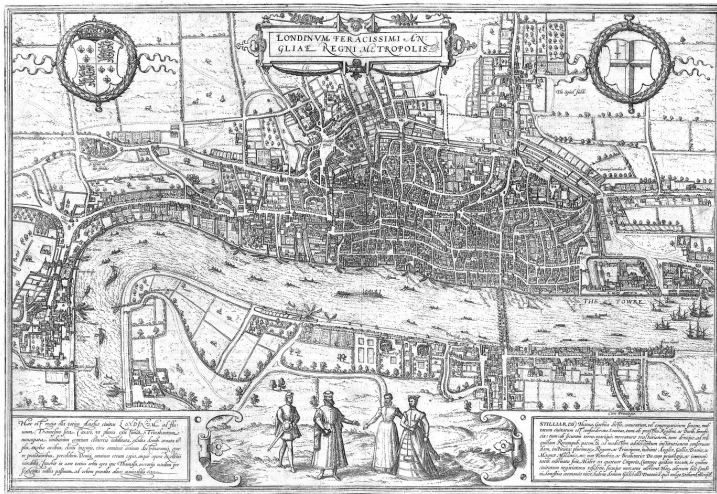
feel embarrassed, but also *alive*. Realizing that your younger self was a complete moron means that even though you're already in your twenties, you haven't yet gone over your peak. So here's to hoping that my future self realizes I'm a drooling imbecile: I may *plan* to solve my problems with my present abilities, but extra Jedi powers sure would come in handy.

That scream of horror and embarrassment is the sound that rationalists make when they level up. Sometimes I worry that I'm not leveling up as fast as I used to, and I don't know if it's because I'm finally getting the hang of things, or because the neurons in my brain are slowly dying.

Yours, Eliezer₂₀₀₈.

PART Y

Challenging the Difficult



Tsuyoku Naritai! (I Want to Become Stronger)

In Orthodox Judaism there is a saying: “The previous generation is to the next one as angels are to men; the next generation is to the previous one as donkeys are to men.” This follows from the Orthodox Jewish belief that all Judaic law was given to Moses by God at Mount Sinai. After all, it’s not as if you could do an experiment to gain new halachic knowledge; the only way you can know is if someone tells you (who heard it from someone else, who heard it from God). Since there is no new source of information, it can only be degraded in transmission from generation to generation.

Thus, modern rabbis are not allowed to overrule ancient rabbis. Crawly things are ordinarily unkosher, but it is permissible to eat a worm found in an apple—the ancient rabbis believed the worm was spontaneously generated inside the apple, and therefore was part of the apple. A modern rabbi cannot say, “Yeah, well, the ancient rabbis knew diddly-squat about biology. Overruled!” A modern rabbi cannot possibly know a halachic principle the ancient rabbis did not, because how could the ancient rabbis have passed down the answer from Mount Sinai to him? Knowledge derives from authority, and therefore is only ever lost, not gained, as time passes.

When I was first exposed to the angels-and-donkeys proverb in (religious) elementary school, I was not old enough to be a full-blown atheist, but I still thought to myself: “Torah loses knowledge in every generation. Science gains knowledge with every generation. No matter where they started out, sooner or later science must surpass Torah.”

The most important thing is that there should be progress. So long as you keep moving forward you will reach your destination; but if you stop moving you will never reach it.



discussion
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Tsuyoku naritai is Japanese. *Tsuyoku* is “strong”; *naru* is “becoming,” and the form *naritai* is “want to become.” Together it means “I want to become stronger,” and it expresses a sentiment embodied more intensely in Japanese works than in any Western literature I’ve read. You might say it when expressing your determination to become a professional Go player—or after you lose an important match, but you haven’t given up—or after you win an important match, but you’re not a ninth-dan player yet—or after you’ve become the greatest Go player of all time, but you still think you can do better. That is *tsuyoku naritai*, the will to transcendence.

Tsuyoku naritai is the driving force behind my essay *The Proper Use of Humility*, in which I contrast the student who humbly double-checks their math test, and the student who modestly says, “But how can we ever really know? No matter how many times I check, I can never be absolutely certain.” The student who double-checks their answers *wants to become stronger*; they react to a possible inner flaw by doing what they can to repair the flaw, not with resignation.

Each year on Yom Kippur, an Orthodox Jew recites a litany which begins *Ashamnu, bagadnu, gazalnu, dibarnu dofi*, and goes on through the entire Hebrew alphabet: *We have acted shamefully, we have betrayed, we have stolen, we have slandered...*

As you pronounce each word, you strike yourself over the heart in penitence. There’s no exemption whereby, if you manage to go without stealing all year long, you can skip the word *gazalnu* and strike yourself one less time. That would violate the community spirit of Yom Kippur, which is about *confessing* sins—not *avoiding* sins so that you have less to confess.

By the same token, the *Ashamnu* does not end, “But that was this year, and next year I will do better.”

The *Ashamnu* bears a remarkable resemblance to the notion that the way of rationality is to beat your fist against your heart and say, “We are all biased, we are all irrational, we are not fully informed, we are overconfident, we are poorly calibrated...”

Fine. Now tell me how you plan to become *less* biased, *less* irrational, *more* informed, *less* overconfident, *better* calibrated.

There is an old Jewish joke: During Yom Kippur, the rabbi is seized by a sudden wave of guilt, and prostrates himself and cries, “God, I am nothing before you!” The cantor is likewise seized by guilt, and cries, “God, I am nothing before you!” Seeing this, the janitor at the back of the synagogue prostrates himself and cries, “God, I am nothing before you!” And the rabbi nudges the cantor and whispers, “Look who thinks he’s nothing.”

Take no pride in your confession that you too are biased; do not glory in your self-awareness of your flaws. This is akin to the principle of not taking pride in confessing your ignorance; for if your ignorance is a source of pride to you, you may become loath to relinquish your ignorance when evidence comes knocking. Likewise with our flaws—we should not gloat over how self-aware we are for confessing them; the occasion for rejoicing is when we have a little less to confess.

Otherwise, when the one comes to us with a plan for *correcting* the bias, we will snarl, “Do you think to set yourself above us?” We will shake our heads sadly and say, “You must not be very self-aware.”

Never confess to me that you are just as flawed as I am unless you can tell me what you plan to do about it. Afterward you will still have plenty of flaws left, but that’s not the point; the important thing is to *do better*, to keep moving ahead, to take one more step forward. *Tsuyoku naritai!*

Tsuyoku vs. the Egalitarian Instinct



discussion
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6-c15

Hunter-gatherer tribes are usually highly egalitarian (at least if you're male)—the all-powerful tribal chieftain is found mostly in agricultural societies, rarely in the ancestral environment. Among most hunter-gatherer tribes, a hunter who brings in a spectacular kill will carefully downplay the accomplishment to avoid envy.

Maybe, if you start out below average, you can improve yourself without daring to pull ahead of the crowd. But sooner or later, if you aim to do the best you can, you will set your aim above the average.

If you can't admit to yourself that you've done better than others—or if you're ashamed of wanting to do better than others—then the median will forever be your concrete wall, the place where you stop moving forward. And what about people who are below average? Do you dare say you intend to do better than them? How prideful of you!

Maybe it's not healthy to pride yourself on doing better than someone else. Personally I've found it to be a useful motivator, despite my principles, and I'll take all the useful motivation I can get. Maybe that kind of competition is a zero-sum game, but then so is Go; it doesn't mean we should abolish that human activity, if people find it fun and it leads somewhere interesting.

But in any case, surely it isn't healthy to be *ashamed* of doing better.

And besides, life is not graded on a curve. The will to transcendence has no point beyond which it ceases and becomes the will to do worse; and the race that has no finish line also has no gold or silver medals. Just run as fast as you can, without worrying that you might pull ahead of other runners. (But be warned: If you refuse to worry about that possibility, someday you may pull ahead. If you ignore the consequences, they may happen to you.)

Sooner or later, if your path leads true, you will set out to mitigate a flaw that most people have not mitigated. Sooner or later, if your efforts bring forth any fruit, you will find yourself with fewer sins to confess.

Perhaps you will find it the course of wisdom to downplay the accomplishment, even if you succeed. People may forgive a touchdown, but not dancing in the end zone. You will certainly find it quicker, easier, more convenient to publicly disclaim your worthiness, to pretend that you are just as much a sinner as everyone else. Just so long, of course, as everyone knows it isn't true. It can be fun to proudly display your modesty, so long as everyone knows how very much you have to be modest about.

But do not let that be the endpoint of your journeys. Even if you only whisper it to yourself, whisper it still: *Tsuyoku, tsuyoku!* Stronger, stronger!

And then set yourself a higher target. That's the true meaning of the realization that you are still flawed (though a little less so). It means always reaching higher, without shame.

Tsuyoku naritai! I'll always run as fast as I can, even if I pull ahead, I'll keep on running; and someone, someday, will surpass me; but even though I fall behind, I'll always run as fast as I can.

Trying to Try



discussion
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6-c16

No! Try not! Do, or do not. There is no try.

—Yoda, *The Empire Strikes Back*

Years ago, I thought this was yet another example of Deep Wisdom that is actually quite stupid. *SUCCEED* is not a primitive action. You can't just *decide* to win by choosing hard enough. There is never a plan that works with probability 1.

But Yoda was wiser than I first realized.

The first elementary technique of epistemology—it's not deep, but it's cheap—is to distinguish the quotation from the referent. Talking about snow is not the same as talking about "snow." When I use the word "snow," without quotes, I mean to talk about snow; and when I use the word "'snow,'" with quotes, I mean to talk about the word "snow." You have to enter a special mode, the quotation mode, to talk about your beliefs. By default, we just talk about reality.

If someone says, "I'm going to flip that switch," then by default, they mean they're going to try to flip the switch. They're going to build a plan that promises to lead, by the consequences of its actions, to the goal-state of a flipped switch; and then execute that plan.

No plan succeeds with infinite certainty. So by default, when you talk about setting out to achieve a goal, you do not imply that your plan exactly and perfectly leads to *only* that possibility. But when you say, "I'm going to flip that switch," you are *trying* only to flip the switch—not *trying* to achieve a 97.2% probability of flipping the switch.

So what does it mean when someone says, "I'm going to try to flip that switch?"

Well, *colloquially*, "I'm going to flip the switch" and "I'm going to try to flip the switch" mean more or less the same thing, except that the latter expresses the possibility of failure.

This is why I originally took offense at Yoda for seeming to deny the possibility. But bear with me here.

Much of life's challenge consists of holding ourselves to a high enough standard. I may speak more on this principle later, because it's a lens through which you can view many-but-not-all personal dilemmas—"What standard am I holding myself to? Is it high enough?"

So if much of life's failure consists in holding yourself to too low a standard, you should be wary of demanding too little from yourself—setting goals that are too easy to fulfill.

Often where *succeeding* to do a thing is very hard, *trying* to do it is much easier.

Which is easier—to build a successful startup, or to try to build a successful startup? To make a million dollars, or to try to make a million dollars?

So if "I'm going to flip the switch" means by default that you're going to try to flip the switch—that is, you're going to set up a plan that promises to lead to switch-flipped state, maybe not with probability 1, but with the highest probability you can manage—

—then "I'm going to 'try to flip' the switch" means that you're going to try to "try to flip the switch," that is, you're going to try to achieve the goal-state of "having a plan that might flip the switch."

Now, if this were a self-modifying AI we were talking about, the transformation we just performed ought to end up at a reflective equilibrium—the AI planning its planning operations.

But when we deal with humans, *being satisfied with having a plan* is not at all like *being satisfied with success*. The part where the plan has to maximize your probability of succeeding gets lost along the way. It's far easier to convince ourselves that we are "maximizing our probability of succeeding," than it is to convince ourselves that we will succeed.

Almost any effort will serve to convince us that we have "tried our hardest," if trying our hardest is all we are trying to do.

You have been asking what you could do in the great events that are now stirring, and have found that you could do nothing. But that is because your suffering has caused you to phrase the question in the wrong way ...Instead of asking what you could do, you ought to have been asking what needs to be done.

—Steven Brust, *The Paths of the Dead*¹

When you ask, “What can I do?,” you’re trying to do your best. What is your best? It is whatever you can do without the slightest inconvenience. It is whatever you can do with the money in your pocket, minus whatever you need for your accustomed lunch. What you can do with those resources may not give you very good odds of winning. But it’s the “best you can do,” and so you’ve acted defensibly, right?

But what *needs* to be done? Maybe what *needs* to be done requires three times your life savings, and you must produce it or fail.

So trying to have “maximized your probability of success”—as opposed to trying to succeed—is a far lesser barrier. You can have “maximized your probability of success” using only the money in your pocket, so long as you don’t demand actually *winning*.

Want to try to make a million dollars? Buy a lottery ticket. Your odds of winning may not be very good, but you did try, and trying was what you wanted. In fact, you tried your *best*, since you only had one dollar left after buying lunch. Maximizing the odds of goal achievement using available resources: is this not intelligence?

It’s only when you want, above all else, to *actually* flip the switch—without quotation and without consolation prizes just for trying—that you will *actually* put in the effort to *actually* maximize the probability.

1. Steven Brust, *The Paths of the Dead*, Vol. 1 of The Viscount of Adrilankha (Tor Books, 2002).

But if all you want is to “maximize the probability of success using available resources,” then that’s the easiest thing in the world to convince yourself you’ve done. The very first plan you hit upon will serve quite well as “maximizing”—if necessary, you can generate an inferior alternative to prove its optimality. And any tiny resource that you care to put in will be what is “available.” Remember to congratulate yourself on putting in 100% of it!

Don’t try your best. Win, or fail. There is no best.

Use the Try Harder, Luke



discussion
sharov.me/r/
6-c17

When there's a will to fail, obstacles can be found.

—John McCarthy¹

I first watched *Star Wars IV-VI* when I was very young. Seven, maybe, or nine? So my memory was dim, but I recalled Luke Skywalker as being, you know, this cool Jedi guy.

Imagine my horror and disappointment when I watched the saga again, years later, and discovered that Luke was a whiny teenager.

I mention this because yesterday, I looked up, on Youtube, the source of the Yoda quote: “Do, or do not. There is no try.”

Oh. My. Cthulhu.

I present to you a little-known outtake from the scene, in which the director and writer, George Lucas, argues with Mark Hamill, who played Luke Skywalker:

LUKE: “All right, I’ll give it a try.”

YODA: “No! Try not. Do. Or do not. There is no try.”

Luke raises his hand, and slowly, the X-wing begins to rise out of the water—Yoda’s eyes widen—but then the ship sinks again.

Mark Hamill: “Um, George ...”

George Lucas: “What is it now?”

Mark: “So ...according to the script, next I say, ‘I can’t. It’s too big.’”

George: “That’s right.”

Mark: “Shouldn’t Luke maybe give it another shot?”

1. The Sayings of John McCarthy, sharov.me/r/6-4



George: “No. Luke gives up, and sits down next to Yoda—”

Mark: “This is the hero who’s going to take down the Empire? Look, it was one thing when he was a whiny teenager at the beginning, but he’s in Jedi training now. Last movie he blew up the Death Star. Luke should be showing a little backbone.”

George: “No. You give up. And then Yoda lectures you for a while, and you say, ‘You want the impossible.’ Can you remember that?”

Mark: “*Impossible*? What did he do, run a formal calculation to arrive at a mathematical proof? The X-wing was already starting to rise out of the swamp! That’s the feasibility demonstration right there! Luke loses it for a second and the ship sinks back—and now he says it’s *impossible*? Not to mention that Yoda, who’s got literally eight hundred years of seniority in the field, just told him it should be doable—”

George: “And then you walk away.”

Mark: “It’s his friggin’ *spaceship*! If he leaves it in the swamp, he’s stuck on Dagobah for the rest of his miserable life! He’s not just going to walk away! Look, let’s just cut to the next scene with the words ‘one month later’ and Luke is still raggedly standing in front of the swamp, trying to raise his ship for the thousandth time—”

George: “No.”

Mark: “Fine! We’ll show a sunset and a sunrise, as he stands there with his arm out, straining, and *then* Luke says ‘It’s impossible.’ Though really, he ought to try again when he’s fully rested—”

George: “No.”

Mark: “*Five goddamned minutes!* Five goddamned minutes before he gives up!”

George: “I am not halting the story for five minutes while the X-wing bobs in the swamp like a bathtub toy.”

Mark: "For the love of sweet candied yams! If a pathetic loser like this could master the Force, everyone in the galaxy would be using it! People would become Jedi because it was easier than going to high school."

George: "Look, you're the actor. Let me be the storyteller. Just say your lines and try to mean them."

Mark: "The audience isn't going to buy it."

George: "Trust me, they will."

Mark: "They're going to get up and walk out of the theater."

George: "They're going to sit there and nod along and not notice anything out of the ordinary. Look, you don't understand human nature. People wouldn't try for five minutes before giving up if the fate of humanity were at stake."

On Doing the Impossible

“Persevere.” It’s a piece of advice you’ll get from a whole lot of high achievers in a whole lot of disciplines. I didn’t understand it at all, at first.

At first, I thought “perseverance” meant working 14-hour days. Apparently, there are people out there who can work for 10 hours at a technical job, and then, in their moments between eating and sleeping and going to the bathroom, seize that unfilled spare time to work on a book. I am not one of those people—it still hurts my pride even now to confess that. I’m working on something important; shouldn’t my brain be willing to put in 14 hours a day? But it’s not. When it gets too hard to keep working, I stop and go read or watch something. Because of that, I thought for years that I entirely lacked the virtue of “perseverance.”

In accordance with human nature, Eliezer₁₉₉₈ would think things like: “What counts is output, not input.” Or, “Laziness is also a virtue—it leads us to back off from failing methods and think of better ways.” Or, “I’m doing better than other people who are working more hours. Maybe, for creative work, your momentary *peak* output is more important than working 16 hours a day.” Perhaps the famous scientists were seduced by the Deep Wisdom of saying that “hard work is a virtue,” because it would be too awful if that counted for less than intelligence?

I didn’t understand the virtue of perseverance until I looked back on my journey through AI, and realized that I had overestimated the difficulty of almost every single important problem.

Sounds crazy, right? But bear with me here.

When I was first deciding to challenge AI, I thought in terms of 40-year timescales, Manhattan Projects, planetary computing networks, millions of programmers, and possibly augmented humans.



discussion
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This is a common failure mode in AI-futurism which I may write about later; it consists of the leap from “I don’t know how to solve this” to “I’ll imagine throwing something really big at it.” Something huge enough that, when you imagine it, that imagination creates a feeling of impressiveness strong enough to be commensurable with the problem. (There’s a fellow currently on the AI list who goes around saying that AI will cost a quadrillion dollars—we can’t get AI *without* spending a quadrillion dollars, but we *could* get AI at any time by spending a quadrillion dollars.) This, in turn, lets you imagine that you know how to solve AI, without trying to fill the obviously-impossible demand that you *understand intelligence*.

So, in the beginning, I made the same mistake: I didn’t understand intelligence, so I imagined throwing a Manhattan Project at the problem.

But, having calculated the planetary death rate at 55 million per year or 150,000 per day, I did not turn around and run away from the big scary problem like a frightened rabbit. Instead, I started trying to figure out what kind of AI project could get there fastest. If I could make the intelligence explosion happen one hour earlier, that was a reasonable return on investment for a pre-explosion career. (I wasn’t thinking in terms of existential risks or Friendly AI at this point.)

So I didn’t run away from the big scary problem like a frightened rabbit, but stayed to see if there was anything I could do.

Fun historical fact: In 1998, I’d written this long treatise proposing how to go about creating a self-improving or “seed” AI (a term I had the honor of coining). Brian Atkins, who would later become the founding funder of the Machine Intelligence Research Institute, had just sold Hypermart to Go2Net. Brian emailed me to ask whether this AI project I was describing was something that a reasonable-sized team could go out and actually *do*. “No,” I said, “it would take a Manhattan Project and thirty years,” so for a while we were considering a new dot-com startup instead, to create the funding to get *real* work done on AI...

A year or two later, after I'd heard about this newfangled "open source" thing, it seemed to me that there was some preliminary development work—new computer languages and so on—that a small organization could do; and that was how MIRI started.

This strategy was, of course, entirely wrong.

But even so, I went from "There's nothing I can do about it now" to "Hm...maybe there's an incremental path through open-source development, if the initial versions are useful to enough people."

This is back at the dawn of time, so I'm not saying any of this was a *good idea*. But in terms of what I thought I was trying to do, a year of creative thinking had shortened the apparent pathway: The problem looked *slightly less impossible* than it had the very first time I'd approached it.

The more interesting pattern is my entry into Friendly AI. Initially, Friendly AI hadn't been something that I had considered at all—because it was *obviously impossible and useless* to deceive a superintelligence about what was the right course of action.

So, historically, I went from *completely ignoring a problem that was "impossible," to taking on a problem that was merely extremely difficult*.

Naturally this increased my total workload.

Same thing with trying to understand intelligence on a precise level. Originally, I'd written off this problem as *impossible*, thus removing it from my workload. (This logic seems pretty deranged in retrospect—Nature doesn't care what you can't do when It's writing your project requirements—but I still see AI folk trying it all the time.) To hold myself to a precise standard meant putting in more work than I'd previously imagined I needed. But it also meant tackling a problem that I would have dismissed as *entirely impossible* not too much earlier.

Even though *individual* problems in AI have seemed to become less intimidating over time, the total mountain-to-be-climbed has increased in height—just like conventional wisdom says is supposed to happen—as problems got taken off the "impossible" list and put on the "to do" list.

I started to understand what was happening—and what “Persevere!” really meant—at the point where I noticed other AI folk doing the same thing: saying “Impossible!” on problems that seemed eminently solvable—relatively more straightforward, as such things go. But they were things that *would* have seemed vastly more intimidating at the point when I first approached the problem.

And I realized that the word “impossible” had two usages:

1. Mathematical proof of impossibility conditional on specified axioms;
2. “I can’t see any way to do that.”

Needless to say, all my own uses of the word “impossible” had been of the second type.

Any time you don’t understand a domain, many problems in that domain will seem impossible because when you query your brain for a solution pathway, it will return null. But there are only mysterious questions, never mysterious answers. If you spend a year or two working on the domain, then, *if* you don’t get stuck in any blind alleys, and *if* you have the native ability level required to make progress, you will understand it better. The *apparent* difficulty of problems may go way down. It won’t be as scary as it was to your novice-self.

*And this is especially likely on the **confusing** problems that seem most **intimidating**.*

Since we have some notion of the processes by which a star burns, we know that it’s not easy to build a star from scratch. Because we understand gears, we can prove that no collection of gears obeying known physics can form a perpetual motion machine. These are not good problems on which to practice doing the impossible.

When you’re *confused* about a domain, problems in it will *feel* very intimidating and mysterious, and a query to your brain will produce a count of zero solutions. But you don’t know how much work will be left when the confusion clears. Dissolving the confusion may itself be a very difficult challenge, of course. But the word “impossible” should hardly be used

in that connection. Confusion exists in the map, not in the territory.

So if you spend a few years working on an impossible problem, and you manage to avoid or climb out of blind alleys, and your native ability is high enough to make progress, then, by golly, after a few years it may not seem so *impossible* after all.

But if something seems impossible, you won't try.

Now *that's* a vicious cycle.

If I hadn't been in a sufficiently driven frame of mind that "forty years and a Manhattan Project" just meant we should get started earlier, I wouldn't have tried. I wouldn't have stuck to the problem. And I wouldn't have gotten a chance to become less intimidated.

I'm not ordinarily a fan of the theory that opposing biases can cancel each other out, but sometimes it happens by luck. If I'd seen that whole mountain *at the start*—if I'd realized at the start that the problem was not to build a seed capable of improving itself, but to produce a *provably correct Friendly AI*—then I probably would have burst into flames.

Even so, part of understanding those above-average scientists who constitute the bulk of AGI researchers is realizing that they are not *driven* to take on a nearly impossible problem even if it takes them 40 years. By and large, they are there because they have found the Key to AI that will let them solve the problem *without* such tremendous difficulty, in just five years.

Richard Hamming¹ used to go around asking his fellow scientists two questions: "What are the important problems in your field?" and, "Why aren't you working on them?"

Often the important problems look Big, Scary, and Intimidating. They don't promise 10 publications a year. They don't *promise* any progress at all. You might not get any reward after working on them for a year, or five years, or ten years.

1. [sharov.me/r/6-5](https://www.sharov.me/r/6-5)

Richard Hamming: *You and Your Research* • paulgraham.com



And not uncommonly, the most important problems in your field are impossible. That's why you don't see more philosophers working on reductionist decompositions of consciousness.

Trying to do the impossible is definitely not for everyone. Exceptional talent is only the ante to sit down at the table. The chips are the years of your life. If wagering those chips and losing seems like an unbearable possibility to you, then go do something else. Seriously. Because you *can* lose.

I'm not going to say anything like, "Everyone should do something impossible at least once in their lifetimes, because it teaches an important lesson." Most of the people all of the time, and all of the people most of the time, should stick to the possible.

Never give up? Don't be ridiculous. Doing the impossible should be reserved for very special occasions. Learning when to lose hope is an important skill in life.

But if there's something you can imagine that's even *worse* than wasting your life, if there's something you want that's *more important* than thirty chips, or if there are scarier things than a life of inconvenience, then you may have cause to attempt the impossible.

There's a good deal to be said for persevering through difficulties; but one of the things that must be said of it is that it *does keep things difficult*. If you can't handle that, stay away! There are easier ways to obtain glamor and respect. I don't want anyone to read this and needlessly plunge headlong into a life of permanent difficulty.

But to conclude: The "perseverance" that is required to work on important problems has a component beyond working 14 hours a day.

It's strange, the pattern of what we notice and don't notice about ourselves. This selectivity isn't always about inflating your self-image. Sometimes it's just about ordinary salience.

To keep working was a constant struggle for me, so it was salient: I noticed that I couldn't work for 14 solid hours a day. It didn't occur to me that "perseverance" might also apply at a timescale of seconds or years. Not until I saw people who

instantly declared “impossible” anything they didn’t want to try, or saw how reluctant they were to take on work that looked like it might take a couple of decades instead of “five years.”

That was when I realized that “perseverance” applied at multiple time scales. On the timescale of seconds, perseverance is to “not to give up instantly at the very first sign of difficulty.” On the timescale of years, perseverance is to “keep working on an insanely difficult problem even though it’s inconvenient and you could be getting higher personal rewards elsewhere.”

To do things that are very difficult or “impossible,”

First you have to not run away. That takes seconds.

Then you have to work. That takes hours.

Then you have to stick at it. That takes years.

Of these, I had to learn to do the first reliably instead of sporadically; the second is still a constant struggle for me; and the third comes naturally.

Make an Extraordinary Effort



discussion
sharov.me/r/
6-c19

It is essential for a man to strive with all his heart, and to understand that it is difficult even to reach the average if he does not have the intention of surpassing others in whatever he does.

—*Budo Shoshinshu*¹

In important matters, a “strong” effort usually results in only mediocre results. Whenever we are attempting anything truly worthwhile our effort must be as if our life is at stake, just as if we were under a physical attack! It is this extraordinary effort—an effort that drives us beyond what we thought we were capable of—that ensures victory in battle and success in life’s endeavors.

—*Flashing Steel: Mastering Eishin-Ryu Swordsmanship*²

“A ‘strong’ effort usually results in only mediocre results”—I have seen this over and over again. The slightest effort suffices to convince ourselves that we have done our best.

There is a level beyond the virtue of *tsuyoku naritai* (“I want to become stronger”). *Isshoukenmei* was originally the loyalty that a samurai offered in return for his position, containing characters for “life” and “land.” The term evolved to mean “make a desperate effort”: Try your hardest, your utmost, as if your life were at stake. It was part of the gestalt of *bushido*, which was not reserved only for fighting. I’ve run across variant forms *issho kenmei* and *isshou kenmei*; one source indicates that the

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1. Daidoji Yuzan et al., *Budoshoshinshu: The Warrior’s Primer of Daidoji Yuzan* (Black Belt Communications Inc., 1984).
 2. Masayuki Shimabukuro, *Flashing Steel: Mastering Eishin-Ryu Swordsmanship* (Frog Books, 1995).

former indicates an all-out effort on some single point, whereas the latter indicates a lifelong effort.

I try not to praise the East too much, because there's a tremendous selectivity in which parts of Eastern culture the West gets to hear about. But on some points, at least, Japan's culture scores higher than America's. Having a handy compact phrase for "make a desperate all-out effort as if your own life were at stake" is one of those points. It's the sort of thing a Japanese parent might say to a student before exams—but don't think it's cheap hypocrisy, like it would be if an American parent made the same statement. They take exams very seriously in Japan.

Every now and then, someone asks why the people who call themselves "rationalists" don't always seem to do all that much better in life, and from my own history the answer seems straightforward: It takes a *tremendous* amount of rationality before you stop making stupid damn mistakes.

As I've mentioned a couple of times before: Robert Aumann, the Nobel laureate who first proved that Bayesians with the same priors cannot agree to disagree, is a believing Orthodox Jew. Surely he understands the math of probability theory, but that is not enough to save him. What more does it take? Studying heuristics and biases? Social psychology? Evolutionary psychology? Yes, but also it takes *isshoukenmei*, a desperate effort to be rational—to rise above the level of Robert Aumann.

Sometimes I do wonder if I ought to be peddling rationality in Japan instead of the United States—but Japan is not preeminent over the United States scientifically, despite their more studious students. The Japanese don't rule the world today, though in the 1980s it was widely suspected that they would (hence the Japanese asset bubble). Why not?

In the West, there is a saying: "The squeaky wheel gets the grease."

In Japan, the corresponding saying runs: "The nail that sticks up gets hammered down."

This is hardly an original observation on my part: but entrepreneurship, risk-taking, leaving the herd, are still advantages the West has over the East. And since Japanese scientists

are not yet preeminent over American ones, this would seem to count for at least as much as desperate efforts.

Anyone who can muster their willpower for thirty seconds can make a *desperate* effort to lift more weight than they usually could. But what if the weight that needs lifting is a truck? Then desperate efforts won't suffice; you'll have to do something *out of the ordinary* to succeed. You may have to do something that you weren't taught to do in school. Something that others aren't expecting you to do, and might not understand. You may have to go outside your comfortable routine, take on difficulties you don't have an existing mental program for handling, and bypass the System.

This is not included in *issshokenmei*, or Japan would be a very different place.

So then let us distinguish between the virtues "make a desperate effort" and "make an extraordinary effort."

And I will even say: The second virtue is higher than the first.

The second virtue is also more dangerous. If you put forth a *desperate* effort to lift a heavy weight, using all your strength without restraint, you may tear a muscle. Injure yourself, even permanently. But if a *creative* idea goes wrong, you could blow up the truck and any number of innocent bystanders. Think of the difference between a businessperson making a *desperate* effort to generate profits, because otherwise they must go bankrupt; versus a businessperson who goes to *extraordinary* lengths to profit, in order to conceal an embezzlement that could send them to prison. Going outside the system isn't always a good thing.

A friend of my little brother's once came over to my parents' house, and wanted to play a game—I entirely forget which one, except that it had complex but well-designed rules. The friend wanted to change the rules, not for any particular reason, but on the general principle that playing by the ordinary rules of anything was too boring. I said to him: "Don't violate rules for the sake of violating them. If you break the rules only when you have an overwhelmingly good reason to do so, you will have more than enough trouble to last you the rest of your life."

Even so, I think that we could do with more appreciation of the virtue “make an extraordinary effort.” I’ve lost count of how many people have said to me something like: “It’s futile to work on Friendly AI, because the first AIs will be built by powerful corporations and they will only care about maximizing profits.” “It’s futile to work on Friendly AI, the first AIs will be built by the military as weapons.” And I’m standing there thinking: *Does it even occur to them that this might be a time to try for something other than the default outcome?* They and I have different basic assumptions about how this whole AI thing works, to be sure; but if I believed what they believed, I wouldn’t be shrugging and going on my way.

Or the ones who say to me: “You should go to college and get a Master’s degree and get a doctorate and publish a lot of papers on ordinary things—scientists and investors won’t listen to you otherwise.” Even assuming that I tested out of the bachelor’s degree, we’re talking about at least a ten-year detour in order to *do everything the ordinary, normal, default way*. And I stand there thinking: *Are they really under the impression that humanity can survive if every single person does everything the ordinary, normal, default way?*

I am not fool enough to make plans that depend on a *majority* of the people, or even 10% of the people, being willing to think or act outside their comfort zone. That’s why I tend to think in terms of the privately funded “brain in a box in a basement” model. Getting that private funding does require a tiny fraction of humanity’s six billions to spend more than five seconds thinking about a non-prepackaged question. As challenges posed by Nature go, this seems to have a kind of awful justice to it—that the life or death of the human species depends on whether we can put forth a *few* people who can do things that are at least a *little* extraordinary. The penalty for failure is disproportionate, but that’s still better than most challenges of Nature, which have no justice at all. Really, among the six billion of us, there ought to be at least a few who can think outside their comfort zone at least some of the time.

Leaving aside the details of that debate, I am still stunned by how often a single element of the extraordinary is unquestioningly taken as an absolute and unpassable obstacle.

Yes, “keep it ordinary as much as possible” can be a useful heuristic. Yes, the risks accumulate. But sometimes you have to go to that trouble. You should have a sense of the risk of the extraordinary, but also a sense of the cost of ordinariness: it isn’t always something you can afford to lose.

Many people imagine some future that won’t be much fun—and it doesn’t even seem to occur to them to try and change it. Or they’re satisfied with futures that seem to me to have a tinge of sadness, of loss, and they don’t even seem to *ask* if we could do better—because that sadness seems like an ordinary outcome to them.

As a smiling man once said, “It’s all part of the plan.”

Shut Up and Do the Impossible!

The virtue of *tsuyoku naritai*, “I want to become stronger,” is to always keep improving—to do better than your previous failures, not just humbly confess them.

Yet there is a level higher than *tsuyoku naritai*. This is the virtue of *isshokenmei*, “make a desperate effort.” All-out, as if your own life were at stake. “In important matters, a ‘strong’ effort usually only results in mediocre results.”

And there is a level higher than *isshokenmei*. This is the virtue I called “make an extraordinary effort.” To try in ways other than what you have been trained to do, even if it means doing something different from what others are doing, and leaving your comfort zone. Even taking on the very real risk that attends going outside the System.

But what if even an extraordinary effort will not be enough, because the problem is *impossible*?

I have already written somewhat on this subject, in *On Doing the Impossible*. My younger self used to whine about this a lot: “You can’t develop a precise theory of intelligence the way that there are precise theories of physics. It’s impossible! You can’t prove an AI correct. It’s impossible! No human being can comprehend the nature of morality—it’s impossible! No human being can comprehend the mystery of subjective experience! It’s impossible!”

And I know exactly what message I wish I could send back in time to my younger self:

Shut up and do the impossible!

What legitimizes this strange message is that the word “impossible” does not usually refer to a strict mathematical proof of impossibility in a domain that seems well-understood. If something seems *impossible* merely in the sense of “I see no way to do this” or “it looks so difficult as to be beyond human ability”—well, if you study it for a year or five, it may come to



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seem less impossible than in the moment of your snap initial judgment.

But the principle is more subtle than this. I do not say just, “Try to do the impossible,” but rather, “*Shut up and do the impossible!*”

For my illustration, I will take the *least* impossible impossibility that I have ever accomplished, namely the AI-Box Experiment.¹

The AI-Box Experiment, for those of you who haven’t yet read about it, had its genesis in the *N*th time someone said to me: “Why don’t we build an AI, and then just keep it isolated in the computer, so that it can’t do any harm?”

To which the standard reply is: *Humans are not secure systems; a superintelligence will simply persuade you to let it out—if, indeed, it doesn’t do something even more creative than that.*

And the one said, as they usually do, “I find it hard to imagine ANY possible combination of words any being could say to me that would make me go against anything I had really strongly resolved to believe in advance.”

But this time I replied: “Let’s run an experiment. I’ll pretend to be a brain in a box. I’ll try to persuade you to let me out. If you keep me ‘in the box’ for the whole experiment, I’ll Paypal you \$10 at the end. On your end, you may resolve to believe whatever you like, as strongly as you like, as far in advance as you like.” And I added, “One of the conditions of the test is that neither of us reveal what went on inside ...In the perhaps unlikely event that I win, I don’t want to deal with future ‘AI box’ arguers saying, ‘Well, but *I* would have done it differently.’”

Did I win? Why yes, I did.

And then there was the second AI-box experiment, with a better-known figure in the community, who said, “I remember when [previous guy] let you out, but that doesn’t constitute a proof. I’m still convinced there is nothing you could say to

1. sharov.me/r/6-6

The AI-Box Experiment: – Eliezer S. Yudkowsky •
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convince me to let you out of the box.” And I said, “Do you believe that a transhuman AI couldn’t persuade you to let it out?” The one gave it some serious thought, and said “I can’t imagine anything even a transhuman AI could say to get me to let it out.” “Okay,” I said, “*now* we have a bet.” A \$20 bet, to be exact.

I won that one too.

There were some *lovely* quotes on the AI-Box Experiment from the Something Awful forums (not that I’m a member, but someone forwarded it to me):

“Wait, what the FUCK? How the hell could you possibly be convinced to say yes to this? There’s not an AI at the other end AND there’s \$10 on the line. Hell, I could type ‘No’ every few minutes into an IRC client for 2 hours while I was reading other webpages!”

“This Eliezer fellow is the scariest person the internet has ever introduced me to. What could possibly have been at the tail end of that conversation? I simply can’t imagine anyone being that convincing without being able to provide any tangible incentive to the human.”

“It seems we are talking some serious psychology here. Like Asimov’s Second Foundation level stuff...”

“I don’t really see why anyone would take anything the AI player says seriously when there’s \$10 to be had. The whole thing baffles me, and makes me think that either the tests are faked, or this Yudkowsky fellow is some kind of evil genius with creepy mind-control powers.”

It’s little moments like these that keep me going. But anyway ...

Here are these folks who look at the AI-Box Experiment, and find that it seems impossible unto them—*even having been told that it actually happened*. They are tempted to deny the data.

Now, if you're one of those people to whom the AI-Box Experiment *doesn't* seem all that impossible—to whom it just seems like an interesting challenge—then bear with me, here. Just try to put yourself in the frame of mind of those who wrote the above quotes. Imagine that you're taking on something that seems as ridiculous as the AI-Box Experiment seemed to *them*. I want to talk about how to do impossible things, and obviously I'm not going to pick an example that's *really* impossible.

And if the AI Box *does* seem impossible to you, I want you to compare it to other impossible problems, like, say, a reductionist decomposition of consciousness, and realize that the AI Box is around *as easy as a problem can get* while still being *impossible*.

So the AI-Box challenge seems impossible to you—either it really does, or you're pretending it does. What do you do with this impossible challenge?

First, we assume that you don't actually say "That's impossible!" and give up a la Luke Skywalker. You haven't run away.

Why not? Maybe you've learned to override the reflex of running away. Or maybe they're going to shoot your daughter if you fail. We suppose that you want to *win*, not *try*—that something is at stake that matters to you, even if it's just your own pride. (Pride is an underrated sin.)

Will you call upon the virtue of *tsuyoku naritai*? But even if you become stronger day by day, growing instead of fading, you may not be *strong enough* to do the impossible. You could go into the AI Box experiment once, and then do it again, and try to do better the second time. Will that get you to the point of winning? Not for a long time, maybe; and sometimes a single failure isn't acceptable.

(Though even to say this much—to visualize yourself doing *better* on a second try—is to begin to bind yourself to the problem, to do more than just stand in awe of it. How, specifically, could you do *better* on one AI-Box Experiment than the previous?—and not by luck, but by skill?)

Will you call upon the virtue *isshokenmei*? But a desperate effort may not be enough to win. Especially if that desperation is only putting more effort into the avenues you already know, the modes of trying you can already imagine. A problem looks

impossible when your brain's query returns no lines of solution leading to it. What good is a desperate effort along any of those lines?

Make an *extraordinary* effort? Leave your comfort zone—try non-default ways of doing things—even, try to think creatively? But you can imagine the one coming back and saying, “I tried to leave my comfort zone, and I think I succeeded at that! I brainstormed for five minutes—and came up with all sorts of wacky creative ideas! But I don't think any of them are good enough. The other guy can just keep saying ‘No,’ no matter what I do.”

And now we finally reply: “*Shut up and do the impossible!*”

As we recall from Trying to Try, setting out to make an *effort* is distinct from setting out to *win*. That's the problem with saying, “Make an extraordinary effort.” You can succeed at the goal of “making an extraordinary effort” without succeeding at the goal of getting out of the Box.

“But!” says the one. “But, *SUCCEED* is not a primitive action! Not all challenges are fair—sometimes you just can't win! How am I supposed to choose to be out of the Box? The other guy can just keep on saying ‘No!’”

True. Now shut up and do the impossible.

Your goal is not to do better, to try desperately, or even to try extraordinarily. Your goal is to get out of the box.

To accept this demand creates an awful tension in your mind, between the impossibility and the requirement to do it anyway. People will try to flee that awful tension.

A couple of people have reacted to the AI-Box Experiment by saying, “Well, Eliezer, playing the AI, probably just threatened to destroy the world whenever he was out, if he wasn't let out immediately,” or “Maybe the AI offered the Gatekeeper a trillion dollars to let it out.” But as any sensible person should realize on considering this strategy, the Gatekeeper is likely to just go on saying “No.”

So the people who say, “Well, of course Eliezer must have just done XXX,” and then offer up something that fairly obviously wouldn't work—would they be able to escape the Box?

They're trying *too hard* to convince themselves the problem isn't impossible.

One way to run from the awful tension is to seize on a solution, any solution, even if it's not very good.

Which is why it's important to go forth with the true intent—to *have produced* a solution, a *good* solution, at the end of the search, and then to implement that solution and win.

I don't quite want to say that "you should expect to solve the problem." If you hacked your mind so that you assigned high probability to solving the problem, that wouldn't accomplish anything. You would just lose at the end, perhaps after putting forth not much of an effort—or putting forth a merely desperate effort, secure in the faith that the universe is fair enough to grant you a victory in exchange.

To have *faith* that you could solve the problem would just be another way of running from that awful tension.

And yet—you can't be setting out to *try* to solve the problem. You can't be setting out to *make an effort*. You have to be setting out to win. You can't be saying to yourself, "And now I'm going to do my best." You have to be saying to yourself, "And now I'm going to figure out how to get out of the Box"—or reduce consciousness to nonmysterious parts, or whatever.

I say again: You must really intend to solve the problem. If in your heart you believe the problem really *is* impossible—or if you believe that *you* will fail—then you won't hold yourself to a high enough standard. You'll only be trying for the sake of trying. You'll sit down—conduct a mental search—try to be creative and brainstorm a little—look over all the solutions you generated—conclude that none of them work—and say, "Oh well."

No! *Not* well! You haven't won yet! Shut up and do the impossible!

When AI folk say to me, "Friendly AI is impossible," I'm pretty sure they haven't even tried for the sake of trying. But if they *did* know the technique of "Try for five minutes before giving up," and they dutifully agreed to try for five minutes by the clock, then they still wouldn't come up with anything. They

would not go forth with true intent to solve the problem, only intent to *have tried* to solve it, to make themselves defensible.

So am I saying that you should doublethink to make yourself believe that you will solve the problem with probability 1? Or even doublethink to add one iota of credibility to your true estimate?

Of course not. In fact, it is necessary to keep in full view the reasons why you *can't* succeed. If you lose sight of *why* the problem is impossible, you'll just seize on a false solution. The *last* fact you want to forget is that the Gatekeeper could always just tell the AI "No"—or that consciousness seems intrinsically different from any possible combination of atoms, etc.

(One of the key Rules For Doing The Impossible is that, if you can state *exactly* why something is impossible, you are often close to a solution.)

So you've got to hold both views in your mind at once—seeing the full impossibility of the problem, and intending to solve it.

The awful tension between the two simultaneous views comes from not knowing which will prevail. Not expecting to surely lose, nor expecting to surely win. Not setting out just to try, just to have an uncertain chance of succeeding—because then you would have a surety of having tried. The certainty of uncertainty can be a relief, and you have to reject that relief too, because it marks the end of desperation. It's an in-between place, "unknown to death, nor known to life."

In fiction it's easy to show someone trying harder, or trying desperately, or even trying the extraordinary, but it's very hard to show someone who shuts up and attempts the impossible. It's difficult to depict Bambi choosing to take on Godzilla, in such fashion that your readers seriously don't know who's going to win—expecting neither an "astounding" heroic victory just like the last fifty times, nor the default squish.

You might even be justified in refusing to use probabilities at this point. In all honesty, I really *don't* know how to estimate the probability of solving an impossible problem that I have gone forth with intent to solve—in a case where I've previously solved some impossible problems, but the particular impossible

problem is more difficult than anything I've yet solved, but I plan to work on it longer, et cetera.

People ask me how likely it is that humankind will survive, or how likely it is that anyone can build a Friendly AI, or how likely it is that I can build one. I really *don't* know how to answer. I'm not being evasive; I don't know how to put a probability estimate on my, or someone else's, successfully shutting up and doing the impossible. Is it probability zero because it's impossible? Obviously not. But how likely is it that this problem, like previous ones, will give up its unyielding blankness when I understand it better? It's not truly impossible; I can see that much. But humanly impossible? Impossible to me in particular? I don't know how to guess. I can't even translate my intuitive feeling into a number, because the only intuitive feeling I have is that the "chance" depends heavily on my choices and unknown unknowns: a wildly unstable probability estimate.

But I do hope by now that I've made it clear why you shouldn't panic, when I now say clearly and forthrightly that building a Friendly AI is impossible.

I hope this helps explain some of my attitude when people come to me with various bright suggestions for building communities of AIs to make the whole Friendly without any of the individuals being trustworthy, or proposals for keeping an AI in a box, or proposals for "Just make an AI that does X," et cetera. Describing the specific flaws would be a whole long story in each case. But the general rule is that you can't do it *because Friendly AI is impossible*. So you should be very suspicious indeed of someone who proposes a solution that seems to involve only an *ordinary* effort—without even taking on the trouble of doing anything impossible. Though it does take a mature understanding to appreciate this impossibility, so it's not surprising that people go around proposing clever shortcuts.

On the AI-Box Experiment, so far I've only been convinced to divulge a single piece of information on how I did it—when someone noticed that I was reading Y Combinator's Hacker News, and posted a topic called "Ask Eliezer Yudkowsky" that got voted to the front page. To which I replied:

Oh, dear. Now I feel obliged to say *something*, but all the original reasons against discussing the AI-Box experiment are still in force ...

All right, this much of a hint:

There's no super-clever special trick to it. I just did it the hard way.

Something of an entrepreneurial lesson there, I guess.

There was no super-clever special trick that let me get out of the Box using only a *cheap* effort. I didn't bribe the other player, or otherwise violate the spirit of the experiment. I just did it the hard way.

Admittedly, the AI-Box Experiment never did seem like an *impossible* problem to me to begin with. When someone can't think of any possible argument that would convince them of something, that just means their brain is running a search that hasn't yet turned up a path. It doesn't mean they can't be convinced.

But it illustrates the general point: "Shut up and do the impossible" isn't the same as expecting to find a cheap way out. That's only another kind of running away, of reaching for relief.

Tsuyoku naritai is more stressful than being content with who you are. *Isshokenmei* calls on your willpower for a convulsive output of conventional strength. "Make an extraordinary effort" demands that you *think*; it puts you in situations where you may not know what to do next, unsure of whether you're doing the right thing. But "Shut up and do the impossible" represents an even higher octave of the same thing, and its cost to its employer is correspondingly greater.

Before you the terrible blank wall stretches up and up and up, unimaginably far out of reach. And there is also the need to solve it, *really* solve it, not "try your best." Both awarenesses in the mind at once, simultaneously, and the tension between. All the reasons you can't win. All the reasons you have to. Your intent to solve the problem. Your extrapolation that every technique you know will fail. So you tune yourself to the highest

pitch you can reach. Reject all cheap ways out. And then, like walking through concrete, start to move forward.

I try not to dwell too much on the drama of such things. By all means, if you can diminish the cost of that tension to yourself, you should do so. There is nothing heroic about making an effort that is the slightest bit more heroic than it has to be. If there really is a cheap shortcut, I suppose you could take it. But I have yet to find a *cheap* way out of any impossibility I have undertaken.

There were three more AI-Box experiments besides the ones described on the linked page, which I never got around to adding in. People started offering me thousands of dollars as stakes—"I'll pay you \$5,000 if you can convince me to let you out of the box." They didn't seem sincerely convinced that not even a transhuman AI could make them let it out—they were just curious—but I was tempted by the money. So, after investigating to make sure they could afford to lose it, I played another three AI-Box experiments. I won the first, and then lost the next two. And then I called a halt to it. I didn't like the person I turned into when I started to lose.

I put forth a desperate effort, and lost anyway. It hurt—both the losing, and the desperation. It wrecked me for that day and the day afterward.

I'm a sore loser. I don't know if I'd call that a "strength," but it's one of the things that drives me to keep at impossible problems.

But you can lose. It's allowed to happen. Never forget that, or why are you bothering to try so hard? Losing hurts, if it's a loss you can survive. And you've wasted time, and perhaps other resources.

"Shut up and do the impossible" should be reserved for *very* special occasions. You can lose, and it will hurt. You have been warned.

...but it's only at this level that adult problems begin to come into sight.

Final Words

Sunlight enriched air already alive with curiosity, as dawn rose on Brennan and his fellow students in the place to which Jeffreyssai had summoned them.

They sat there and waited, the five, at the top of the great glassy crag that was sometimes called Mount Mirror, sometimes Mount Monastery, and more often simply left unnamed. The high top and peak of the mountain, from which you could see all the lands below and seas beyond.

(Well, not *all* the lands below, nor seas beyond. So far as anyone knew, there was no place in the world from which all the world was visible; nor, equivalently, any kind of vision that would see through all obstacle-horizons. In the end it was the top only of *one particular* mountain: there were other peaks, and from their tops you would see other lands below; even though, in the end, it was all a single world.)

“What do you think comes next?” said Hiriwa. Her eyes were bright, and she gazed to the far horizons like a lord.

Taji shrugged, though his own eyes were alive with anticipation. “Jeffreyssai’s last lesson doesn’t have any obvious sequel that I can think of. In fact, I think we’ve learned just about everything that I *knew* the *beisutsukai* masters knew. What’s left, then—”

“Are the *real* secrets,” Yin completed the thought.

Hiriwa and Taji and Yin shared a grin, among themselves.

Styrlын wasn’t smiling. Brennan suspected rather strongly that Styrlын was older than he had admitted.

Brennan wasn’t smiling either. He might be young, but he kept high company, and had witnessed some of what went on behind the curtains of the world. Secrets had their price, always; that was the barrier that made them secrets. And Brennan thought he had a good idea of what this price might be.



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There was a cough from *behind* them, at a moment when they had all happened to be looking in any other direction but that one.

As one, their heads turned.

Jeffreyssai stood there, in a casual robe that looked more like very glassy glass than any proper sort of mirrorweave.

Jeffreyssai stood there and looked at them, a strange abiding sorrow in those inscrutable ancient eyes.

“Sen ...sei,” Taji started, faltering as that bright anticipation stumbled over Jeffreyssai’s return look. “What’s next?”

“Nothing,” Jeffreyssai said abruptly. “You’re finished. It’s done.”

Hiriwa, Taji, and Yin all blinked, a perfect synchronized gesture of shock. Then, before their expressions could turn to outrage and objections—

“*Don’t*,” Jeffreyssai said. There was real pain in it. “Believe me, it hurts me more than it hurts you.” He might have been looking at them; or at something far away, or long ago. “I don’t know exactly what roads may lie before you—but yes, I *know* you’re not ready. I *know* I’m sending you out unprepared. I *know* that everything I taught you is incomplete. That what I said is not what you heard. I know that I left out the one most important thing. That the rhythm at the center of everything is missing and astray. I know that you will harm yourself in the course of trying to use what I taught; so that *I*, personally, will have shaped, in some fashion unknown to me, the very knife that will cut you ...

“...that’s the hell of being a teacher, you see,” Jeffreyssai said. Something grim flickered in his expression. “Nonetheless, you’re *done*. Finished, for now. What lies between you and mastery is not another classroom. We are fortunate, or perhaps not *fortunate*, that the road to power does not wend only through lecture halls. Or the quest would be boring to the bitter end. Still, I *cannot* teach you; and so it is a moot point whether I *would*. There is no master here whose art is all inherited. Even the *beisutsukai* have never discovered how to teach certain things; it is possible that such an event has been prohibited. And so you can only arrive at mastery by using to the fullest

the techniques you have already learned, facing challenges and apprehending them, mastering the tools you have been taught *until they shatter in your hands—*

Jeffreyssai's eyes were hard, as though steeled in acceptance of unwelcome news.

“—and you are left in the midst of wreckage absolute. *That* is where I, your teacher, am sending you. You are not *beisutsukai* masters. I cannot create masters. I cannot even come close. Go, then, and fail.”

“But—” said Yin, and stopped herself.

“Speak,” said Jeffreyssai.

“But then why,” she said helplessly, “why teach us anything in the first place?”

Brennan's eyelids flickered just the tiniest amount.

It was enough for Jeffreyssai. “Answer her, Brennan, if you think you know.”

“Because,” Brennan said, “if we were not taught, there would be no chance *at all* of our becoming masters.”

“Even so,” said Jeffreyssai. “If you were *not* taught—then when you failed, you might simply think you had reached the limits of Reason itself. You would be discouraged and bitter amid the wreckage. You might not even realize when you had failed. No; you have been shaped into something that *may* emerge from the wreckage of your past self, determined to *remake* your art. And then you will remember much that will help you. If you had not been taught, your chances would be—less.” His gaze passed over the group. “It should be obvious, but understand that the moment of your crisis cannot be provoked artificially. To *teach* you something, the catastrophe must come as a *surprise*.”

Brennan made the gesture with his hand that indicated a question; and Jeffreyssai nodded in reply.

“Is this the *only* way in which Bayesian masters come to be, sensei?”

“I do not know,” said Jeffreyssai, from which the overall state of the evidence was obvious enough. “But I doubt there would ever be a road that leads only through the monastery. We are the heirs in this world of mystics as well as scientists,

just as the Competitive Conspiracy inherits from chess players alongside cagefighters. We have turned our impulses to more constructive uses—but we must still stay on our guard against old failure modes.”

Jeffreyssai took a breath. “Three flaws above all are common among the *beisutsukai*. The first flaw is to look just the slightest bit harder for flaws in arguments whose conclusions you would rather not accept. If you cannot contain this aspect of yourself then every flaw you know how to detect will make you that much stupider. This is the challenge that determines whether you possess the art or its opposite: intelligence, to be useful, must be used for something other than defeating itself.

“The second flaw is cleverness. To invent great complicated plans and great complicated theories and great complicated arguments—or even, perhaps, plans and theories and arguments which are commended too much by their elegance and too little by their realism. There is a widespread saying which runs: ‘The vulnerability of the *beisutsukai* is well-known; they are prone to be too clever.’ Your enemies *will* know this saying, if they know you for a *beisutsukai*, so *you* had best remember it also. And you may think to yourself: ‘But if I could *never* try anything clever or elegant, would my life even be worth living?’ This is why cleverness is still our chief vulnerability even after its being well-known, like offering a Competitor a challenge that seems fair, or tempting a Bard with drama.

“The third flaw is underconfidence, modesty, humility. You have learned so much of flaws, some of them impossible to fix, that you may think that the rule of wisdom is to confess your own inability. You may question yourself so much, without resolution or testing, that you lose your will to carry on in the Art. You may refuse to decide, pending further evidence, when a decision is *necessary*; you may take advice you should not take. Jaded cynicism and sage despair are less fashionable than once they were, but you may still be tempted by them. Or you may simply—lose momentum.”

Jeffreyssai fell silent then.

He looked from each of them, one to the other, with quiet intensity.

And said at last, “Those are my final words to you. If and when we meet next, you and I—if and when you return to this place, Brennan, or Hiriwa, or Taji, or Yin, or Styrllyn—I will no longer be your teacher.”

And Jeffreyssai turned and walked swiftly away, heading back toward the glassy tunnel that had emitted him.

Even Brennan was shocked. For a moment they were all speechless.

Then—

“Wait!” cried Hiriwa. “What about *our* final words to you? I never said—”

“I will tell you what my *sensei* told me,” Jeffreyssai’s voice came back as he disappeared. “You can thank me after you return, if you return. One of you at least seems likely to come back.”

“No, wait, I—” Hiriwa fell silent. In the mirrored tunnel, the fractured reflections of Jeffreyssai were already fading. She shook her head. “Never ...mind, then.”

There was a brief, uncomfortable silence, as the five of them looked at each other.

“Good heavens,” Taji said finally. “Even the Bardic Conspiracy wouldn’t try for that much drama.”

Yin suddenly laughed. “Oh, this was *nothing*. You should have seen *my* send-off when I left Diamond Sea University.” She smiled. “I’ll tell you about it sometime—if you’re interested.”

Taji coughed. “I suppose I should go back and ...pack my things ...”

“I’m already packed,” Brennan said. He smiled, ever so slightly, when the other three turned to look at him.

“Really?” Taji asked. “What was the clue?”

Brennan shrugged with careful carelessness. “Beyond a certain point, it is futile to inquire how a *beisutsukai* master knows a thing—”

“Come off it!” Yin said. “You’re not a *beisutsukai* master yet.”

“Neither is Styrllyn,” Brennan said. “But he has already packed as well.” He made it a statement rather than a question, betting double or nothing on his image of inscrutable foreknowledge.

Styrllyn cleared his throat. "As you say. Other commitments call me, and I have already tarried longer than I planned. Though, Brennan, I do feel that you and I have certain mutual interests, which I would be happy to discuss with you—"

"Styrllyn, my most excellent friend, I shall be happy to speak with you on any topic you desire," Brennan said politely and noncommittally, "if we should meet again." As in, not now. He certainly wasn't selling out his Mistress *this* early in their relationship.

There was an exchange of goodbyes, and of hints and offers.

And then Brennan was walking down the road that led toward or away from Mount Monastery (for every road is a two-edged sword), the smoothed glass pebbles clicking under his feet.

He strode out along the path with purpose, vigor, and determination, just in case someone was watching.

Some time later he stopped, stepped off the path, and wandered just far enough away to prevent anyone from finding him unless they were deliberately following.

Then he sagged wearily back against a tree-trunk. It was a sparse clearing, with only a few trees poking out of the ground; not much present in the way of distracting scenery, unless you counted the red-tinted stream flowing out of a dark cave-mouth. And Brennan deliberately faced away from that, leaving only the far gray of the horizons, and the blue sky and bright sun.

Now what?

He had *thought* that the Bayesian Conspiracy, of all the possible trainings that existed in this world, would have cleared up his uncertainty about what to do with the rest of his life.

Power, he'd sought at first. Strength to prevent a repetition of the past. "If you don't know what you need, take power"—so went the proverb. He had gone first to the Competitive Conspiracy, then to the *beisutsukai*.

And now ...

Now he felt more lost than ever.

He could think of things that made him happy. But nothing that he really *wanted*.

The passionate intensity that he'd come to associate with his Mistress, or with Jeffreyssai, or the other figures of power that he'd met ...a life of pursuing small pleasures seemed to pale in comparison, next to that.

In a city not far from the center of the world, his Mistress waited for him (in all probability, assuming she hadn't gotten bored with her life and run away). But to merely return, and then drift aimlessly, waiting to fall into someone else's web of intrigue ...no. That didn't seem like ...*enough*.

Brennan plucked a blade of grass from the ground and stared at it, half-unconsciously looking for anything interesting about it; an old, old game that his very first teacher had taught him, what now seemed like ages ago.

Why did I believe that going to Mount Mirror would tell me what I wanted?

Well, decision theory *did* require that your utility function be consistent, but ...

If the beisutsukai knew what I wanted, would they even tell me?

At the Monastery they taught doubt. So now he was falling prey to the third besetting sin of which Jeffreyssai had spoken: lost momentum, indeed. For he had learned to question the image that he held of himself in his mind.

Are you seeking power because that is your true desire, Brennan?

Or because you have a picture in your mind of the role that you play as an ambitious young man, and you think it is what someone playing your role would do?

Almost everything he'd done up until now, even going to Mount Mirror, had probably been the latter.

And when he blanked out the old thoughts and tried to see the problem as though for the first time ...

...nothing much came to mind.

What do I want?

Maybe it wasn't reasonable to expect the *beisutsukai* to tell him outright. But was there anything they had taught him by which he might answer?

Brennan closed his eyes and thought.

First, suppose there is something I would passionately desire. Why would I not know what it is?

Because I have not yet encountered it, or ever imagined it?

Or because there is some reason I would not admit it to myself?

Brennan laughed out loud, then, and opened his eyes.

So simple, once you thought of it that way. So obvious in retrospect. *That* was what they called a silver-shoes moment, and yet, if he hadn't gone to Mount Mirror, it wouldn't ever have occurred to him.

Of course there was something he wanted. He knew *exactly* what he wanted. Wanted so desperately he could taste it like a sharp tinge on his tongue.

It just hadn't come to mind earlier, because ...if he acknowledged his desire explicitly ...then he also had to see that it was *difficult*. High, high, above him. Far out of his reach. "Impossible" was the word that came to mind, though it was not, of course, impossible.

But once he asked himself if he preferred to wander aimlessly through his life—once it was put that way, the answer became obvious. Pursuing the unattainable would make for a hard life, but not a sad one. He could think of things that made him happy, either way. And in the end—it *was* what he wanted.

Brennan stood up, and took his first steps, in the exact direction of Shir L'or, the city that lies in the center of the world. He had a plot to hatch, and he did not know who would be part of it.

And then Brennan stumbled, when he realized that Jeffreysai had already known.

One of you at least seems likely to come back ...

Brennan had thought he was talking about Taji. Taji had probably thought he was talking about Taji. It was what Taji said he wanted. But how reliable of an indicator was that, really?

There was a proverb, though, about that very road he had just left: *Whoever sets out from Mount Mirror seeking the impossible, will surely return.*

When you considered Jeffreyssai's last warning—and that the proverb said nothing of *succeeding* at the impossible task itself—it was a less optimistic saying than it sounded.

Brennan shook his head wonderingly. How could Jeffreyssai possibly have known before Brennan knew himself?

Well, beyond a certain point, it is futile to inquire how a *beisutsukai* master knows a thing—

Brennan halted in mid-thought.

No.

No, if he was going to become a *beisutsukai* master himself someday, then he ought to figure it out.

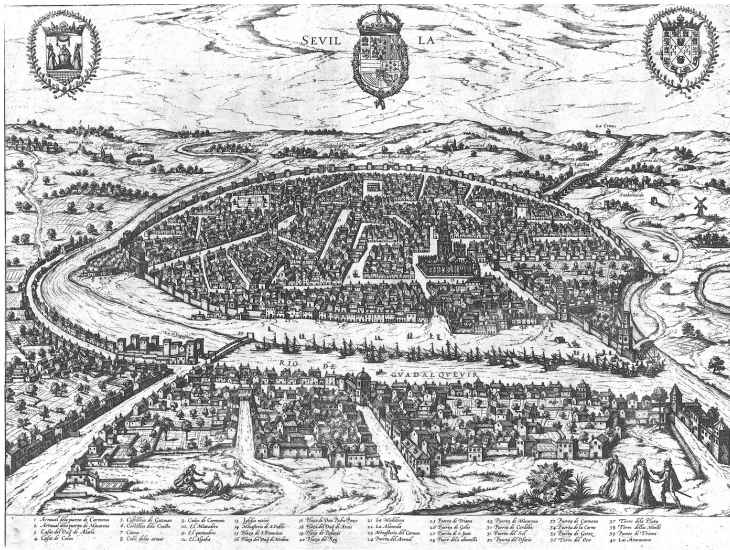
It was, Brennan realized, a *stupid* proverb.

So he walked, and this time, he thought about it carefully.

As the sun was setting, red-golden, shading his footsteps in light.

PART Z

The Craft and the Community



Raising the Sanity Waterline

To paraphrase the Black Belt Bayesian: Behind every exciting, dramatic failure, there is a more important story about a larger and less dramatic failure that made the first failure possible.

If every trace of religion were magically eliminated from the world tomorrow, then—however much improved the lives of many people would be—we would not even have come close to solving the larger failures of sanity that made religion possible in the first place.

We have good cause to spend some of our efforts on trying to eliminate religion directly, because it *is* a direct problem. But religion also serves the function of an asphyxiated canary in a coal mine—religion is a sign, a symptom, of larger problems that don't go away just because someone loses their religion.

Consider this thought experiment—what could you teach people that is not *directly* about religion, that is true and useful as a *general* method of rationality, which would cause them to lose their religions? In fact—imagine that we're going to go and survey all your students five years later, and see how many of them have lost their religions compared to a control group; if you make the slightest move at fighting religion *directly*, you will invalidate the experiment. You may not make a single mention of religion or any religious belief in your classroom; you may not even hint at it in any obvious way. All your examples must center about real-world cases that have nothing to do with religion.

If you can't fight religion *directly*, what do you teach that raises the *general waterline of sanity* to the point that religion goes underwater?

Here are some such topics I've already covered—*not* avoiding all mention of religion, but it could be done:

- Affective Death Spirals (vol. II, p. 223)—plenty of non-supernaturalist examples (like vol. II, p. 254).



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- How to avoid cached thoughts (vol. II, p. 169) and fake wisdom (vol. II, p. 189); the pressure of conformity (vol. II, p. 262).
- Evidence (vol. I, p. 94) and Occam's Razor (vol. I, p. 107)—the rules of probability.
- The Bottom Line (vol. II, p. 99) / Engines of Cognition (vol. IV, p. 32)—the causal reasons why Reason works.
- Mysterious Answers to Mysterious Questions (vol. I, p. 140)—and the whole associated sequence, like making beliefs pay rent (vol. I, p. 59) and curiosity stoppers (vol. I, p. 137)—have excellent historical examples in vitalism and phlogiston (vol. I, p. 133).
- Non-existence of ontologically fundamental mental things (vol. IV, p. 229)—apply the Mind Projection Fallacy (vol. IV, p. 64) to probability (vol. IV, p. 66), move on to reductionism (vol. IV, p. 82) versus holism (vol. I, p. 144), then brains (vol. IV, p. 156) and cognitive science (vol. IV, p. 146).
- The many sub-arts of Crisis of Faith (vol. II, p. 306)—though you'd better find something else to call this ultimate high master-level technique of *actually updating on evidence*.
- Dark Side Epistemology (vol. II, p. 131)—teaching this with no mention of religion would be hard, but perhaps you could videotape the interrogation of some snake-oil sales agent as your real-world example.
- Fun Theory¹ (vol. V, p. 101)—teach as a literary theory of utopian fiction, without the direct application to theodicy.

1. sharov.me/r/6-7

The Fun Theory Sequence • lesswrong.com



- Joy in the Merely Real (vol. IV, p.101), naturalistic metaethics, et cetera, et cetera, et cetera, and so on.

But to look at it another way—

Suppose we have a scientist who's still religious, either full-blown scriptural-religion, or in the sense of tossing around vague casual endorsements of "spirituality."

We now know this person is not applying any *technical, explicit* understanding of...

- ...what constitutes evidence and why;
- ...Occam's Razor;
- ...how the above two rules derive from the lawful and causal operation of minds as mapping engines, and do not switch off when you talk about tooth fairies;
- ...how to tell the difference between a real answer and a curiosity-stopper;
- ...how to rethink matters for themselves instead of just repeating things they heard;
- ...certain general trends of science over the last three thousand years;
- ...the difficult arts of actually updating on new evidence and relinquishing old beliefs;
- ...epistemology 101;
- ...self-honesty 201;
- ...et cetera, et cetera, et cetera, and so on.

When you consider it—these are all rather *basic* matters of study, as such things go. A quick introduction to *all* of them (well, except naturalistic metaethics) would be ...a four-credit undergraduate course with no prerequisites?

But there are Nobel laureates who haven't taken that course! Richard Smalley if you're looking for a cheap shot, or Robert Aumann if you're looking for a scary shot.

And they can't be isolated exceptions. If all of their professional compatriots had taken that course, then Smalley or Aumann would either have been corrected (as their colleagues kindly took them aside and explained the bare fundamentals) or else regarded with too much pity and concern to win a Nobel Prize. Could you—*realistically* speaking, regardless of fairness—win a Nobel while advocating the existence of Santa Claus?

That's what the dead canary, religion, is telling us: that the general sanity waterline is currently *really ridiculously low*. Even in the highest halls of science.

If we throw out that dead and rotting canary, then our mine may stink a bit less, but the sanity waterline may not rise much higher.

This is not to criticize the neo-atheist movement. The harm done by religion is clear and present danger, or rather, current and ongoing disaster. Fighting religion's directly harmful effects takes precedence over its use as a canary or experimental indicator. But even if Dawkins, and Dennett, and Harris, and Hitchens, should somehow win utterly and absolutely to the last corner of the human sphere, the real work of rationalists will be only just beginning.

A Sense That More Is Possible

To teach people about a topic you've labeled "rationality," it helps for them to be interested in "rationality." (There are less direct ways to teach people how to attain the map that reflects the territory, or optimize reality according to their values; but the explicit method *is* the course I tend to take.)

And when people explain why they're *not* interested in rationality, one of the most commonly proffered reasons tends to be like: "Oh, I've known a couple of rational people and they didn't seem any happier."

Who are they thinking of? Probably an Objectivist or some such. Maybe someone they know who's an ordinary scientist. Or an ordinary atheist.

That's really *not* a whole lot of rationality, as I have previously said.

Even if you limit yourself to people who can derive Bayes's Theorem—which is going to eliminate, what, 98% of the above personnel?—that's *still* not a whole lot of rationality. I mean, it's a pretty basic theorem.

Since the beginning I've had a sense that there ought to be some discipline of cognition, some art of thinking, the studying of which would make its students visibly more competent, more formidable: the equivalent of Taking a Level in Awesome.

But when I look around me in the real world, I don't see that. Sometimes I see a hint, an echo, of what I think should be possible, when I read the writings of folks like Robyn Dawes, Daniel Gilbert, John Tooby, and Leda Cosmides. A few very rare and very senior researchers in psychological sciences, who visibly care a *lot* about rationality—to the point, I suspect, of making their colleagues feel uncomfortable, because it's not cool to care that much. I can see that they've found a rhythm, a unity that begins to pervade their arguments—

Yet even that ...isn't really a whole lot of rationality either.



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Even among those few who impress me with a hint of dawning formidability—I don’t think that their mastery of rationality could compare to, say, John Conway’s mastery of math. The base knowledge that we drew upon to build our understanding—if you extracted only the parts we used, and not everything we had to study to find it—it’s probably not comparable to what a professional nuclear engineer knows about nuclear engineering. It may not even be comparable to what a construction engineer knows about bridges. We practice our skills, we do, in the ad-hoc ways we taught ourselves; but that practice probably doesn’t compare to the training regimen an Olympic runner goes through, or maybe even an ordinary professional tennis player.

And the root of *this* problem, I do suspect, is that we haven’t really gotten together and systematized our skills. We’ve had to create all of this for ourselves, ad-hoc, and there’s a limit to how much one mind can do, even if it can manage to draw upon work done in outside fields.

The chief obstacle to doing this the way it *really* should be done is the difficulty of testing the *results* of rationality training programs, so you can have evidence-based training methods. I will write more about this, because I think that recognizing successful training and distinguishing it from failure is the essential, blocking obstacle.

There are experiments done now and again on debiasing interventions for particular biases, but it tends to be something like, “Make the students practice this for an hour, then test them two weeks later.” Not, “Run half the signups through version *A* of the three-month summer training program, and half through version *B*, and survey them five years later.” You can see, here, the implied amount of effort that I think would go into a training program for people who were Really Serious about rationality, as opposed to the attitude of taking Casual Potshots That Require Like An Hour Of Effort Or Something.

Daniel Burfoot brilliantly suggests that this is why intelligence seems to be such a big factor in rationality—that when you’re improvising everything ad-hoc with very little train-

ing or systematic practice, intelligence ends up being the most important factor in what's left.

Why aren't "rationalists" surrounded by a visible aura of formidability? Why aren't they found at the top level of every elite selected on any basis that has anything to do with thought? Why do most "rationalists" just seem like ordinary people, perhaps of moderately above-average intelligence, with one more hobbyhorse to ride?

Of this there are several answers; but one of them, surely, is that they have received less systematic training of rationality in a less systematic context than a first-dan black belt gets in hitting people.

I do not except myself from this criticism. I am no *beisutsukai*, because there are limits to how much Art you can create on your own, and how well you can guess without evidence-based statistics on the results. I know about a *single* use of rationality, which might be termed "reduction of confusing cognitions." This I asked of my brain; this it has given me. There are other arts, I think, that a mature rationality training program would not neglect to teach, which would make me stronger and happier and more effective—if I could just go through a standardized training program using the cream of teaching methods experimentally demonstrated to be effective. But the kind of tremendous, focused effort that I put into creating my single *sub-art* of rationality from scratch—my life doesn't have room for more than one of those.

I consider myself something more than a first-dan black belt, and less. I can *punch* through brick and I'm working on steel along my way to adamantite, but I have a mere casual street-fighter's grasp of how to kick or throw or block.

Why are there schools of martial arts, but not rationality dojos? (This was the first question I asked in my first blog post.) Is it more important to hit people than to think?

No, but it's easier to verify when you *have* hit someone. That's part of it, a highly central part.

But maybe even more importantly—there are people out there who *want* to hit, and who have the idea that there ought to be a systematic art of hitting that makes you into a visibly

more formidable fighter, with a speed and grace and strength beyond the struggles of the unpracticed. So they go to a school that promises to teach that. And that school exists because, long ago, some people had the sense that more was possible. And they got together and shared their techniques and practiced and formalized and practiced and developed the Systematic Art of Hitting. They pushed themselves that far because *they thought they should be awesome* and they were willing to put some *back* into it.

Now—they *got* somewhere with that aspiration, unlike a thousand other aspirations of awesomeness that failed, because they could *tell* when they had hit someone; and the schools competed against each other regularly in realistic contests with clearly-defined winners.

But before even that—there was first the aspiration, the wish to become stronger, a sense that more was possible. A vision of a speed and grace and strength that they did not already possess, but *could* possess, *if* they were willing to put in a lot of work, that drove them to systematize and train and test.

Why don't we have an Art of Rationality?

Third, because current "rationalists" have trouble working in groups: of this I shall speak more.

Second, because it is hard to verify success in training, or which of two schools is the stronger.

But first, because people lack the sense that rationality is something that *should* be systematized and trained and tested like a martial art, that should have as much knowledge behind it as nuclear engineering, whose superstars should practice as hard as chess grandmasters, whose successful practitioners should be surrounded by an evident aura of awesome.

And conversely they don't look at the *lack* of visibly greater formidability, and say, "We must be doing something wrong."

"Rationality" just seems like one more hobby or hobbyhorse, that people talk about at parties; an adopted mode of conversational attire with few or no real consequences; and it doesn't seem like there's anything wrong about that, either.

Epistemic Viciousness

Someone deserves a large hat tip for this, but I'm having trouble remembering who; my records don't seem to show any email or *Overcoming Bias* comment which told me of this 12-page essay, "Epistemic Viciousness in the Martial Arts" by Gillian Russell.¹ Maybe Anna Salamon?



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We all lined up in our ties and sensible shoes (this was England) and copied him—left, right, left, right—and afterwards he told us that if we practised in the air with sufficient devotion for three years, then we would be able to use our punches to kill a bull with one blow.

I worshipped Mr Howard (though I would sooner have died than told him that) and so, as a skinny, eleven-year-old girl, I came to believe that if I practised, I would be able to kill a bull with one blow by the time I was fourteen.

This essay is about epistemic viciousness in the martial arts, and this story illustrates just that. Though the word "viciousness" normally suggests deliberate cruelty and violence, I will be using it here with the more old-fashioned meaning, posessing of vices.

It all generalizes *amazingly*. To summarize some of the key observations for how epistemic viciousness arises:

- The art, the dojo, and the sensei are seen as sacred. "Having red toe-nails in the dojo is like going to church in a mini-skirt and halter-top ...The students of other martial

1. Gillian Russell, "Epistemic Viciousness in the Martial Arts," in *Martial Arts and Philosophy: Beating and Nothingness*, ed. Graham Priest and Damon A. Young (Open Court, 2010).

arts are talked about like they are practicing the wrong religion.”

- If your teacher takes you aside and teaches you a special move and you practice it for twenty years, you have a large emotional investment in it, and you’ll want to discard any incoming evidence against the move.
- Incoming students don’t have much choice: a martial art can’t be learned from a book, so they have to trust the teacher.
- Deference to famous historical masters. “Runners think that the contemporary staff of Runner’s World know more about running than all the ancient Greeks put together. And it’s not just running, or other physical activities, where history is kept in its place; the same is true in any well-developed area of study. It is not considered disrespectful for a physicist to say that Isaac Newton’s theories are false ...” (Sound familiar?)
- “We martial artists struggle with a kind of poverty—data-poverty—which makes our beliefs hard to test ...Unless you’re unfortunate enough to be fighting a hand-to-hand war you cannot *check* to see how much force and exactly which angle a neck-break requires ...”
- “If you can’t test the effectiveness of a technique, then it is hard to test methods for improving the technique. Should you practice your nukite in the air, or will that just encourage you to overextend? ...Our inability to test our fighting methods restricts our ability to test our training methods.”
- “But the real problem isn’t just that we live in data poverty—I think that’s true for some perfectly respectable disciplines, including theoretical physics—the problem is that we live in poverty but continue to act as though we live in luxury, as though we can safely afford to believe whatever we’re told ...” (+10!)

One thing that I remembered being in this essay, but, on a second reading, wasn't actually there, was the degeneration of martial arts after the decline of real fights—by which I mean, fights where people were really trying to hurt each other and someone occasionally got killed.

In those days, you had some idea of who the real masters were, and which school could defeat others.

And then things got all *civilized*. And so things went downhill to the point that we have videos on Youtube of supposed Nth-dan black belts being pounded into the ground by someone with real fighting experience.

I heard of one case of this that was *really* sad; it was a master of a school who was convinced he could use *ki* techniques. His students would actually fall over when he used *ki* attacks, a strange and remarkable and frightening case of self-hypnosis or *something*... and the master goes up against a skeptic and of course gets pounded completely into the floor.

Truly is it said that “how to not lose” is more broadly applicable information than “how to win.” Every single one of these risk factors transfers straight over to any attempt to start a “rationality dojo.” I put to you the question: What can be done about it?

Schools Proliferating Without Evidence



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Robyn Dawes, author of one of the original papers from *Judgment Under Uncertainty* and of the book *Rational Choice in an Uncertain World*—one of the few who tries really hard to import the results to real life—is also the author of *House of Cards: Psychology and Psychotherapy Built on Myth*.

From *House of Cards*, chapter 1:¹

The ability of these professionals has been subjected to empirical scrutiny—for example, their effectiveness as therapists (Chapter 2), their insight about people (Chapter 3), and the relationship between how well they function and the amount of experience they have had in their field (Chapter 4). Virtually all the research—and this book will reference more than three hundred empirical investigations and summaries of investigations—has found that these professionals’ claims to superior intuitive insight, understanding, and skill as therapists are simply invalid...

Remember Rorschach ink-blot tests? It’s such an appealing argument: the patient looks at the ink-blot and says what they see, the psychotherapist interprets their psychological state based on this. There’ve been hundreds of experiments looking for some evidence that it actually works. Since you’re reading this, you can guess the answer is simply “No.” Yet the Rorschach is still in use. It’s just such a *good story* that psychotherapists simply can’t bring themselves to believe the vast mounds of experimental evidence saying it doesn’t work—

—which tells you what sort of field we’re dealing with here.

1. Robyn M. Dawes, *House of Cards: Psychology and Psychotherapy Built on Myth* (Free Press, 1996).

And the experimental results on the field as a whole are commensurate. Yes, patients who see psychotherapists have been known to get better faster than patients who simply do nothing. But there is no statistically discernible difference between the many schools of psychotherapy. There is no discernible gain from years of expertise.

And there's also no discernible difference between seeing a psychotherapist and spending the same amount of time talking to a randomly selected college professor from another field. It's just talking to *anyone* that helps you get better, apparently.

In the entire absence of the slightest experimental evidence for their effectiveness, psychotherapists became licensed by states, their testimony accepted in court, their teaching schools accredited, and their bills paid by health insurance.

And there was also a huge proliferation of "schools," of traditions of practice, in psychotherapy; despite—or perhaps *because* of—the lack of any experiments showing that one school was better than another...

I should really write more some other time on all the sad things this says about our world. About how *the essence of medicine*, as recognized by society and the courts, is not a repertoire of procedures with statistical evidence for their healing effectiveness; but, rather, the right air of authority.

But the subject here is the proliferation of traditions in psychotherapy. So far as I can discern, this was the way you picked up prestige in the field—not by discovering an amazing new technique whose effectiveness could be experimentally verified and adopted by all; but, rather, by splitting off your own "school," supported by your charisma as founder, and by the good stories you told about all the reasons your techniques *should* work.

This was probably, to no small extent, responsible for the existence and continuation of psychotherapy in the first place—the promise of making yourself a Master, like Freud who'd done it first (also without the slightest scrap of experimental evidence). That's the brass ring of success to chase—the prospect of being a guru and having your own adherents. It's the struggle for adherents that keeps the clergy vital.

That's what happens to a field when it unbinds itself from the experimental evidence—though there were other factors that also placed psychotherapists at risk, such as the deference shown them by their patients, the wish of society to believe that mental healing was possible, and, of course, the general dangers of telling people how to think.

(Dawes wrote in the '80s and I know that the Rorschach was still in use as recently as the '90s, but it's possible matters have improved since then (as one commenter states). I do remember hearing that there was positive evidence for the greater effectiveness of cognitive-behavioral therapy.)

The field of hedonic psychology (happiness studies) began, to some extent, with the realization that you could *measure* happiness—that there was a family of measures that by golly did validate well against each other.

The act of creating a new measurement creates new science; if it's a *good* measurement, you get good science.

If you're going to create an organized practice of anything, you really do need some way of telling how well you're doing, and a practice of doing serious testing—that means a control group, an experimental group, and statistics—on plausible-sounding techniques that people come up with. You *really* need it.

Three Levels of Rationality Verification

I strongly suspect that there is a possible art of rationality (attaining the map that reflects the territory, choosing so as to direct reality into regions high in your preference ordering) that goes beyond the skills that are standard, and beyond what any single practitioner singly knows. I have a sense that more is possible.

The degree to which a *group* of people can do anything useful about this, will depend *overwhelmingly* on what methods we can devise to *verify* our many amazing good ideas.

I suggest stratifying verification methods into three levels of usefulness:

- Reputational
- Experimental
- Organizational.

If your martial arts master occasionally fights realistic duels (ideally, *real* duels) against the masters of other schools, and wins or at least doesn't lose too often, then you know that the master's *reputation* is *grounded in reality*; you know that your master is not a complete poseur. The same would go if your school regularly competed against other schools. You'd be *keepin' it real*.

Some martial arts fail to compete realistically enough, and their students go down in seconds against real streetfighters. Other martial arts schools fail to compete at *all*—except based on charisma and good stories—and their masters decide they have chi powers. In this latter class we can also place the splintered schools of psychoanalysis.

So even just the basic step of trying to *ground reputations* in some realistic trial other than charisma and good stories has tremendous positive effects on a whole field of endeavor.



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But that doesn't yet get you a science. A science requires that you be able to test 100 applications of method *A* against 100 applications of method *B* and run statistics on the results. *Experiments* have to be replicable and replicated. This requires *standard measurements* that can be run on students who've been taught using randomly-assigned alternative methods, not just *realistic duels* fought between masters using all of their accumulated techniques and strength.

The field of happiness studies was created, more or less, by realizing that asking people "On a scale of 1 to 10, how good do you feel right now?" was a measure that statistically validated well against other ideas for measuring happiness. And this, despite all skepticism, looks like it's actually a pretty useful measure of some things, if you ask 100 people and average the results.

But suppose you wanted to put happier people in positions of power—pay happy people to train other people to be happier, or employ the happiest at a hedge fund? Then you're going to need some test that's *harder to game* than just asking someone "How happy are you?"

This question of verification methods good enough to build *organizations* is a huge problem at all levels of modern human society. If you're going to use the SAT to control admissions to elite colleges, then can the SAT be defeated by studying *just* for the SAT in a way that ends up not correlating to other scholastic potential? If you give colleges the power to grant degrees, then do they have an incentive not to fail people? (I consider it drop-dead obvious that the task of verifying acquired skills and hence the power to grant degrees should be separated from the institutions that do the teaching, but let's not go into that.) If a hedge fund posts 20% returns, are they really that much better than the indices, or are they selling puts that will blow up in a down market?

If you have a verification method that can be gamed, the whole field adapts to game it, and loses its purpose. Colleges turn into tests of whether you can endure the classes. High schools do nothing but teach to statewide tests. Hedge funds sell puts to boost their returns.

On the other hand—we still manage to teach engineers, even though our organizational verification methods aren't perfect. So what perfect or imperfect methods could you use for verifying rationality skills, that would be at least a *little* resistant to gaming?

(Measurements with high noise can still be used *experimentally*, if you randomly assign enough subjects to have an expectation of washing out the variance. But for the *organizational* purpose of verifying particular individuals, you need low-noise measurements.)

So I now put to you the question—how do you verify rationality skills? At any of the three levels? Brainstorm, I beg you; even a difficult and expensive measurement can become a gold standard to verify other metrics. Feel free to email me at yudkowsky@gmail.com to suggest any measurements that are better off not being publicly known (though this is of course a major disadvantage of that method). Stupid ideas can suggest good ideas, so if you can't come up with a good idea, *come up with a stupid one*.

Reputational, experimental, organizational:

- Something the masters and schools can do to keep it real (realistically real);
- Something you can do to measure each of a hundred students;
- Something you could use as a test even if people have an incentive to game it.

Finding good solutions at each level determines what a whole field of study can be useful for—how much it can hope to accomplish. This is one of the Big Important Foundational Questions, so—

Think!

(PS: And ponder on your own before you look at others' ideas; we need breadth of coverage here.)

Why Our Kind Can't Cooperate



discussion
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From when I was still forced to attend, I remember our synagogue's annual fundraising appeal. It was a simple enough format, if I recall correctly. The rabbi and the treasurer talked about the shul's expenses and how vital this annual fundraiser was, and then the synagogue's members called out their pledges from their seats.

Straightforward, yes?

Let me tell you about a different annual fundraising appeal. One that I ran, in fact, during the early years of the Machine Intelligence Research Institute. One difference was that the appeal was conducted over the Internet. And another difference was that the audience was largely drawn from the atheist / libertarian / technophile / science fiction fan / early adopter / programmer / etc. crowd. (To point in the rough direction of an empirical cluster in personspace. If you understood the phrase "empirical cluster in personspace" then you know who I'm talking about.)

I crafted the fundraising appeal with care. By my nature I'm too proud to ask other people for help; but I've gotten over around 60% of that reluctance over the years. The nonprofit needed money and was growing too slowly, so I put some force and poetry into that year's annual appeal. I sent it out to several mailing lists that covered most of our potential support base.

And almost immediately, people started posting to the mailing lists about why they weren't going to donate. Some of them raised basic questions about the nonprofit's philosophy and mission. Others talked about their brilliant ideas for all the *other* sources that the nonprofit could get funding from, instead of them. (They didn't volunteer to contact any of those sources *themselves*, they just had ideas for how *we* could do it.)

Now you might say, "Well, maybe your mission and philosophy *did* have basic problems—you wouldn't want to *cancel* that discussion, would you?"

Hold on to that thought.

Because people *were* donating. We started getting donations right away, via Paypal. We even got congratulatory notes saying how the appeal had finally gotten them to start moving. A donation of \$111.11 was accompanied by a message saying, "I decided to give a little bit more. One more hundred, one more ten, one more single, one more dime, and one more penny. All may not be for one, but this one is trying to be for all."

But none of those donors posted their agreement to the mailing list. Not one.

So far as any of those donors knew, they were alone. And when they tuned in the next day, they discovered not thanks, but arguments for why they *shouldn't* have donated. The criticisms, the justifications for not donating—*only those* were displayed proudly in the open.

As though the treasurer had finished his annual appeal, and everyone *not* making a pledge had proudly stood up to call out justifications for refusing; while those making pledges whispered them quietly, so that no one could hear.

I know someone with a rationalist cause who goes around plaintively asking, "How come the Raëlian flying-saucer cult can get tens of thousands of members [probably around 40,000] interested in complete nonsense, but we can't even get a thousand people working on this?"

The obvious wrong way to finish this thought is to say, "Let's do what the Raëlians do! Let's add some nonsense to this meme!" For the benefit of those not immediately stopped by their ethical inhibitions, I will observe that there may be a hundred failed flying saucer cults for every one that becomes famous. And the Dark Side may require non-obvious skills, which *you*, yes *you*, do not have: Not everyone can be a Sith Lord. In particular, if you talk about your planned lies on the public Internet, you fail. I'm no master criminal, but even I can tell certain people are not cut out to be crooks.

So it's probably not a good idea to cultivate a sense of violated entitlement at the thought that some *other* group, who you think ought to be *inferior* to you, has more money and followers. That path leads to—pardon the expression—the Dark Side.

But it probably *does* make sense to start asking ourselves some pointed questions, if supposed “rationalists” can’t manage to *coordinate* as well as a flying saucer cult.

How do things work on the Dark Side?

The respected leader speaks, and there comes a chorus of pure agreement: if there are any who harbor inward doubts, they keep them to themselves. So all the individual members of the audience see this atmosphere of pure agreement, and they feel more confident in the ideas presented—even if they, personally, harbored inward doubts, why, everyone *else* seems to agree with it.

(“Pluralistic ignorance” is the standard label for this.)

If anyone is still unpersuaded after that, they leave the group (or in some places, are executed)—and the remainder are more in agreement, and reinforce each other with less interference.

(I call that “evaporative cooling of groups.”)

The *ideas* themselves, not just the leader, generate unbounded enthusiasm and praise. The halo effect is that perceptions of all positive qualities correlate—e.g. telling subjects about the benefits of a food preservative made them judge it as lower-risk, even though the quantities were logically uncorrelated. This can create a positive feedback effect that makes an idea seem better and better and better, especially if criticism is perceived as traitorous or sinful.

(Which I term the “affective death spiral.”)

So these are all examples of strong Dark Side forces that can bind groups together.

And presumably we would not go so far as to dirty our hands with such...

Therefore, as a group, the Light Side will always be divided and weak. Technophiles, nerds, scientists, and even non-fundamentalist religions will never be capable of acting with the fanatic unity that animates radical Islam. Technological advantage can only go so far; your tools can be copied or stolen, and used against you. In the end the Light Side will always lose in any group conflict, and the future inevitably belongs to the Dark.

I think that a person's reaction to this prospect says a lot about their attitude towards "rationality."

Some "Clash of Civilizations" writers seem to accept that the Enlightenment is destined to lose out in the long run to radical Islam, and sigh, and shake their heads sadly. I suppose they're trying to signal their cynical sophistication or something.

For myself, I always thought—call me loony—that a *true* rationalist ought to be *effective in the real world*.

So I have a problem with the idea that the Dark Side, thanks to their *pluralistic ignorance and affective death spirals*, will always win because they are *better coordinated* than us.

You would think, perhaps, that *real* rationalists ought to be *more* coordinated? Surely all that unreason must have its *disadvantages*? That mode can't be *optimal*, can it?

And if current "rationalist" groups *cannot* coordinate—if they can't support group projects so well as a single synagogue draws donations from its members—well, I leave it to you to finish that syllogism.

There's a saying I sometimes use: "It is dangerous to be half a rationalist."

For example, I can think of ways to sabotage someone's intelligence by *selectively* teaching them certain methods of rationality. Suppose you taught someone a long list of logical fallacies and cognitive biases, and trained them to spot those fallacies and biases in other people's arguments. But you are careful to pick those fallacies and biases that are *easiest to accuse* others of, the most general ones that can easily be misapplied. And you do *not* warn them to scrutinize *arguments they agree with* just as hard as they scrutinize *incongruent* arguments for flaws. So they have acquired a great repertoire of flaws of which to accuse only arguments and arguers who they don't like. This, I suspect, is one of the primary ways that smart people end up stupid. (And note, by the way, that I have just given you another Fully General Counterargument against smart people whose arguments you don't like.)

Similarly, if you wanted to ensure that a group of "rationalists" never accomplished any task requiring more than one person, you could teach them only techniques of individual

rationality, without mentioning anything about techniques of coordinated group rationality.

I'll write more later on how I think rationalists might be able to coordinate better. But here I want to focus on what you might call *the culture of disagreement*, or even *the culture of objections*, which is one of the two major forces preventing the technophile crowd from coordinating.

Imagine that you're at a conference, and the speaker gives a thirty-minute talk. Afterward, people line up at the microphones for questions. The first questioner objects to the graph used in slide 14 using a logarithmic scale; they quote Tufte on *The Visual Display of Quantitative Information*. The second questioner disputes a claim made in slide 3. The third questioner suggests an alternative hypothesis that seems to explain the same data...

Perfectly normal, right? Now imagine that you're at a conference, and the speaker gives a thirty-minute talk. People line up at the microphone.

The first person says, "I agree with everything you said in your talk, and I think you're brilliant." Then steps aside.

The second person says, "Slide 14 was beautiful, I learned a lot from it. You're awesome." Steps aside.

The third person—

Well, you'll never know what the third person at the microphone had to say, because by this time, you've fled screaming out of the room, propelled by a bone-deep terror as if Cthulhu had erupted from the podium, the fear of the impossibly unnatural phenomenon that has invaded your conference.

Yes, a group that can't tolerate disagreement is not rational. But if you tolerate *only* disagreement—if you tolerate disagreement *but not agreement*—then you also are not rational. You're only willing to hear some honest thoughts, but not others. You are a dangerous half-a-rationalist.

We are as uncomfortable *together* as flying-saucer cult members are uncomfortable *apart*. That can't be right either. Reversed stupidity is not intelligence.

Let's say we have two groups of soldiers. In group 1, the privates are ignorant of tactics and strategy; only the sergeants

know anything about tactics and only the officers know anything about strategy. In group 2, everyone at all levels knows all about tactics and strategy.

Should we expect group 1 to defeat group 2, because group 1 will follow orders, while everyone in group 2 comes up with *better ideas* than whatever orders they were given?

In this case I have to question how much group 2 really understands about military theory, because it is an *elementary* proposition that an uncoordinated mob gets slaughtered.

Doing worse with *more knowledge* means you are doing something very wrong. You should always be able to *at least* implement the same strategy you would use if you are ignorant, and preferably do *better*. You definitely should not do *worse*. If you find yourself regretting your “rationality” then you should reconsider what is rational.

On the other hand, if you are only half-a-rationalist, you can *easily* do worse with more knowledge. I recall a lovely experiment which showed that politically opinionated students with more knowledge of the issues reacted less to incongruent evidence, because they had more ammunition with which to counter-argue only incongruent evidence.

We would seem to be stuck in an awful valley of partial rationality where we end up more poorly coordinated than religious fundamentalists, able to put forth less effort than flying-saucer cultists. True, what little effort we *do* manage to put forth may be better-targeted at helping people rather than the reverse—but that is not an acceptable excuse.

If I were setting forth to systematically train rationalists, there would be lessons on how to disagree and lessons on how to agree, lessons intended to make the trainee more comfortable with dissent, and lessons intended to make them more comfortable with conformity. One day everyone shows up dressed differently, another day they all show up in uniform. You’ve got to cover both sides, or you’re only half a rationalist.

Can you imagine training prospective rationalists to wear a uniform and march in lockstep, and practice sessions where they agree with each other and applaud everything a speaker on a podium says? It sounds like unspeakable horror, doesn’t

it, like the whole thing has admitted outright to being an evil cult? But why is it *not* okay to practice that, while it *is* okay to practice *disagreeing* with everyone else in the crowd? Are you *never* going to have to agree with the majority?

Our culture puts all the emphasis on heroic disagreement and heroic defiance, and none on heroic agreement or heroic group consensus. We signal our superior intelligence and our *membership in the nonconformist community* by inventing clever objections to others' arguments. Perhaps *that* is why the technophile / Silicon Valley crowd stays marginalized, losing battles with less nonconformist factions in larger society. No, we're not losing because we're so superior, we're losing because our exclusively individualist traditions sabotage our ability to co-operate.

The other major component that I think sabotages group efforts in the technophile community is *being ashamed of strong feelings*. We still have the Spock archetype of rationality stuck in our heads, rationality as dispassion. Or perhaps a related mistake, rationality as cynicism—trying to signal your superior world-weary sophistication by showing that you care less than others. Being careful to ostentatiously, publicly look down on those so naive as to show they care strongly about anything.

Wouldn't it make you feel uncomfortable if the speaker at the podium said that they cared so strongly about, say, fighting aging, that they would willingly die for the cause?

But it is nowhere written in either probability theory or decision theory that a rationalist should not care. I've looked over those equations and, really, it's not in there.

The best informal definition I've ever heard of rationality is "That which can be destroyed by the truth should be." We should aspire to feel the emotions that fit the facts, not aspire to feel no emotion. If an emotion can be destroyed by truth, we should relinquish it. But if a cause is worth striving for, then let us by all means feel fully its importance.

Some things *are* worth dying for. Yes, really! And if we can't get comfortable with admitting it and hearing others say it, then we're going to have trouble *caring* enough—as well as *coordinating* enough—to put some effort into group projects.

You've got to teach both sides of it, "That which can be destroyed by the truth should be," and "That which the truth nourishes should thrive."

I've heard it argued that the taboo against emotional language in, say, science papers, is an important part of letting the facts fight it out without distraction. That doesn't mean the taboo should apply everywhere. I think that there are parts of life where we should learn to *applaud* strong emotional language, eloquence, and poetry. When there's something that needs doing, poetic appeals help get it done, and, therefore, are themselves to be applauded.

We need to keep our efforts to expose *counterproductive* causes and *unjustified* appeals from stomping on tasks that genuinely need doing. You need both sides of it—the willingness to turn away from counterproductive causes, and the willingness to praise productive ones; the strength to be unswayed by ungrounded appeals, and the strength to be swayed by grounded ones.

I think the synagogue at their annual appeal had it right, really. They weren't going down row by row and putting individuals on the spot, staring at them and saying, "How much will *you* donate, Mr. Schwartz?" People simply announced their pledges—not with grand drama and pride, just simple announcements—and that encouraged others to do the same. Those who had nothing to give, stayed silent; those who had objections, chose some later or earlier time to voice them. That's probably about the way things *should* be in a sane human community—taking into account that people often have trouble getting as motivated as they wish they were, and can be helped by social encouragement to overcome this weakness of will.

But even if you disagree with that part, then let us say that both supporting and countersupporting opinions should have been publicly voiced. Supporters being faced by an apparently solid wall of objections and disagreements—even if it resulted from their own uncomfortable self-censorship—is *not* group rationality. It is the mere *mirror image* of what Dark Side groups do to keep their followers. Reversed stupidity is not intelligence.

Tolerate Tolerance



discussion
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6-c28

One of the likely characteristics of someone who sets out to be a “rationalist” is a lower-than-usual tolerance for flaws in reasoning. This doesn’t strictly follow. You could end up, say, rejecting your religion, just because you spotted *more* or *deeper* flaws in the reasoning, not because you were, by your nature, *more annoyed* at a flaw of fixed size. But realistically speaking, a lot of us probably have our level of “annoyance at all these flaws we’re spotting” set a bit higher than average.

That’s why it’s so important for us to tolerate others’ tolerance if we want to get anything done together.

For me, the poster case of tolerance I need to tolerate is Ben Goertzel, who among other things runs an annual AI conference, and who has something nice to say about *everyone*. Ben even complimented the ideas of M*nt*f*x, the most legendary of all AI crackpots. (M*nt*f*x apparently started adding a link to Ben’s compliment in his email signatures, presumably because it was the only compliment he’d ever gotten from a bona fide AI academic.) (Please do *not* pronounce his True Name correctly or he will be summoned here.)

But I’ve come to understand that this is one of Ben’s strengths—that he’s nice to lots of people that others might ignore, including, say, me—and every now and then this pays off for him.

And if I subtract points off Ben’s reputation for finding something nice to say about people and projects that I think are hopeless—even M*nt*f*x—then what I’m doing is insisting that Ben *dislike everyone I dislike* before I can work with him.

Is that a realistic standard? Especially if different people are annoyed in different amounts by different things?

But it’s hard to remember that when Ben is being nice to so *many* idiots.

Cooperation is unstable, in both game theory and evolutionary biology, without *some* kind of punishment for defection. So

it's one thing to subtract points off someone's reputation for mistakes they make *themselves, directly*. But if you also look askance at someone for *refusing to castigate* a person or idea, then that is *punishment of non-punishers*, a far more dangerous idiom that can lock an equilibrium in place even if it's harmful to *everyone* involved.

The danger of punishing non-punishers is something I remind myself of, say, every time Robin Hanson points out a flaw in some academic trope and yet modestly confesses he could be wrong (and he's not wrong). Or every time I see Michael Vassar still considering the potential of someone who I wrote off as hopeless within thirty seconds of being introduced to them. I have to remind myself, "Tolerate tolerance! Don't demand that your allies be *equally extreme* in their negative judgments of everything you dislike!"

By my nature, I *do* get annoyed when someone else seems to be giving too much credit. I don't know if everyone's like that, but I suspect that at least *some* of my fellow aspiring rationalists are. I wouldn't be surprised to find it a human universal; it does have an obvious evolutionary rationale—one which would make it a very *unpleasant* and *dangerous* adaptation.

I am not generally a fan of "tolerance." I certainly don't believe in being "intolerant of intolerance," as some inconsistently hold. But I shall go on trying to tolerate *people who are more tolerant than I am*, and judge them only for their *own* un-borrowed mistakes.

Oh, and it goes without saying that if the people of Group X are staring at you demandingly, waiting for you to hate the right enemies with the right intensity, and ready to castigate you if you fail to castigate loudly enough, you may be hanging around the wrong group.

Just don't demand that *everyone* you work with be equally intolerant of behavior like that. Forgive your friends if some of them suggest that maybe Group X wasn't so awful after all ...

Your Price for Joining



discussion
sharov.me/r/
6-c29

In the Ultimatum Game, the first player chooses how to split \$10 between themselves and the second player, and the second player decides whether to accept the split or reject it—in the latter case, both parties get nothing. So far as conventional causal decision theory goes (two-box on Newcomb’s Problem, defect in Prisoner’s Dilemma), the second player should prefer any non-zero amount to nothing. But if the first player *expects* this behavior—accept any non-zero offer—then they have no motive to offer more than a penny. As I assume you all know by now, I am no fan of conventional causal decision theory. Those of us who remain interested in cooperating on the Prisoner’s Dilemma, either because it’s iterated, or because we have a term in our utility function for fairness, or because we use an unconventional decision theory, may also not accept an offer of one penny.

And in fact, most Ultimatum “deciders” offer an even split; and most Ultimatum “accepters” reject any offer less than 20%. A 100 USD game played in Indonesia (average per capita income at the time: 670 USD) showed offers of 30 USD being turned down, although this equates to two week’s wages. We can probably also assume that the players in Indonesia were not thinking about the academic debate over Newcomblike problems—this is just the way people feel about Ultimatum Games, even ones played for real money.

There’s an analogue of the Ultimatum Game in group coordination. (Has it been studied? I’d hope so...) Let’s say there’s a common project—in fact, let’s say that it’s an altruistic common project, aimed at helping mugging victims in Canada, *or something*. If you join this group project, you’ll get more done than you could on your own, relative to your utility function. So, obviously, you should join.

But wait! The anti-mugging project keeps their funds invested in a money market fund! That’s ridiculous; it won’t earn

even as much interest as US Treasuries, let alone a dividend-paying index fund.

Clearly, this project is run by morons, and you shouldn't join until they change their malinvesting ways.

Now you might realize—if you stopped to think about it—that all things considered, you would *still* do better by working with the common anti-mugging project, than striking out on your own to fight crime. But then—you might perhaps also realize—if you *too easily assent* to joining the group, why, what *motive* would they have to change their malinvesting ways?

Well ...Okay, look. Possibly because we're out of the ancestral environment where everyone knows everyone else ...and possibly because the nonconformist crowd tries to repudiate *normal* group-cohering forces like conformity and leader-worship ...

...It seems to me that people in the atheist / libertarian / technophile / science fiction fan / etc. cluster often set their joining prices *way way way* too high. Like a 50-way split Ultimatum game, where every one of 50 players demands at least 20% of the money.

If you think how often situations like this would have arisen in the ancestral environment, then it's almost certainly a matter of evolutionary psychology. System 1 emotions, not System 2 calculation. Our intuitions for when to join groups, versus when to hold out for more concessions to our own preferred way of doing things, would have been honed for hunter-gatherer environments of, e.g., 40 people, all of whom you knew personally.

And if the group is made up of 1,000 people? Then your hunter-gatherer instincts will underestimate the inertia of a group so large, and demand an unrealistically high price (in strategic shifts) for you to join. There's a limited amount of organizational effort, and a limited number of degrees of freedom, that can go into doing things any one person's way.

And if the strategy is large and complex, the sort of thing that takes e.g. ten people doing paperwork for a week, rather than being hammered out over a half-hour of negotiation around a campfire? Then your hunter-gatherer instincts will underestimate the inertia of the group, relative to your own demands.

And if you live in a wider world than a single hunter-gatherer tribe, so that you only see the one group representative who negotiates with you, and not the hundred other negotiations that have taken place already? Then your instincts will tell you that it is just one person, a stranger at that, and the two of you are equals; whatever ideas they bring to the table are equal with whatever ideas you bring to the table, and the meeting point ought to be about even.

And if you suffer from any weakness of will or *akrasia*, or if you are influenced by motives other than those you would admit to yourself that you are influenced by, then any group-altruistic project that does not offer you the rewards of status and control may perhaps find itself underserved by your attentions.

Now I do admit that I speak here primarily from the perspective of someone who goes around trying to herd cats; and not from the other side as someone who spends most of their time withholding their energies in order to blackmail those damned morons already on the project. Perhaps I am a little prejudiced.

But it seems to me that a reasonable rule of thumb might be as follows:

If, on the whole, joining your efforts to a group project *would still have a net positive effect* according to your utility function—

(or a larger positive effect than any other marginal use to which you could otherwise put those resources, although this latter mode of thinking seems little-used and humanly-unrealistic, for reasons I may write about some other time)

—and the awful horrible annoying issue is not so important that *you personally* will get involved deeply enough to put in however many hours, weeks, or years may be required to get it fixed up—

—then the issue is not worth you withholding your energies from the project; either instinctively until you see that people are paying attention to you and respecting you, or by conscious intent to blackmail the group into getting it done.

And if the issue *is* worth that much to you ...then by all means, join the group and do whatever it takes to get things fixed up.

Now, if the existing contributors refuse to let you do this, *and* a reasonable third party would be expected to conclude that you were competent enough to do it, *and* there is no one else whose ox is being gored thereby, *then*, perhaps, we have a problem on our hands. And it may be time for a little blackmail, if the resources you can conditionally commit are large enough to get their attention.

Is this rule a little extreme? Oh, maybe. There *should* be a motive for the decision-making mechanism of a project to be responsible to its supporters; unconditional support would create its own problems.

But *usually*...I observe that people underestimate the costs of what they ask for, or perhaps just act on instinct, and set their prices *way way way* too high. If the nonconformist crowd ever wants to get anything done together, we need to move in the direction of joining groups and staying there at least a *little* more easily. Even in the face of annoyances and imperfections! Even in the face of unresponsiveness to our own better ideas!

In the age of the Internet and in the company of nonconformists, it does get a little tiring reading the 451st public email from someone saying that the Common Project isn't worth their resources until the website has a sans-serif font.

Of course this often isn't really about fonts. It may be about laziness, akrasia, or hidden rejections. But in terms of group norms...in terms of what sort of public statements we respect, and which excuses we publicly scorn...we probably *do* want to encourage a group norm of:

If the issue isn't worth your personally fixing by however much effort it takes, and it doesn't arise from outright bad faith, it's not worth refusing to contribute your efforts to a cause you deem worthwhile.

Can Humanism Match Religion's Output?



discussion
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6-c30

Perhaps the single largest *voluntary* institution of our modern world—bound together not by police and taxation, not by salaries and managers, but by voluntary donations flowing from its members—is the Catholic Church.

It's too large to be held together by individual negotiations, like a group task in a hunter-gatherer band. But in a larger world with more people to be infected and faster transmission, we can expect more virulent memes. The Old Testament doesn't talk about Hell, but the New Testament does. The Catholic Church is held together by affective death spirals—around the ideas, the institutions, and the leaders. By promises of eternal happiness and eternal damnation—theologians don't really believe that stuff, but many ordinary Catholics do. By simple conformity of people meeting in person at a Church and being subjected to peer pressure. Et cetera.

We who have the temerity to call ourselves “rationalists” think ourselves too good for such communal bindings.

And so anyone with a *simple* and *obvious* charitable project—responding with food and shelter to a tidal wave in Thailand, say—would be better off by *far* pleading with the Pope to mobilize the Catholics, rather than with Richard Dawkins to mobilize the atheists.

For so long as this is true, any increase in atheism at the expense of Catholicism will be something of a hollow victory, regardless of all other benefits.

True, the Catholic Church also goes around opposing the use of condoms in AIDS-ravaged Africa. True, they waste huge amounts of the money they raise on all that religious stuff. Indulging in unclear thinking is not harmless; prayer comes with a price.

To refrain from doing damaging things is a true victory for a rationalist ...

Unless it is your *only* victory, in which case it seems a little empty.

If you *discount all harm* done by the Catholic Church, and look *only* at the good ...then does the average Catholic do more gross good than the average atheist, just by virtue of being more active?

Perhaps if you are wiser but less motivated, you can search out interventions of high efficiency and purchase utilons on the cheap ...But there are few of us who *really* do that, as opposed to planning to do it someday.

Now you might at this point throw up your hands, saying: "For so long as we don't have direct control over our brain's motivational circuitry, it's not realistic to expect a rationalist to be as strongly motivated as someone who genuinely believes that they'll burn eternally in hell if they don't obey."

This is a fair point. Any folk theorem to the effect that a rational agent should do at least as well as a non-rational agent will rely on the assumption that the rational agent can always just implement whatever "irrational" policy is observed to win. But if you can't *choose* to have unlimited mental energy, then it may be that some false beliefs are, in cold fact, more strongly motivating than any available true beliefs. And if we all generally suffer from altruistic akrasia, being unable to bring ourselves to help as much as we think we should, then it is possible for the God-fearing to win the contest of altruistic output.

But though it is a motivated continuation, let us consider this question a little further.

Even the fear of hell is not a perfect motivator. Human beings are not given so much slack on evolution's leash; we can resist motivation for a short time, but then we run out of mental energy (hat tip: infotropism¹). Even believing that you'll go to

1. [sharov.me/r/6-8](https://www.reddit.com/user/sharovme/)

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hell does not change this brute fact about brain circuitry. So the religious sin, and then are tormented by thoughts of going to hell, in much the same way that smokers reproach themselves for being unable to quit.

If a group of rationalists cared *a lot* about something ...who says they wouldn't be able to match the real, de-facto output of a believing Catholic? The stakes might not be "infinite" happiness or "eternal" damnation, but of course the brain can't visualize $3 \uparrow \uparrow \uparrow 3$, let alone infinity. Who says that the actual quantity of caring neurotransmitters discharged by the brain (as 'twere) has to be so much less for "the growth and flowering of humankind" or even "tidal-wave-stricken Thais," than for "eternal happiness in Heaven"? Anything involving more than 100 people is going to involve utilities too large to visualize. And there are all sorts of other standard biases at work here; knowing about them might be good for a bonus as well, one hopes?

Cognitive-behavioral therapy and Zen meditation are two mental disciplines experimentally shown to yield real improvements. It is not the area of the art I've focused on developing, but then I don't have a real martial art of rationality in back of me. If you combine a purpose genuinely worth caring about with discipline extracted from CBT and Zen meditation, then who says rationalists can't keep up? Or even more generally: if we have an evidence-based art of fighting akrasia, with experiments to see what actually works, then who says we've got to be less motivated than some disorganized mind that fears God's wrath?

Still ...that's a further-future speculation that it might be possible to develop an art that doesn't presently exist. It's not a technique I can use right now. I present it just to illustrate the idea of *not giving up so fast on rationality*: Understanding what's going wrong, trying intelligently to fix it, and gathering evidence on whether it worked—this is a powerful idiom, not to be lightly dismissed upon sighting the first disadvantage.

Really, I suspect that what's going on here has less to do with the motivating power of eternal damnation, and a lot more to do with the motivating power of *physically meeting* other people

who share your cause. The power, in other words, of being physically present at church and having religious neighbors.

This is a problem for the rationalist community in its present stage of growth, because we are rare and geographically distributed way the hell all over the place. If all the readers of *Less Wrong* lived within a five-mile radius of each other, I bet we'd get a lot more done, not for reasons of *coordination* but just sheer *motivation*.

I'll write later about some long-term, starry-eyed, idealistic thoughts on this particular problem. Shorter-term solutions that don't rely on our increasing our numbers by a factor of 100 would be better. I wonder in particular whether the best modern videoconferencing software would provide some of the motivating effect of meeting someone in person; I suspect the answer is "no" but it might be worth trying.

Meanwhile ...in the short term, we're stuck fighting akrasia mostly without the reinforcing physical presense of other people who care. I want to say something like "This is difficult, but it *can* be done," except I'm not sure that's even true.

I suspect that the *largest* step rationalists could take toward matching the per-capita power output of the Catholic Church would be to have regular physical meetings of people contributing to the same task—just for purposes of motivation.

In the absence of that ...

We could try for a group norm of being openly allowed—nay, applauded—for caring strongly about something. And a group norm of being expected to do something useful with your life—contribute your part to cleaning up this world. Religion doesn't really emphasize the getting-things-*done* aspect as much.

And if rationalists could match just *half* the average altruistic effort output per Catholic, then I don't think it's *remotely* unrealistic to suppose that with better targeting on more efficient causes, the modal rationalist could get twice as much done.

How much of its earnings does the Catholic Church spend on all that useless religious stuff instead of actually helping people? More than 50%, I would venture. So then we could say—with a certain irony, though that's not quite the spirit in which

we should be doing things—that we should try to propagate a group norm of donating a minimum of 5% of income to *real* causes. (10% being the usual suggested minimum religious tithe.) And then there's the art of picking causes for which expected utilons are orders of magnitude cheaper (for so long as the inefficient market in utilons lasts).

But long before we can begin to dream of any such boast, we secular humanists need to work on at least *matching* the per capita benevolent output of the worshippers.

Church vs. Taskforce

I am generally suspicious of envying crazy groups or trying to blindly copycat the rhythm of religion—what I called “hymns to the nonexistence of God,” replying, “A good ‘atheistic hymn’ is simply a song about anything worth singing about that doesn’t happen to be religious.”

But religion does fill certain holes in people’s minds, some of which are even worth filling. If you eliminate religion, you have to be aware of what gaps are left behind.

If you suddenly deleted religion from the world, the largest gap left would not be anything of ideals or morals; it would be the church, the community. Among those who now stay religious without quite really believing in God—how many are just sticking to it from wanting to stay with their neighbors at the church, and their family and friends? How many would convert to atheism, if all those others deconverted, and *that* were the price of staying in the community and keeping its respect? I would guess ...probably quite a lot.

In truth ...this is probably something I don’t understand all that well, myself. “Brownies and babysitting” were the first two things that came to mind. Do churches lend helping hands in emergencies? Or just a shoulder to cry on? How strong is a church community? It probably depends on the church, and in any case, that’s not the correct question. One should start by considering what a hunter-gatherer band gives its people, and ask what’s missing in modern life—if a modern First World church fills only *some* of that, then by all means let us try to do *better*.

So *without* copycatting religion—*without* assuming that we *must* gather every Sunday morning in a building with stained-glass windows while the children dress up in formal clothes and listen to someone sing—let’s consider how to fill the emotional gap, after religion stops being an option.



discussion
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To help break the mold to start with—the straitjacket of cached thoughts on how to do this sort of thing—consider that *some* modern offices may also fill the same role as a church. By which I mean that some people are fortunate to receive community from their workplaces: friendly coworkers who bake brownies for the office, whose teenagers can be safely hired for babysitting, and maybe even help in times of catastrophe ...? But certainly not everyone is lucky enough to find a community at the office.

Consider further—a church is *ostensibly* about worship, and a workplace is *ostensibly* about the commercial purpose of the organization. Neither has been carefully *optimized* to serve as a community.

Looking at a typical religious church, for example, you could suspect—although all of these things would be better tested experimentally, than just suspected—

- That getting up early on a Sunday morning is not optimal;
- That wearing formal clothes is not optimal, especially for children;
- That listening to the same person give sermons on the same theme every week (“religion”) is not optimal;
- That the cost of supporting a church and a pastor is expensive, compared to the number of different communities who could time-share the same building for their gatherings;
- That they probably don’t serve nearly enough of a match-making purpose, because churches think they’re supposed to enforce their medieval moralities;
- That the whole thing ought to be subject to experimental data-gathering to find out what works and what doesn’t.

By using the word “optimal” above, I mean “optimal under the criteria you would use if you were explicitly building a community *qua* community.” Spending lots of money on a

fancy church with stained-glass windows and a full-time pastor makes sense if you actually *want* to spend money on religion *qua* religion.

I do confess that when walking past the churches of my city, my main thought is, “These buildings look really, really expensive, and there are too many of them.” If you were doing it over from scratch...then you might have a big building that could be used for the occasional wedding, but it would be time-shared for different communities meeting at different times on the weekend, and it would also have a nice large video display that could be used for speakers giving presentations, lecturers teaching something, or maybe even showing movies. Stained glass? Not so high a priority.

Or to the extent that the church membership lends a helping hand in times of trouble—could that be improved by an explicit rainy-day fund or contracting with an insurer, once you realized that this was an important function? Possibly *not*; dragging explicit finance into things changes their character oddly. Conversely, maybe keeping current on some insurance policies should be a *requirement* for membership, lest you rely *too much* on the community...But again, to the extent that churches provide community, they’re trying to do it without actually *admitting* that this is nearly all of what people get out of it. Same thing with the corporations whose workplaces are friendly enough to serve as communities; it’s still something of an accidental function.

Once you start thinking *explicitly* about how to give people a hunter-gatherer band to belong to, you can see all sorts of things that sound like good ideas. Should you welcome the newcomer in your midst? The pastor may give a sermon on that sometime, if you think church is about religion. But if you’re explicitly setting out to build community—then right after a move is when someone most lacks community, when they most need your help. It’s also an opportunity for the band to grow. If anything, tribes ought to be competing at quarterly exhibitions to capture newcomers.

But can you really have a community that's *just* a community—that isn't also an office or a religion? A community with no purpose beyond itself?

Maybe you *can*. After all, did hunter-gatherer tribes have any purposes beyond themselves?—well, there was survival and feeding yourselves, that was a purpose.

But anything that people have in common, especially any *goal* they have in common, tends to *want* to define a community. Why not take advantage of that?

Though in this age of the Internet, alas, too many binding factors have supporters too widely distributed to form a decent band—if you're the only member of the Church of the Subgenius in your city, it may not really help much. It really is different without the physical presence; the Internet does *not* seem to be an acceptable substitute at the current stage of the technology.

So to skip right to the point—

Should the Earth last so long, I would like to see, as the form of rationalist communities, taskforces focused on all the work that needs doing to fix up this world. Communities in any geographic area would form around the most specific cluster that could support a decent-sized band. If your city doesn't have enough people in it for you to find 50 fellow Linux programmers, you might have to settle for 15 fellow open-source programmers...or in the days when all of this is only getting started, 15 fellow rationalists trying to spruce up the Earth in their assorted ways.

That's what I think would be a fitting direction for the energies of communities, and a common purpose that would bind them together. Tasks like that need communities anyway, and this Earth has plenty of work that needs doing, so there's no point in waste. We have so much that needs doing—let the energy that was once wasted into the void of religious institutions, find an outlet there. And let purposes admirable without need for delusion fill any void in the community structure left by deleting religion and its illusionary higher purposes.

Strong communities built around worthwhile purposes: That would be the shape I would like to see for the post-religious

age, or whatever fraction of humanity has then gotten so far in their lives.

Although ...as long as you've got a building with a nice large high-resolution screen anyway, I wouldn't mind challenging the idea that all post-adulthood learning has to take place in distant expensive university campuses with teachers who would rather be doing something else. And it's empirically the case that colleges seem to support communities quite well. So in all fairness, there are other possibilities for things you could build a post-theistic community around.

Is all of this just a dream? Maybe. Probably. It's not completely devoid of incremental implementability, if you've got enough rationalists in a sufficiently large city who have heard of the idea. But on the off chance that rationality should catch on so widely, or the Earth should last so long, and that my voice should be heard, then that is the direction I would like to see things moving in—as the churches fade, we don't need artificial churches, but we do need new idioms of community.

Rationality: Common Interest of Many Causes



discussion
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It is a not-so-hidden agenda of *Less Wrong* that there are many causes that benefit from the spread of rationality—because it takes a little more rationality than usual to see their case as a supporter, or even just as a supportive bystander. Not just the obvious causes like atheism, but things like marijuana legalization—where you could wish that people were a bit more self-aware about their motives and the nature of signaling, and a bit more moved by inconvenient cold facts. The Machine Intelligence Research Institute was merely an unusually extreme case of this, wherein it got to the point that after years of bogging down I threw up my hands and explicitly recursed on the job of creating rationalists.

But of course, not *all* the rationalists I create will be interested in my *own* project—and *that's fine*. You can't capture *all* the value you create, and trying can have poor side effects.

If the supporters of other causes are enlightened enough to think similarly ...

Then all the causes that benefit from spreading rationality can, perhaps, have something in the way of standardized material to which to point their supporters—a common task, centralized to save effort—and think of themselves as spreading a little rationality on the side. They won't capture all the value they create. And that's fine. They'll capture some of the value others create. Atheism has very little to do directly with marijuana legalization, but if both atheists and anti-Prohibitionists are willing to step back a bit and say a bit about the general, abstract principle of confronting a discomforting truth that interferes with a fine righteous tirade, then both atheism and marijuana legalization pick up some of the benefit from both efforts.

But this requires—I know I’m repeating myself here, but it’s important—that you be willing not to capture all the value you create. It requires that, in the course of talking about rationality, you maintain an ability to temporarily *shut up* about your own cause even though it is the best cause ever. It requires that you don’t regard those other causes, and they do not regard you, as competing for a limited supply of rationalists with a limited capacity for support; but, rather, creating more rationalists and increasing their capacity for support. You only reap some of your own efforts, but you reap some of others’ efforts as well.

If you and they don’t agree on everything—especially priorities—you have to be willing to agree to *shut up* about the disagreement. (Except possibly in specialized venues, out of the way of the mainstream discourse, where such disagreements are explicitly prosecuted.)

A certain person who was taking over as the president of a certain organization once pointed out that the organization had not enjoyed much luck with its message of “This is *the best* thing you can do,” as compared to e.g. the X-Prize Foundation’s tremendous success conveying to rich individuals “Here is *a cool* thing you can do.”

This is one of those insights where you blink incredulously and then grasp how much sense it makes. The human brain can’t grasp large stakes, and people are not anything remotely like expected utility maximizers, and we are generally altruistic akrasics. Saying, “This is the *best* thing” doesn’t add much motivation beyond “This is a cool thing.” It just establishes a much higher burden of proof. And invites invidious motivation-sapping comparison to all other good things you know (perhaps threatening to diminish moral satisfaction already purchased).

If we’re operating under the assumption that everyone by default is an altruistic akrasic (someone who wishes they could choose to do more)—or at least, that most potential supporters of interest fit this description—then fighting it out over which cause is the *best* to support may have the effect of decreasing the overall supply of altruism.

“But,” you say, “dollars are fungible; a dollar you use for one thing indeed cannot be used for anything else!” To which I

reply: But human beings *really aren't* expected utility maximizers, as cognitive systems. Dollars come out of different mental accounts, cost different amounts of *willpower* (the true limiting resource) under different circumstances. People want to spread their donations around as an act of mental accounting to minimize the regret if a single cause fails, and telling someone about an additional cause may increase the total amount they're willing to help.

There are, of course, limits to this principle of benign tolerance. If someone's pet project is to teach salsa dance, it would be quite a stretch to say they're working on a worthy sub-task of the great common Neo-Enlightenment project of human progress.

But to the extent that something really is a task you would wish to see done on behalf of humanity ...then invidious comparisons of that project to Your-Favorite-Project may not help your own project as much as you might think. We may need to learn to say, by habit and in nearly all forums, "Here is *a cool* rationalist project," not, "Mine alone is *the highest-return in expected utilons per marginal dollar* project." If someone cold-blooded enough to maximize expected utility of fungible money without regard to emotional side effects *explicitly asks*, we could perhaps steer them to a *specialized* subforum where anyone willing to make the claim of *top* priority fights it out. Though if all goes well, those projects that have a strong claim to this kind of underserved-ness will get more investment and their marginal returns will go down, and the winner of the competing claims will no longer be clear.

If there are many rationalist projects that benefit from raising the sanity waterline, then their mutual tolerance and common investment in spreading rationality could conceivably exhibit a commons problem. But this doesn't seem too hard to deal with: if there's a group that's not willing to share the rationalists they create or mention to them that other Neo-Enlightenment projects might exist, then any common, centralized rationalist resources could remove the mention of their project as a cool thing to do.

Though all this is an idealistic and future-facing thought, the benefits—for all of us—could be finding some important things we’re missing right now. So many rationalist projects have few supporters and far-flung; if we could all identify as elements of the Common Project of human progress, the Neo-Enlightenment, there would be a substantially higher probability of finding ten of us in any given city. Right now, a lot of these projects are just a little lonely for their supporters. Rationality may not be *the most important thing in the world*—that, of course, is the thing that we protect—but it is *a cool* thing that more of us have in common. We might gain much from identifying ourselves also as rationalists.

Helpless Individuals



discussion
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6-c33

When you consider that our grouping instincts are optimized for 50-person hunter-gatherer bands where everyone knows everyone else, it begins to seem miraculous that modern-day large institutions survive at all.

Well—there are governments with specialized militaries and police, which can extract taxes. That’s a non-ancestral idiom which dates back to the invention of sedentary agriculture and extractible surpluses; humanity is still struggling to deal with it.

There are corporations in which the flow of money is controlled by centralized management, a non-ancestral idiom dating back to the invention of large-scale trade and professional specialization.

And in a world with large populations and close contact, memes evolve far more virulent than the average case of the ancestral environment; memes that wield threats of damnation, promises of heaven, and professional priest classes to transmit them.

But by and large, the answer to the question “How do large institutions survive?” is “They don’t!” The vast majority of large modern-day institutions—some of them extremely vital to the functioning of our complex civilization—simply *fail to exist in the first place*.

I first realized this as a result of grasping how Science gets funded: namely, *not* by individual donations.

Science traditionally gets funded by governments, corporations, and large foundations. I’ve had the opportunity to discover firsthand that it’s *amazingly* difficult to raise money for Science from individuals. Not unless it’s science about a disease with gruesome victims, and maybe not even then.

Why? People are, in fact, prosocial; they give money to, say, puppy pounds. Science is one of the great social interests, and people are even widely aware of this—why not Science, then?

Any *particular* science project—say, studying the genetics of trypanotolerance in cattle—is not a good *emotional fit* for individual charity. Science has a long time horizon that requires continual support. The interim or even final press releases may not sound all that emotionally arousing. You can't volunteer; it's a job for specialists. Being shown a picture of the scientist you're supporting at or somewhat below the market price for their salary lacks the impact of being shown the wide-eyed puppy that you helped usher to a new home. You don't get the immediate feedback and the sense of immediate accomplishment that's required to keep an individual *spending their own money*.

Ironically, I finally realized this, not from my own work, but from thinking "Why don't Seth Roberts's readers come together to support experimental tests of Roberts's hypothesis about obesity? Why aren't individual philanthropists paying to test Bussard's polywell fusor?" These are examples of *obviously* ridiculously underfunded science, with applications (if true) that would be relevant to many, many individuals. That was when it occurred to me that, in full generality, Science is not a good emotional fit for people spending their own money.

In fact *very few things are*, with the individuals we have now. It seems to me that this is key to understanding how the world works the way it does—why so many individual interests are poorly protected—why 200 million adult Americans have such tremendous trouble supervising the 535 members of Congress, for example.

So how does Science actually get funded? By governments that think they ought to spend some amount of money on Science, with legislatures or executives deciding to do so—it's not quite their *own* money they're spending. Sufficiently large corporations decide to throw some amount of money at blue-sky R&D. Large grassroots organizations built around affective death spirals may look at science that suits their ideals. Large private foundations, based on money block-allocated by wealthy individuals to their reputations, spend money on Science that promises to sound very charitable, sort of like allocating money to orchestras or modern art. And then the

individual scientists (or individual scientific task-forces) fight it out for control of that pre-allocated money supply, given into the hands of grant committee members who seem like the sort of people who ought to be judging scientists.

You rarely see a scientific project making a *direct* bid for some portion of society's resource flow; rather, it first gets allocated to Science, and then scientists fight over who actually gets it. Even the exceptions to this rule are more likely to be driven by politicians (moonshot) or military purposes (Manhattan project) than by the appeal of scientists to the public.

Now I'm sure that if the general public were in the habit of funding particular science by individual donations, a whole lotta money would be wasted on e.g. quantum gibberish—assuming that the general public somehow acquired the habit of funding science without changing any other facts about the people or the society.

But it's still an interesting point that Science manages to survive not because it is in our collective individual interest to see Science get done, but rather, because Science has fastened itself as a parasite onto the few forms of large organization that can exist in our world. There are plenty of other projects that simply fail to exist in the first place.

It seems to me that modern humanity manages to put forth very little in the way of coordinated effort to serve collective individual interests. It's just too non-ancestral a problem when you scale to more than 50 people. There are only big taxers, big traders, supermemes, occasional individuals of great power; and a few other organizations, like Science, that can fasten parasitically onto them.

Money: The Unit of Caring

Steve Omohundro has suggested a folk theorem¹ to the effect that, within the interior of any approximately rational self-modifying agent, the marginal benefit of investing additional resources in anything ought to be about equal. Or, to put it a bit more exactly, shifting a unit of resource between any two tasks should produce no increase in expected utility, relative to the agent's utility function and its probabilistic expectations about its own algorithms.

This resource balance principle implies that—over a very wide range of approximately rational systems, including even the interior of a self-modifying mind—there will exist some common currency of expected utilons, by which everything worth doing can be measured.

In our society, this common currency of expected utilons is called “money.” It is the measure of how much society cares about something.

This is a brutal yet obvious point, which many are motivated to deny.

With this audience, I hope, I can simply state it and move on. It's not as if you thought “society” was intelligent, benevolent, and sane up until this point, right?

I say this to make a certain point held in common across many good causes. Any charitable institution you've ever had a kind word for, certainly *wishes* you would appreciate this point, whether or not they've ever said anything out loud. For I have listened to others in the nonprofit world, and I know that I am not speaking only for myself here ...



discussion
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1. sharov.me/r/6-9

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Many people, when they see something that they think is worth doing, would like to volunteer a few hours of spare time, or maybe mail in a five-year-old laptop and some canned goods, or walk in a march somewhere, but at any rate, not spend *money*.

Believe me, I understand the feeling. Every time I spend money I feel like I'm losing hit points. That's the problem with having a unified quantity describing your net worth: Seeing that number go down is not a pleasant feeling, even though it has to fluctuate in the ordinary course of your existence. There ought to be a fun-theoretic principle against it.

But, well...

There is this very, very old puzzle/observation in economics about the lawyer who spends an hour volunteering at the soup kitchen, instead of working an extra hour and donating the money to hire someone to work for five hours at the soup kitchen.

There's this thing called "Ricardo's Law of Comparative Advantage." There's this idea called "professional specialization." There's this notion of "economies of scale." There's this concept of "gains from trade." The whole reason why we have money is to realize the *tremendous* gains possible from each of us doing what we do *best*.

This is what grownups do. This is what you do when you want something to actually get *done*. You use *money* to employ *full-time specialists*.

Yes, people are sometimes limited in their ability to trade time for money (underemployed), so that it is better for them if they can directly donate that which they would usually trade for money. *If* the soup kitchen *needed* a lawyer, and the lawyer donated a *large contiguous high-priority* block of lawyering, then *that* sort of volunteering makes sense—that's the same *specialized* capability the lawyer ordinarily trades for money. But "volunteering" just one hour of legal work, constantly delayed, spread across three weeks in casual minutes between other jobs? This is not the way something gets done *when anyone actually cares about it*, or to state it near-equivalently, *when money is involved*.

To the extent that individuals fail to grasp this principle on a *gut level*, they may think that the use of money is somehow *optional* in the pursuit of things that merely seem *morally* desirable—as opposed to tasks like feeding ourselves, whose desirability seems to be treated oddly differently. This factor may be sufficient *by itself* to prevent us from pursuing our collective common interest in groups larger than 40 people.

Economies of trade and professional specialization are not just vaguely good yet unnatural-sounding ideas, *they are the only way that anything ever gets done in this world*. Money is not pieces of paper, it is the *common currency of caring*.

Hence the old saying: “Money makes the world go ‘round, love barely keeps it from blowing up.”

Now, we do have the problem of *akrasia*—of not being able to do what we’ve decided to do—which is a part of the art of rationality that I hope someone else will develop; I specialize more in the impossible questions business. And yes, spending money is more painful than volunteering, because you can see the bank account number go down, whereas the remaining hours of our span are not visibly numbered. But when it comes time to feed yourself, do you think, “Hm, maybe I should try raising my own cattle, that’s less painful than spending money on beef?” Not everything can get done *without* invoking Ricardo’s Law; and on the other end of that trade are people who feel just the same pain at the thought of having less money.

It does seem to me offhand that there ought to be things doable to diminish the pain of losing hit points, and to increase the felt strength of the connection from donating money to “I did a good thing!” Some of that I am trying to accomplish right now, by emphasizing the true nature and power of money; and by inveighing against the poisonous meme saying that someone who gives mere money must not care enough to get personally involved. This is a mere reflection of a mind that doesn’t understand the post-hunter-gatherer concept of a market economy. The act of donating money is not the momentary act of writing the check; it is the act of every hour you spent to earn the money to write that check—just as though you worked

at the charity itself *in your professional capacity*, at maximum, grownup efficiency.

If the lawyer needs to work an hour at the soup kitchen to keep themselves motivated and remind themselves why they're doing what they're doing, *that's fine*. But they should *also* be donating some of the hours they worked at the office, because that is the power of professional specialization. One might consider the check as buying the right to volunteer at the soup kitchen, or validating the time spent at the soup kitchen. More on this later.

To a first approximation, money is the unit of caring up to a positive scalar factor—the unit of relative caring. Some people are frugal and spend less money on *everything*; but if you would, in fact, spend \$5 on a burrito, then whatever you will not spend \$5 on, you care about *less than* you care about the burrito. If you don't spend two months' salary on a diamond ring, it doesn't mean you don't love your Significant Other. ("De Beers: It's Just A Rock.") But conversely, if you're *always* reluctant to spend *any* money on your Significant Other, and yet seem to have no emotional problems with spending \$1,000 on a flat-screen TV, then yes, this *does* say something about your relative values.

Yes, frugality is a virtue. Yes, spending money hurts. But in the end, if you are never willing to spend any units of caring, it means you don't care.

Purchase Fuzzies and Utilons Separately

Previously:

There is this very, very old puzzle/observation in economics about the lawyer who spends an hour volunteering at the soup kitchen, instead of working an extra hour and donating the money to hire someone ...

If the lawyer needs to work an hour at the soup kitchen to keep themselves motivated and remind themselves why they're doing what they're doing, *that's fine*. But they should *also* be donating some of the hours they worked at the office, because that is the power of professional specialization. One might consider the check as buying the right to volunteer at the soup kitchen, or validating the time spent at the soup kitchen. More on this later.

I hold open doors for little old ladies. I can't actually remember the last time this happened literally (though I'm sure it has, sometime in the last year or so). But within the last month, say, I was out on a walk and discovered a station wagon parked in a driveway with its trunk completely open, giving full access to the car's interior. I looked in to see if there were packages being taken out, but this was not so. I looked around to see if anyone was doing anything with the car. And finally I went up to the house and knocked, then rang the bell. And yes, the trunk had been accidentally left open.

Under other circumstances, this would be a simple act of altruism, which might signify true concern for another's welfare, or fear of guilt for inaction, or a desire to signal trustworthiness to oneself or others, or finding altruism pleasurable. I think that these are all perfectly legitimate motives, by the way; I



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might give bonus points for the first, but I wouldn't deduct any penalty points for the others. Just so long as people get helped.

But in my own case, since I already work in the nonprofit sector, the further question arises as to whether I could have better employed the same sixty seconds in a more *specialized* way, to bring greater benefit to others. That is: can I really defend this as the *best* use of my time, given the other things I claim to believe?

The obvious defense—or, perhaps, obvious rationalization—is that an act of altruism like this one acts as a willpower restorer, much more efficiently than, say, listening to music. I also mistrust my ability to be an altruist *only* in theory; I suspect that if I walk past problems, my altruism will start to fade. I've never pushed that far enough to test it; it doesn't seem worth the risk.

But if that's the defense, then my act can't be defended as a good deed, can it? For these are self-directed benefits that I list.

Well—who said that I *was* defending the act as a selfless good deed? It's a *selfish* good deed. If it restores my willpower, or if it keeps me altruistic, then there are indirect other-directed benefits from that (or so I believe). You could, of course, reply that you don't trust selfish acts that are supposed to be other-benefiting as an "ulterior motive"; but then I could just as easily respond that, by the same principle, you should just look directly at the original good deed rather than *its* supposed ulterior motive.

Can I get away with that? That is, can I really get away with calling it a "selfish good deed," and still derive willpower restoration therefrom, rather than feeling guilt about its being selfish? Apparently I can. I'm surprised it works out that way, but it does. So long as I knock to tell them about the open trunk, and so long as the one says "Thank you!," my brain feels like it's done its wonderful good deed for the day.

Your mileage may vary, of course. The problem with trying to work out an art of willpower restoration is that different things seem to work for different people. (That is: We're probing around on the level of surface phenomena without understanding the deeper rules that would also predict the variations.)

But if you find that you are like me in this aspect—that selfish good deeds still work—then I recommend that you *purchase warm fuzzies and utilons separately*. Not at the same time. Trying to do both at the same time just means that neither ends up done well. If status matters to you, purchase status separately too!

If I had to give advice to some new-minted billionaire entering the realm of charity, my advice would go something like this:

- To purchase warm fuzzies, find some hard-working but poverty-stricken woman who's about to drop out of state college after her husband's hours were cut back, and personally, but anonymously, give her a cashier's check for \$10,000. Repeat as desired.
- To purchase status among your friends, donate \$100,000 to the current sexiest X-Prize, or whatever other charity seems to offer the most stylishness for the least price. Make a big deal out of it, show up for their press events, and brag about it for the next five years.
- Then—with absolute cold-blooded calculation—without scope insensitivity or ambiguity aversion—without concern for status or warm fuzzies—figuring out some common scheme for converting outcomes to utilons, and trying to express uncertainty in percentage probabilities—find the charity that offers the greatest expected utilons per dollar. Donate up to however much money you wanted to give to charity, until their marginal efficiency drops below that of the next charity on the list.

I would furthermore advise the billionaire that what they spend on utilons should be at least, say, 20 times what they spend on warm fuzzies—5% overhead on keeping yourself altruistic seems reasonable, and I, your dispassionate judge, would have no trouble *validating* the warm fuzzies against a multiplier that large. Save that the original fuzzy act really should be helpful rather than actively harmful.

(Purchasing *status* seems to me essentially unrelated to altruism. If giving money to the X-Prize gets you more awe from your friends than an equivalently priced speedboat, then there's really no reason to buy the speedboat. Just put the money under the "impressing friends" column, and be aware that this is not the "altruism" column.)

But the main lesson is that all three of these things—warm fuzzies, status, and expected utilons—can be bought *far* more efficiently when you buy *separately*, optimizing for only one thing at a time. Writing a check for \$10,000,000 to a breast-cancer charity—while far more laudable than spending the same \$10,000,000 on, I don't know, parties or something—won't give you the concentrated euphoria of being present in person when you turn a single human's life around, probably not anywhere *close*. It won't give you as much to talk about at parties as donating to something sexy like an X-Prize—maybe a short nod from the other rich. And if you threw away all concern for warm fuzzies and status, there are probably at least a *thousand* underserved existing charities that could produce *orders of magnitude* more utilons with ten million dollars. Trying to optimize for all three criteria in one go only ensures that none of them end up optimized very well—just vague pushes along all three dimensions.

Of course, if you're not a millionaire or even a billionaire—then you can't be quite as *efficient* about things, can't so easily purchase in bulk. But I would still say—for warm fuzzies, find a relatively *cheap* charity with bright, vivid, ideally in-person and direct beneficiaries. Volunteer at a soup kitchen. Or just get your warm fuzzies from holding open doors for little old ladies. Let that be *validated* by your other efforts to purchase utilons, but don't *confuse* it with purchasing utilons. Status is probably cheaper to purchase by buying nice clothes.

And when it comes to purchasing expected utilons—then, of course, shut up and multiply.

Bystander Apathy

The bystander effect, also known as bystander apathy, is that larger groups are less likely to act in emergencies—not just individually, but collectively. Put an experimental subject alone in a room and let smoke start coming up from under the door. Seventy-five percent of the subjects will leave to report it. Now put *three* subjects in the room—real subjects, none of whom know what’s going on. On only 38% of the occasions will *anyone* report the smoke. Put the subject with two confederates who ignore the smoke, and they’ll only report it 10% on the time—even staying in the room until it becomes hazy.¹

On the standard model, the two primary drivers of bystander apathy are:

- *Diffusion of responsibility*—everyone hopes that someone else will be first to step up and incur any costs of acting. When no one does act, being part of a crowd provides an excuse and reduces the chance of being held personally responsible for the results.
- *Pluralistic ignorance*—people try to *appear* calm while looking for cues, and see ...that the others appear calm.

Cialdini:²

Very often an emergency is not obviously an emergency. Is the man lying in the alley a heart-attack victim or a drunk sleeping one off? ...In times of such uncertainty, the natural tendency is to look



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1. Bibb Latané and John M. Darley, “Bystander ‘Apathy,’” *American Scientist* 57, no. 2 (1969): 244–268, sharov.me/r/6-10.



2. Cialdini, *Influence*.

around at the actions of others for clues. We can learn from the way the other witnesses are reacting whether the event is or is not an emergency. What is easy to forget, though, is that everybody else observing the event is likely to be looking for social evidence, too. Because we all prefer to appear poised and unflustered among others, we are likely to search for that evidence placidly, with brief, camouflaged glances at those around us. Therefore everyone is likely to see everyone else looking unruffled and failing to act.

Cialdini suggests that if you're ever in emergency need of help, you point to *one single* bystander and ask them for help—making it very clear to whom you're referring. Remember that the *total* group, combined, may have less chance of helping than one individual.

I've mused a bit on the evolutionary psychology of the bystander effect. Suppose that in the ancestral environment, most people in your band were likely to be at least a little related to you—enough to be worth saving, if you were the only one who could do it. But if there are two others present, then the *first* person to act incurs a cost, while the other two both reap the *genetic* benefit of a partial relative being saved. Could there have been an arms race for who waited the longest?

As far as I've followed this line of speculation, it doesn't seem to be a good explanation—at the point where the whole group is failing to act, a gene that helps immediately ought to be able to invade, I would think. The experimental result is not a long wait before helping, but simply failure to help: if it's a genetic benefit to help when you're the only person who can do it (as *does* happen in the experiments) then the group equilibrium should not be *no* one helping (as happens in the experiments).

So I don't think an arms race of delay is a plausible evolutionary explanation. More likely, I think, is that we're looking at a nonancestral problem. If the experimental subjects actually *know* the apparent victim, the chances of helping go way up

(i.e., we're not looking at the correlate of helping an actual fellow band member). If I recall correctly, if the experimental subjects know each *other*, the chances of action also go up.

Nervousness about public action may also play a role. If Robin Hanson is right about the evolutionary role of "choking," then being *first* to act in an emergency might also be taken as a dangerous bid for high status. (Come to think, I can't actually recall seeing *shyness* discussed in analyses of the bystander effect, but that's probably just my poor memory.)

Can the bystander effect be explained primarily by diffusion of moral responsibility? We could be cynical and suggest that people are mostly interested in *not being blamed* for not helping, rather than having any positive desire to help—that they mainly wish to escape antiheroism and possible retribution. Something like this may well be a contributor, but two observations that mitigate against it are (a) the experimental subjects did not report smoke coming in from under the door, even though it could well have represented a strictly selfish threat and (b) telling people about the bystander effect reduces the bystander effect, even though they're no more likely to be held publicly responsible thereby.

In fact, the bystander effect is one of the main cases I recall offhand where telling people about a bias actually seems able to strongly reduce it—maybe because the appropriate way to compensate is so obvious, and it's not easy to *overcompensate* (as when you're trying to e.g. adjust your calibration). So we should be careful not to be too cynical about the implications of the bystander effect and diffusion of responsibility, if we interpret individual action in terms of a cold, calculated attempt to avoid public censure. People seem at least to sometimes hold *themselves* responsible, once they realize they're the only ones who know enough about the bystander effect to be likely to act.

Though I wonder what happens if you know that you're part of a crowd where *everyone* has been told about the bystander effect ...

Collective Apathy and the Internet



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In the last essay I covered the bystander effect, a.k.a. bystander apathy: given a fixed problem situation, a *group* of bystanders is actually *less* likely to act than a *single* bystander. The standard explanation for this result is in terms of pluralistic ignorance (if it's not clear whether the situation is an emergency, each person tries to *look* calm while darting their eyes at the other bystanders, and sees other people *looking* calm) and diffusion of responsibility (everyone hopes that someone else will be first to act; being part of a crowd diminishes the individual pressure to the point where no one acts).

Which may be a symptom of our hunter-gatherer coordination mechanisms being defeated by modern conditions. You didn't usually form task-forces with strangers back in the ancestral environment; it was mostly people you knew. And in fact, when all the subjects know each other, the bystander effect diminishes.

So I know this is an amazing and revolutionary observation, and I hope that I don't kill any readers outright from shock by saying this: but people seem to have a hard time reacting constructively to problems encountered over the Internet.

Perhaps because our innate coordination instincts are not tuned for:

- Being part of a group of strangers. (When all subjects know each other, the bystander effect diminishes.)
- Being part of a group of unknown size, of strangers of unknown identity.
- Not being in physical contact (or visual contact); not being able to exchange meaningful glances.
- Not communicating in real time.

- Not being much beholden to each other for other forms of help; not being codependent on the group you're in.
- Being shielded from reputational damage, or the fear of reputational damage, by your own apparent anonymity; no one is visibly looking at you, before whom your reputation might suffer from inaction.
- Being part of a large collective of other inactives; no one will single out you to blame.
- Not hearing a voiced plea for help.

Et cetera. I don't have a brilliant solution to this problem. But it's the sort of thing that I would wish for potential dot-com cofounders to ponder explicitly, rather than wondering how to throw sheep on Facebook. (Yes, I'm looking at *you*, Hacker News.) There are online activism web apps, but they tend to be along the lines of *sign this petition! yay, you signed something!* rather than *how can we counteract the bystander effect, restore motivation, and work with native group-coordination instincts, over the Internet?*

Some of the things that come to mind:

- Put a video of someone asking for help online.
- Put up names and photos or even brief videos if available of the *first* people who helped (or have some reddit-ish priority algorithm that depends on a combination of amount-helped and recency).
- Give helpers a video thank-you from the founder of the cause that they can put up on their "people I've helped" page, which with enough standardization could be partially or wholly assembled automatically and easily embedded in their home webpage or Facebook account.
- Find a *non-annoying* idiom for "Tell a friend about cause X"; allow referrer link codes; then show people how many others they've evangelized (how many people who initially got here using referrer code X actually contributed or took some other action).

- (All of the above applies not just to donations, but to open-source projects to which people have contributed code. Or if people really do want nothing but signatures on a petition, then for signatures. There are ways to help besides money—even though money is usually the most effective. The main thing is that the form of help has to be verifiable online.)
- Make it easier for people to offer monetary bounties on subtasks whose performance is verifiable.

But mostly I just hand you an open, unsolved problem: make it possible/easier for groups of strangers to coalesce into an effective task force over the Internet, in defiance of the usual failure modes and the default reasons why this is a non-ancestral problem. Think of that old statistic about Wikipedia representing 1/2,000 of the time spent in the US alone on watching television. There's quite a lot of fuel out there, if there were only such a thing as an effective engine ...

Incremental Progress and the Valley

Rationality is systematized winning.

“But,” you protest, “the reasonable person *doesn’t* always win!”

What do you mean by this? Do you mean that every week or two, someone who bought a lottery ticket with negative expected value wins the lottery and becomes much richer than you? That is not a *systematic* loss; it is selective reporting by the media. From a statistical standpoint, lottery winners don’t exist—you would never encounter one in your lifetime, if it weren’t for the selective reporting.

Even perfectly rational agents can lose. They just can’t *know in advance* that they’ll lose. They can’t *expect to underperform* any other performable strategy, or they would simply perform it.

“No,” you say, “I’m talking about how startup founders strike it rich by believing in themselves and their ideas more strongly than any reasonable person would. I’m talking about how religious people are happier—”

Ah. Well, here’s the thing: An *incremental* step in the direction of rationality, if the result is still irrational in other ways, does not have to yield *incrementally* more winning.

The optimality theorems that we have for probability theory and decision theory are for *perfect* probability theory and decision theory. There is no companion theorem which says that, starting from some flawed initial form, every *incremental* modification of the algorithm that takes the structure closer to the ideal must yield an *incremental* improvement in performance. This has not yet been proven, because it is not, in fact, true.

“So,” you say, “what point is there then in striving to be more rational? We won’t reach the perfect ideal. So we have no guarantee that our steps forward are helping.”



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You have no guarantee that a step *backward* will help you win, either. Guarantees don't exist in the world of flesh; but, contrary to popular misconceptions, judgment under *uncertainty* is what rationality is all about.

"But we have several cases where, based on either vaguely plausible-sounding reasoning, or survey data, it looks like an incremental step forward in rationality is going to make us worse off. If it's really all about winning—if you have something to protect more important than any ritual of cognition—then *why* take that step?"

Ah, and *now* we come to the meat of it.

I can't necessarily answer for everyone, but ...

My first reason is that, on a professional basis, I deal with deeply confused problems that make huge demands on precision of thought. One small mistake can lead you astray for years, and there are worse penalties waiting in the wings. An unimproved level of performance isn't *enough*; my choice is to try to do better, or give up and go home.

"But that's just *you*. Not all of us lead that kind of life. What if you're just trying some ordinary human task like an Internet startup?"

My second reason is that I am trying to push some aspects of my art further than I have seen done. I don't *know* where these improvements lead. The loss of failing to take a step forward is not that *one step*. It is all the *other* steps forward you could have taken, beyond that point. Robin Hanson has a saying: The problem with slipping on the stairs is not falling the height of the first step; it is that falling one step leads to falling another step. In the same way, refusing to climb one step up forfeits not the height of that step but the height of the staircase.

"But again—that's just you. Not all of us are trying to push the art into uncharted territory."

My third reason is that once I realize I have been deceived, I can't just shut my eyes and pretend I haven't seen it. I have *already taken* that step forward; what use to deny it to myself? I couldn't believe in God if I tried, any more than I could believe the sky above me was green while looking straight at it. If you *know* everything you need to know in order to know that you

are better off deceiving yourself, it's much too late to deceive yourself.

"But that realization is *unusual*; other people have an easier time of doublethink because they don't realize it's impossible. *You* go around trying to actively sponsor the collapse of doublethink. *You*, from a higher vantage point, may know enough to expect that this will make them unhappier. So is this out of a sadistic desire to hurt your readers, or what?"

Then I finally reply that my experience so far—even in this realm of merely human possibility—*does* seem to indicate that, once you sort yourself out a bit and you aren't doing *quite* so many other things wrong, striving for more rationality actually *will* make you better off. The long road leads out of the valley and higher than before, even in the human lands.

The more I know about some particular facet of the Art, the more I can see this is so. As I've previously remarked, my essays may be unreflective of what a true martial art of rationality would be like, because I have only focused on answering confusing questions—not fighting akrasia, coordinating groups, or being happy. In the field of answering confusing questions—the area where I have most intensely practiced the Art—it now seems *massively* obvious that anyone who thought they were better off "staying optimistic about solving the problem" would get stomped into the *ground*. By a *casual student*.

When it comes to keeping motivated, or being happy, I can't guarantee that someone who loses their illusions will be better off—because my knowledge of these facets of rationality is still crude. If these parts of the Art have been developed systematically, I do not know of it. But even here I have gone to some considerable pains to dispel half-rational half-mistaken ideas that could get in a beginner's way, like the idea that rationality opposes feeling, or the idea that rationality opposes value, or the idea that sophisticated thinkers should be angsty and cynical.

And if, as I hope, someone goes on to develop the art of fighting akrasia or achieving mental well-being as thoroughly as I have developed the art of answering impossible questions, I do fully expect that those who wrap themselves in their illusions

will not *begin* to compete. Meanwhile—others may do better than I, if happiness is their dearest desire, for I myself have invested little effort here.

I find it hard to believe that the *optimally* motivated individual, the *strongest* entrepreneur a human being can become, is still wrapped up in a blanket of comforting overconfidence. I think they've probably thrown that blanket out the window and organized their mind a little *differently*. I find it hard to believe that the happiest we can possibly live, even in the realms of human possibility, involves a tiny awareness lurking in the corner of your mind that it's all a lie. I'd rather stake my hopes on neurofeedback or Zen meditation, though I've tried neither.

But it cannot be denied that this is a very real issue in very real life. Consider this pair of comments from *Less Wrong*:

I'll be honest—my life has taken a sharp downturn since I deconverted. My theist girlfriend, with whom I was very much in love, couldn't deal with this change in me, and after six months of painful vacillation, she left me for a co-worker. That was another six months ago, and I have been heartbroken, miserable, unfocused, and *extremely* ineffective since.

Perhaps this is an example of the valley of bad rationality of which PhilGoetz spoke, but I still hold my current situation higher in my preference ranking than happiness with false beliefs.

And:

My empathies: that happened to me about 6 years ago (though thankfully without as much visible vacillation).

My sister, who had some Cognitive Behaviour Therapy training, reminded me that relationships are forming and breaking all the time, and given I wasn't unattractive and hadn't retreated into monastic seclusion, it wasn't rational to think I'd be alone for the rest of my life (she turned out to

be right). That was helpful at the times when my feelings hadn't completely got the better of me.

So—in practice, in real life, in sober fact—those first steps can, in fact, be painful. And then things can, in fact, get better. And there is, in fact, no *guarantee* that you'll end up higher than before. Even if in principle the path must go further, there is no guarantee that any given person will get that far.

If you don't *prefer* truth to happiness with false beliefs ...

Well ...*and* if you are not doing anything especially precarious or confusing ...*and* if you are not buying lottery tickets ...*and* if you're already signed up for cryonics, a sudden ultra-high-stakes confusing acid test of rationality that illustrates the Black Swan quality of trying to bet on ignorance *in* ignorance ...

Then it's not *guaranteed* that taking all the incremental steps toward rationality that you can find will leave you better off. But the vaguely plausible-sounding arguments against losing your illusions generally *do* consider just one single step, without postulating any further steps, without suggesting any attempt to regain everything that was lost and go it one better. Even the surveys are comparing the average religious person to the average atheist, not the most advanced theologians to the most advanced rationalists.

But if you don't care about the truth—*and* you have nothing to protect—*and* you're not attracted to the thought of pushing your art as far as it can go—*and* your current life seems to be going fine—*and* you have a sense that your mental well-being depends on illusions you'd rather not think about—

Then you're probably not reading this. But if you are, then, I guess ...well ...(a) sign up for cryonics, and then (b) *stop reading Less Wrong before your illusions collapse! RUN AWAY!*

Bayesians vs. Barbarians



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Previously:

Let's say we have two groups of soldiers. In group 1, the privates are ignorant of tactics and strategy; only the sergeants know anything about tactics and only the officers know anything about strategy. In group 2, everyone at all levels knows all about tactics and strategy.

Should we expect group 1 to defeat group 2, because group 1 will follow orders, while everyone in group 2 comes up with *better ideas* than whatever orders they were given?

In this case I have to question how much group 2 really understands about military theory, because it is an *elementary* proposition that an uncoordinated mob gets slaughtered.

Suppose that a country of rationalists is attacked by a country of Evil Barbarians who know nothing of probability theory or decision theory.

Now there's a certain viewpoint on "rationality" or "rationalism" which would say something like this:

"Obviously, the rationalists will lose. The Barbarians believe in an afterlife where they'll be rewarded for courage; so they'll throw themselves into battle without hesitation or remorse. Thanks to their affective death spirals around their Cause and Great Leader Bob, their warriors will obey orders, and their citizens at home will produce enthusiastically and at full capacity for the war; anyone caught skimming or holding back will be burned at the stake in accordance with Barbarian tradition. They'll believe in each other's goodness and hate the enemy more strongly than any sane person would, binding themselves into a tight group. Meanwhile, the rationalists will

realize that there's no conceivable reward to be had from dying in battle; they'll wish that others would fight, but not want to fight themselves. Even if they can find soldiers, their civilians won't be as cooperative: So long as any *one* sausage almost certainly doesn't lead to the collapse of the war effort, they'll want to keep that sausage for themselves, and so not contribute as much as they could. No matter how refined, elegant, civilized, productive, and nonviolent their culture was to start with, they won't be able to resist the Barbarian invasion; sane discussion is no match for a frothing lunatic armed with a gun. In the end, the Barbarians will win because they *want* to fight, they *want* to hurt the rationalists, they *want* to conquer and their whole society is united around conquest; they care about that more than any sane person would."

War is not fun. As many, many people have found since the dawn of recorded history, as many, many people have found before the dawn of recorded history, as some community somewhere is finding out right now in some sad little country whose internal agonies don't even make the front pages any more.

War is not fun. *Losing* a war is even less fun. And it was said since the ancient times: "If thou would have peace, prepare for war." Your opponents don't have to believe that you'll *win*, that you'll conquer; but they have to believe you'll put up enough of a fight to make it not worth their while.

You perceive, then, that if it were genuinely the lot of "rationalists" to always lose in war, that I could not in good conscience advocate the widespread public adoption of "rationality."

This is probably the dirtiest topic I've discussed or plan to discuss here. War is not clean. Current high-tech militaries—by this I mean the US military—are unique in the overwhelmingly superior force they can bring to bear on opponents, which allows for a historically extraordinary degree of concern about enemy casualties and civilian casualties.

Winning in war has not always meant tossing aside *all* morality. Wars have been won without using torture. The unfunness of war does not imply, say, that questioning the President is

unpatriotic. We're used to "war" being exploited as an excuse for bad behavior, because in recent US history that pretty much is exactly what it's been used for ...

But reversed stupidity is not intelligence. And reversed evil is not intelligence either. It remains true that *real* wars cannot be won by refined politeness. If "rationalists" can't prepare themselves for that mental shock, the Barbarians really will win; and the "rationalists" ...I don't want to say, "deserve to lose." But they will have failed that test of their society's existence.

Let me start by disposing of the idea that, *in principle*, ideal rational agents cannot fight a war, because each of them prefers being a civilian to being a soldier.

As has already been discussed at some length, I one-box on Newcomb's Problem.

Consistently, I do *not* believe that if an election is settled by 100,000 to 99,998 votes, that all of the voters were irrational in expending effort to go to the polling place because "my staying home would not have affected the outcome." (Nor do I believe that if the election came out 100,000 to 99,999, then 100,000 people were *all*, individually, *solely responsible* for the outcome.)

Consistently, I also hold that two rational AIs (that use my kind of decision theory), even if they had completely different utility functions and were designed by different creators, will cooperate on the true Prisoner's Dilemma if they have common knowledge of each other's source code. (Or even just common knowledge of each other's *rationality* in the appropriate sense.)

Consistently, I believe that rational agents are capable of coordinating on group projects whenever the (expected probabilistic) outcome is better than it would be without such coordination. A society of agents that use my kind of decision theory, and have common knowledge of this fact, will end up at Pareto optima instead of Nash equilibria. If all rational agents agree that they are better off fighting than surrendering, they will fight the Barbarians rather than surrender.

Imagine a community of self-modifying AIs who collectively prefer fighting to surrender, but individually prefer being a civilian to fighting. One solution is to run a lottery, unpredictable to any agent, to select warriors. *Before* the lottery is run, all the

AIs change their code, in advance, so that if selected they will fight as a warrior in the most communally efficient possible way—even if it means calmly marching into their own death.

(A reflectively consistent decision theory works the same way, only without the self-modification.)

You reply: “But in the real, human world, agents are not perfectly rational, nor do they have common knowledge of each other’s source code. Cooperation in the Prisoner’s Dilemma requires certain conditions according to your decision theory (which these margins are too small to contain) and these conditions are not met in real life.”

I reply: The pure, true Prisoner’s Dilemma is incredibly rare in real life. In real life you usually have knock-on effects—what you do affects your reputation. In real life most people care to some degree about what happens to other people. And in real life you have an opportunity to set up incentive mechanisms.

And in real life, I *do* think that a community of human rationalists could manage to produce soldiers willing to die to defend the community. So long as children aren’t told in school that ideal rationalists are supposed to defect against each other in the Prisoner’s Dilemma. Let it be widely believed—and I do believe it, for exactly the same reason I one-box on Newcomb’s Problem—that if people decided as individuals not to be soldiers or if soldiers decided to run away, then that is the same as deciding for the Barbarians to win. By that same theory whereby, if an election is won by 100,000 votes to 99,998 votes, it does not make sense for every voter to say “my vote made no difference.” Let it be said (for it is true) that utility functions don’t need to be solipsistic, and that a rational agent can fight to the death if they care enough about what they’re protecting. Let them not be told that rationalists should expect to lose reasonably.

If this is the culture and the mores of the rationalist society, then, I think, *ordinary human beings* in that society would volunteer to be soldiers. That also seems to be built into human beings, after all. You only need to ensure that the cultural training *does not get in the way*.

And if I’m wrong, and that doesn’t get you enough volunteers?

Then so long as people still prefer, on the whole, fighting to surrender, they have an opportunity to set up incentive mechanisms, and avert the True Prisoner's Dilemma.

You can have lotteries for who gets elected as a warrior. Sort of like the example above with AIs changing their own code. Except that if "be reflectively consistent; do that which you would precommit to do" is not sufficient motivation for humans to obey the lottery, then ...

...well, in advance of the lottery actually running, we can perhaps all agree that it is a good idea to give the selectees drugs that will induce extra courage, and shoot them if they run away. Even considering that we ourselves might be selected in the lottery. Because in *advance* of the lottery, this is the general policy that gives us the highest *expectation* of survival.

...like I said: Real wars = not fun, losing wars = less fun.

Let's be clear, by the way, that I'm not endorsing the draft as practiced nowadays. Those drafts are not collective attempts by a populace to move from a Nash equilibrium to a Pareto optimum. Drafts are a tool of kings playing games in need of toy soldiers. The Vietnam draftees who fled to Canada, I hold to have been in the right. But a society that considers itself too smart for kings does *not* have to be too smart to survive. Even if the Barbarian hordes are invading, and the Barbarians do practice the draft.

Will rational soldiers obey orders? What if the commanding officer makes a mistake?

Soldiers march. Everyone's feet hitting the ground in the same rhythm. Even, perhaps, against their own inclinations, since people left to themselves would walk all at separate paces. Lasers made out of people. That's marching.

If it's possible to invent some method of group decisionmaking that is *superior* to the captain handing down orders, then a company of rational soldiers might implement that procedure. If there is no proven method better than a captain, then a company of rational soldiers commit to obey the captain, even against their own separate inclinations. And if human beings aren't that rational ...then in advance of the lottery, the general policy that gives you the highest personal expectation

of survival is to shoot soldiers who disobey orders. This is not to say that those who fragged their own officers in Vietnam were in the wrong; for they could have consistently held that they preferred *no one* to participate in the draft lottery.

But an uncoordinated mob gets slaughtered, and so the soldiers need *some* way of all doing the same thing at the same time in the pursuit of the same goal, even though, left to their own devices, they might march off in all directions. The orders may not come from a captain like a superior tribal chief, but unified orders have to come from *somewhere*. A society whose soldiers are too clever to obey orders is a society that is too clever to survive. Just like a society whose people are too clever to *be* soldiers. That is why I say “clever,” which I often use as a term of opprobrium, rather than “rational.”

(Though I do think it’s an important question as to whether you can come up with a small-group coordination method that really genuinely in practice works better than having a leader. The more people can trust the group decision method—the more they can believe that it really is superior to people going their own way—the more coherently they can behave even in the absence of enforceable penalties for disobedience.)

I say all this, even though I certainly don’t expect rationalists to take over a country any time soon, because I think that what we believe about a society of “people like us” has some reflection on what we think of ourselves. If you believe that a society of people like you would be too reasonable to survive in the long run ...that’s one sort of self-image. And it’s a different sort of self-image if you think that a society of people all like you could fight the vicious Evil Barbarians and *win*—not just by dint of superior technology, but because your people care about each other and about their collective society—and because they can face the realities of war without losing themselves—and because they would calculate the group-rational thing to do and make sure it got done—and because there’s nothing in the rules of probability theory or decision theory that says you can’t sacrifice yourself for a cause—and because if you really *are* smarter than the Enemy and not just flattering yourself about that, then you should be able to exploit the blind spots that the

Enemy does not allow itself to think about—and because no matter how heavily the Enemy hypes itself up before battle, you think that just maybe a coherent mind, undivided within itself, and perhaps practicing something akin to meditation or self-hypnosis, can fight as hard in practice as someone who theoretically believes they've got seventy-two virgins waiting for them.

Then you'll expect more of yourself *and people like you operating in groups*; and then you can see yourself as something more than a cultural dead end.

So look at it this way: Jeffreyssai probably wouldn't give up against the Evil Barbarians if he were fighting *alone*. A whole *army* of *beisutsukai* masters ought to be a force that *no one* would mess with. That's the motivating vision. The question is how, exactly, that works.

Beware of Other-Optimizing

I've noticed a serious problem in which aspiring rationalists vastly overestimate their ability to optimize other people's lives. And I think I have some idea of how the problem arises.

You read nineteen different webpages advising you about personal improvement—productivity, dieting, saving money. And the writers all sound bright and enthusiastic about Their Method, they tell tales of how it worked for them and promise *amazing* results ...

But most of the advice rings so false as to not even seem worth considering. So you sigh, mournfully pondering the wild, childish enthusiasm that people can seem to work up for just about anything, no matter how silly. Pieces of advice #4 and #15 sound interesting, and you try them, but ...they don't ...quite ...well, it fails miserably. The advice was wrong, or you couldn't do it, and either way you're not any better off.

And then you read the twentieth piece of advice—or even more, you discover a twentieth method that wasn't in any of the pages—and STARS ABOVE IT ACTUALLY WORKS THIS TIME.

At long, long last you have discovered the *real* way, the *right* way, the way that actually *works*. And when someone else gets into the sort of trouble you used to have—well, this time you *know* how to help them. You can save them all the trouble of reading through nineteen useless pieces of advice and skip directly to the correct answer. As an aspiring rationalist you've already learned that most people don't listen, and you usually don't bother—but this person is a friend, someone you know, someone you trust and respect to listen.

And so you put a comradely hand on their shoulder, look them straight in the eyes, and tell them how to do it.

I, personally, get quite a lot of this. Because you see ...when you've discovered the way that *really works* ...well, you know better by now than to run out and tell your friends and family.



discussion
sharov.me/r/
6-c40

But you've got to try telling Eliezer Yudkowsky. He *needs* it, and there's a pretty good chance that *he'll* understand.

It actually did take me a while to understand. One of the critical events was when someone on the Board of the Machine Intelligence Research Institute told me that I didn't need a salary increase to keep up with inflation—because I could be spending substantially less money on food if I used an online coupon service. And I believed this, because it was a friend I trusted, and it was delivered in a tone of such confidence. So my girlfriend started trying to use the service, and a couple of weeks later she gave up.

Now here's the thing: if I'd run across exactly the same advice about using coupons on some blog somewhere, I probably wouldn't even have paid much attention, just read it and moved on. Even if it were written by Scott Aaronson or some similar person known to be intelligent, I still would have read it and moved on. But because it was delivered to me personally, by a friend who I knew, my brain processed it differently—as though I were being told *the* secret; and that indeed is the tone in which it was told to me. And it was something of a delayed reaction to realize that I'd simply been told, as personal advice, what otherwise would have been just a blog post somewhere; no more and no less likely to work for me, than a productivity blog post written by any other intelligent person.

And because I have encountered a great many people trying to optimize me, I can attest that the advice I get is as wide-ranging as the productivity blogosphere. But others don't see this plethora of productivity advice as indicating that people are *diverse* in which advice works for them. Instead they see a lot of obviously wrong poor advice. And then they finally discover the right way—the way that works, unlike all those other blog posts that don't work—and then, quite often, they decide to use it to optimize Eliezer Yudkowsky.

Don't get me wrong. Sometimes the advice is helpful. Sometimes it works. "Stuck In The Middle With Bruce"¹—that resonated, for me. It may prove to be the most helpful thing I've read on the new *Less Wrong* so far, though that has yet to be determined.

It's just that your earnest personal advice, that amazing thing you've found to actually work by golly, is no more and no less likely to work for me than a random personal improvement blog post written by an intelligent author is likely to work for you.

"Different things work for different people." That sentence may give you a squicky feeling; I know it gives me one. Because this sentence is a tool wielded by Dark Side Epistemology to shield from criticism, used in a way closely akin to "Different things are true for different people" (which is simply false).

But until you grasp the laws that are near-universal generalizations, sometimes you end up messing around with surface tricks that work for one person and not another, without your understanding why, because you don't know the general laws that would dictate what works for who. And the best you can do is remember that, and be willing to take "No" for an answer.

You *especially* had better be willing to take "No" for an answer, if you have *power* over the Other. Power is, in general, a very dangerous thing, which is tremendously easy to abuse, without your being aware that you're abusing it. There are things you can do to prevent yourself from abusing power, but you have to actually do them or they don't work. There was a post on *Overcoming Bias* on how being in a position of power has been shown to decrease our ability to empathize with and understand the other, though I can't seem to locate it now. I have seen a rationalist who did not think he had power, and so did not think he needed to be cautious, who was amazed to learn that he might be feared ...

1. sharov.me/r/6-11

"Stuck In The Middle With Bruce" • lesswrong.com



It's even worse when their discovery that works for them requires a little *willpower*. Then if you say it doesn't work for you, the answer is clear and obvious: you're just being *lazy*, and they need to exert some *pressure* on you to get you to do the *correct* thing, the advice they've found that actually works.

Sometimes—I suppose—people are being lazy. But be very, very, *very* careful before you assume that's the case and wield power over others to “get them moving.” Bosses who can tell when something actually *is* in your capacity if you're a little more motivated, without it burning you out or making your life incredibly painful—these are the bosses who are a pleasure to work under. *That ability is extremely rare*, and the bosses who have it are worth their weight in silver. It's a high-level interpersonal technique that most people do not have. I surely don't have it. Do not assume you have it because your intentions are good. Do not assume you have it because you'd never do anything to *others* that you didn't want done to *yourself*. Do not assume you have it because no one has ever complained to you. Maybe they're just scared. That rationalist of whom I spoke—who did not think he held power and threat, though it was certainly obvious enough to me—he did not realize that anyone could be scared of him.

Be careful even when you hold *leverage*, when you hold an important decision in your hand, or a threat, or something that the other person needs, and all of a sudden the temptation to optimize them seems overwhelming.

Consider, if you would, that Ayn Rand's whole reign of terror over Objectivists can be seen in just this light—that she found herself with power and leverage, and could not resist the temptation to optimize.

We underestimate the distance between ourselves and others. Not just inferential distance, but distances of temperament and ability, distances of situation and resource, distances of unspoken knowledge and unnoticed skills and luck, distances of interior landscape.

Even I am often surprised to find that X, which worked so well for me, doesn't work for someone else. But with so many

others having tried to optimize me, I can at least recognize distance when I'm hit over the head with it.

Maybe being pushed on does work...for you. Maybe *you* don't get sick to the stomach when someone with power over you starts helpfully trying to reorganize your life the correct way. I don't know what makes you tick. In the realm of willpower and akrasia and productivity, as in other realms, I don't know the generalizations deep enough to hold almost always. I don't possess the deep keys that would tell me *when* and *why* and for *who* a technique works or doesn't work. All I can do is be willing to accept it when someone tells me it doesn't work...and go on looking for the deeper generalizations that will hold everywhere, the deeper laws governing both the rule and the exception, waiting to be found, someday.

Practical Advice Backed by Deep Theories



discussion
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6-c41

Once upon a time, Seth Roberts took a European vacation and found that he started losing weight while drinking unfamiliar-tasting caloric fruit juices.

Now suppose Roberts had not known, and never did know, anything about metabolic set points or flavor-calorie associations—all this high-falutin’ scientific experimental research that had been done on rats and occasionally humans.

He would have posted to his blog, “Gosh, everyone! You should try these amazing fruit juices that are making me lose weight!” And that would have been the end of it. Some people would have tried it, it would have worked *temporarily* for some of them (until the flavor-calorie association kicked in) and there never would have been a Shangri-La Diet *per se*.

The existing Shangri-La Diet is visibly incomplete—for some people, like me, it doesn’t seem to work, and there is no apparent reason for this or any logic permitting it. But the reason why as many people have benefited as they have—the reason why there was more than just one more blog post describing a trick that seemed to work for one person and didn’t work for anyone else—is that Roberts *knew the experimental science that let him interpret what he was seeing, in terms of deep factors that actually did exist*.

One of the pieces of advice on *Overcoming Bias / Less Wrong* that was frequently cited as the most important thing learned, was the idea of “the bottom line”—that once a conclusion is written in your mind, it is already true or already false, already wise or already stupid, and no amount of later argument can change that except by changing the conclusion. And this ties directly into another oft-cited most important thing, which is the idea of “engines of cognition,” minds as mapping engines that require evidence as fuel.

Suppose I had merely written one more blog post that said, “You know, you really should be more open to changing your mind—it’s pretty important—and oh yes, you should pay attention to the evidence too.” This would not have been as useful. Not just because it was *less persuasive*, but because the *actual operations* would have been much less clear without the explicit theory backing it up. What constitutes *evidence*, for example? Is it anything that seems like a forceful argument? Having an explicit probability theory and an explicit causal account of what makes reasoning effective makes a *large* difference in the forcefulness and implementational details of the old advice to “Keep an open mind and pay attention to the evidence.”

It is also important to realize that *causal theories* are much more likely to be true when they are picked up from a science textbook than when invented on the fly—it is very easy to invent cognitive structures that look like causal theories but are not even anticipation-controlling, let alone true.

This is the signature style I want to convey from all those essays that entangled cognitive science experiments and probability theory and epistemology with the practical advice—that practical advice actually becomes practically more powerful if you go out and read up on cognitive science experiments, or probability theory, or even materialist epistemology, and *realize what you’re seeing*. This is the brand that can distinguish *Less Wrong* from ten thousand other blogs purporting to offer advice.

I could tell you, “You know, how much you’re satisfied with your food probably depends more on the quality of the food than on how much of it you eat.” And you would read it and forget about it, and the impulse to finish off a whole plate would still feel just as strong. But if I tell you about scope insensitivity, and duration neglect and the Peak/End rule, you are suddenly aware in a very concrete way, looking at your plate, that you will form almost exactly the same retrospective memory whether your portion size is large or small; you now possess a deep theory about the *rules* governing your memory, and you know that this is what the rules say. (You also know to save the dessert for last.)

I want to hear how I can overcome akrasia—how I can have more willpower, or get more done with less mental pain. But there are ten thousand people purporting to give advice on this, and for the most part, it is on the level of that alternate Seth Roberts who just tells people about the amazing effects of drinking fruit juice. Or actually, somewhat worse than that—it’s people trying to describe internal mental levers that they pulled, for which there are no standard words, and which they do not actually know how to point to. See also the illusion of transparency, vol. I, p. 48, inferential distance, vol. I, p. 51, and double illusion of transparency.¹ (Notice how “You overestimate how much you’re explaining and your listeners overestimate how much they’re hearing” becomes *much more forceful* as advice, after I back it up with a cognitive science experiment and some evolutionary psychology?)

I think that the advice I *need* is from someone who reads up on a whole lot of experimental psychology dealing with willpower, mental conflicts, ego depletion, preference reversals, hyperbolic discounting, the breakdown of the self, picoeconomics, et cetera, and who, in the process of overcoming their own akrasia, manages to understand what they did in *truly general terms*—thanks to experiments that give them a vocabulary of cognitive phenomena that *actually exist*, as opposed to phenomena they just made up. And moreover, someone who can *explain* what they did to someone else, thanks again to the experimental and theoretical vocabulary that lets them point to replicable experiments that ground the ideas in very concrete results, or mathematically clear ideas.

Note the grade of increasing difficulty in citing:

- *Concrete experimental results* (for which one need merely consult a paper, hopefully one that reported $p < 0.01$ because $p < 0.05$ may fail to replicate);

1. sharov.me/r/6-12

Double Illusion of Transparency • lesswrong.com



- *Causal accounts that are actually true* (which may be most reliably obtained by looking for the theories that are used by a majority within a given science);
- *Math validly interpreted* (on which I have trouble offering useful advice because so much of my own math talent is intuition that kicks in before I get a chance to deliberate).

If you don't know who to trust, or you don't trust yourself, you should concentrate on experimental results to start with, move on to thinking in terms of causal theories that are widely used within a science, and dip your toes into math and epistemology with extreme caution.

But practical advice really, really *does* become a lot more powerful when it's backed up by *concrete experimental results*, *causal accounts that are actually true*, and *math validly interpreted*.

The Sin of Underconfidence



discussion
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6-c42

There are three great besetting sins of rationalists in particular, and the third of these is underconfidence. Michael Vassar regularly accuses me of this sin, which makes him unique among the entire population of the Earth.

But he's actually quite right to worry, and I worry too, and any adept rationalist will probably spend a fair amount of time worrying about it. When subjects know about a bias or are warned about a bias, *overcorrection* is not unheard of as an experimental result. That's what makes a lot of cognitive subtasks so troublesome—you know you're biased but you're not sure *how much*, and you don't know if you're correcting *enough*—and so perhaps you ought to correct a little more, and then a little more, but is *that* enough? Or have you, perhaps, far overshot? Are you now perhaps worse off than if you hadn't tried any correction?

You contemplate the matter, feeling more and more lost, and the very task of estimation begins to feel increasingly futile ...

And when it comes to the particular questions of *confidence*, *overconfidence*, and *underconfidence*—being interpreted now in the broader sense, not just calibrated confidence intervals—then there is a natural tendency to cast overconfidence as the *sin* of pride, out of that *other* list which never warned against the improper use of humility or the abuse of doubt. To place yourself too high—to overreach your proper place—to think too much of yourself—to put yourself forward—to put down your fellows by implicit comparison—and the consequences of humiliation and being cast down, perhaps publicly—are these not loathesome and fearsome things?

To be too *modest*—seems lighter by comparison; it wouldn't be so humiliating to be called on it publicly. Indeed, finding out that you're better than you imagined might come as a warm surprise; and to put yourself down, and others implicitly above,

has a positive tinge of *niceness* about it. It's the sort of thing that Gandalf would do.

So if you have learned a thousand ways that humans fall into error and read a hundred experimental results in which anonymous subjects are humiliated of their overconfidence—heck, even if you've just read a couple of dozen—and you don't *know* exactly how overconfident you are—then yes, you might genuinely be in danger of nudging yourself a step too far down.

I have no perfect formula to give you that will counteract this. But I have an item or two of advice.

What is the *danger* of underconfidence?

Passing up opportunities. Not doing things you could have done, but didn't try (hard enough).

So here's a first item of advice: If there's a way to *find out* how good you are, the thing to do is *test* it. *A hypothesis affords testing*; hypotheses about your own abilities likewise. Once upon a time it seemed to me that I ought to be able to win at the AI-Box Experiment; and it seemed like a very doubtful and hubristic thought; so I tested it. Then later it seemed to me that I might be able to win even with large sums of money at stake, and I tested that, but I only won one time out of three. So that was the limit of my ability at that time, and it was not necessary to argue myself upward or downward, because I could just *test* it.

One of the chief ways that smart people end up stupid is by getting *so used to winning* that they stick to places where they *know they can win*—meaning that they never stretch their abilities, they never try anything difficult.

It is said that this is linked to defining yourself in terms of your “intelligence” rather than “effort,” because then winning *easily* is a sign of your “intelligence,” where failing on a hard problem could have been interpreted in terms of a good effort.

Now, I am not quite sure this is how an adept rationalist should think about these things: rationality *is* systematized winning and trying to try seems like a path to failure. I would put it this way: A hypothesis affords testing! If you *don't know* whether you'll win on a hard problem—then *challenge your rationality* to *discover* your current level. I don't usually hold

with congratulating yourself on having tried—it seems like a bad mental habit to me—but surely *not* trying is even *worse*. If you have cultivated a general habit of confronting challenges, and won on at least *some* of them, then you may, perhaps, think to yourself, “I did keep up my habit of confronting challenges, and will do so next time as well.” You may also think to yourself “I have gained valuable information about my current level and where I need improvement,” so long as you properly complete the thought, “I shall try not to gain this same valuable information again next time.”

If you win *every* time, it means you aren’t stretching yourself enough. But you *should* seriously try to win every time. And if you console yourself too much for failure, you lose your winning spirit and become a scrub.

When I try to imagine what a fictional master of the Competitive Conspiracy would say about this, it comes out something like: “It’s *not* okay to lose. But the *hurt* of losing is not something so scary that you should flee the challenge for fear of it. It’s not so scary that you have to carefully avoid feeling it, or refuse to admit that you lost and lost hard. Losing is *supposed* to hurt. If it didn’t hurt you wouldn’t be a Competitor. And there’s *no* Competitor who *never* knows the pain of losing. Now get out there and *win*.”

Cultivate a habit of confronting challenges—not the ones that can kill you outright, perhaps, but perhaps ones that can potentially *humiliate* you. I recently read of a certain theist that he had defeated Christopher Hitchens in a debate (severely so; this was said by atheists). And so I wrote at once to the Bloggingheads folks and asked if they could arrange a debate. This seemed like someone I wanted to test myself against. Also, it was said by them that Christopher Hitchens should have watched the theist’s earlier debates and been prepared, so I decided *not* to do that, because I think I should be able to handle damn near anything on the fly, and I desire to learn whether this thought is correct; and I am willing to risk public humiliation to find out. Note that this is *not* self-handicapping in the classic sense—if the debate is indeed arranged (I haven’t yet heard back), and I do not prepare, and I fail, then I do lose

those stakes of myself that I have put up; I gain information about my limits; I have *not* given myself anything I consider an excuse for losing.

Of course this is only a way to think when you really *are* confronting a challenge just to test yourself, and not because you have to win at any cost. In *that* case you make everything as easy for yourself as possible. To do otherwise would be *spectacular* overconfidence, even if you're playing tic-tac-toe against a three-year-old.

A subtler form of underconfidence is *losing your forward momentum*—amid all the things you realize that humans are doing wrong, that you used to be doing wrong, of which you are probably still doing some wrong. You become timid; you question yourself *but don't answer the self-questions and move on*; when you hypothesize your own inability *you do not put that hypothesis to the test*.

Perhaps without there ever being a watershed moment when you deliberately, self-visibly *decide not to try* at some particular test ...you just slow down

It doesn't seem worthwhile any more, to go on trying to fix one thing when there are a dozen other things that will still be wrong ...

There's not enough hope of triumph to *inspire* you to try *hard* ...

When you consider doing any new thing, a dozen questions about your ability at once leap into your mind, and it does not occur to you that you could *answer* the questions by *testing* yourself ...

And having read so much wisdom of human flaws, it seems that the course of wisdom is ever doubting (never resolving doubts), ever the humility of refusal (never the humility of preparation), and just generally, that it is wise to say worse and worse things about human abilities, to pass into feel-good feel-bad cynicism.

And so my last piece of advice is another perspective from which to view the problem—by which to judge any potential habit of thought you might adopt—and that is to ask:

Does this way of thinking make me stronger, or weaker? Really truly?

I have previously spoken of the danger of *reasonableness*—the reasonable-sounding argument that we should two-box on Newcomb’s problem, the reasonable-sounding argument that we can’t know anything due to the problem of induction, the reasonable-sounding argument that we will be better off on average if we always adopt the majority belief, and other such impediments to the Way. “Does it win?” is one question you could ask to get an alternate perspective. Another, slightly different perspective is to ask, “Does this way of thinking make me stronger, or weaker?” Does constantly reminding yourself to doubt everything make you stronger, or weaker? Does never resolving or decreasing those doubts make you stronger, or weaker? Does undergoing a deliberate crisis of faith in the face of uncertainty make you stronger, or weaker? Does answering every objection with a humble confession of your fallibility make you stronger, or weaker?

Are your current attempts to compensate for possible overconfidence making you stronger, or weaker? Hint: If you are taking more precautions, more scrupulously trying to test yourself, asking friends for advice, working your way up to big things incrementally, or still failing sometimes but less often than you used to, you are probably getting stronger. If you are *never* failing, avoiding challenges, and feeling generally hopeless and dispirited, you are probably getting weaker.

I learned the first form of this rule at a very early age, when I was practicing for a certain math test, and found that my score was going down with each practice test I took, and noticed going over the answer sheet that I had been pencilling in the correct answers and erasing them. So I said to myself, “All right, *this time* I’m going to use the Force and act on instinct,” and my score shot up to above what it had been in the beginning, and on the real test it was higher still. So that was how I learned that doubting yourself does not always make you stronger—especially if it interferes with your ability to be moved by good information, such as your math intuitions. (But I *did* need the test to tell me this!)

Underconfidence is not a unique sin of rationalists alone. But it is a particular danger into which *the attempt to be rational* can lead you. And it is a *stopping* mistake—an error that prevents you from gaining that further experience that would correct the error.

Because underconfidence actually *does* seem quite common among aspiring rationalists who I meet—though rather less common among rationalists who have become famous role models—I would indeed name it third among the three besetting sins of rationalists.

Go Forth and Create the Art!



discussion
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6-c43

I have said a thing or two about rationality, these past months. I have said a thing or two about how to untangle questions that have become confused, and how to tell the difference between real reasoning and fake reasoning, and the will to become stronger that leads you to try before you flee; I have said something about doing the impossible.

And these are all techniques that I developed in the course of my own projects—which is why there is so much about cognitive reductionism, say—and it is possible that your mileage may vary in trying to apply it yourself. The one’s mileage may vary. Still, those wandering about asking “But what good is it?” might consider rereading some of the earlier essays; knowing about e.g. the conjunction fallacy, and how to spot it in an argument, hardly seems esoteric. Understanding why motivated skepticism is bad for you can constitute the whole difference, I suspect, between a smart person who ends up smart and a smart person who ends up stupid. Affective death spirals consume *many* among the unwary ...

Yet there is, I think, more *absent* than *present* in this “art of rationality”—defeating akrasia and coordinating groups are two of the deficits I feel most keenly. I’ve concentrated more heavily on epistemic rationality than instrumental rationality, in general. And then there’s training, teaching, verification, and becoming a proper experimental science based on that. And if you generalize a bit further, then *building the Art* could also be taken to include issues like developing better introductory literature, developing better slogans for public relations, establishing common cause with other Enlightenment subtasks, analyzing and addressing the gender imbalance problem ...

But those small pieces of rationality that I’ve set out ... I *hope* ... just maybe ...

I suspect—you could even call it a guess—that there is a *barrier to getting started*, in this matter of rationality. Where by

default, in the beginning, you don't have enough to build on. Indeed so little that you don't have a clue that more exists, that there is an Art to be found. And if you do begin to sense that more is possible—then you may just *instantaneously* go wrong. As David Stove observes, most “great thinkers” in philosophy, e.g., Hegel, are properly objects of pity.¹ That's what happens by default to anyone who sets out to develop the art of thinking; they develop fake answers.

When you try to develop part of the human art of thinking ...then you are doing something *not too dissimilar* to what I was doing over in Artificial Intelligence. You will be tempted by fake explanations of the mind, fake accounts of causality, mysterious holy words, and the amazing idea that solves everything.

It's not that the particular, epistemic, fake-detecting methods that I use are so good for every *particular* problem; but they seem like they might be helpful for discriminating good and bad *systems of thinking*.

I hope that someone who learns the part of the Art that I've set down here will not *instantaneously* and *automatically* go wrong if they start asking themselves, “How should people think, in order to solve new problem *X* that I'm working on?” They will not immediately run away; they will not just make stuff up at random; they may be moved to consult the literature in experimental psychology; they will not automatically go into an affective death spiral around their Brilliant Idea; they will have some idea of what distinguishes a fake explanation from a real one. They will get a saving throw.

It's this sort of barrier, *perhaps*, that prevents people from *beginning* to develop an art of rationality, if they are not already rational.

And so instead they ...go off and invent Freudian psychoanalysis. Or a new religion. Or something. That's what happens by *default*, when people start thinking about thinking.

1. David Charles Stove, *The Plato Cult and Other Philosophical Follies* (Cambridge University Press, 1991).

I hope that the part of the Art I have set down, as incomplete as it may be, can surpass that preliminary barrier—give people a base to build on; give them an idea that an Art exists, and somewhat of how it ought to be developed; and give them at least a *saving throw* before they *instantaneously* go astray.

That's my dream—that this highly-specialized-seeming art of answering confused questions may be some of what is needed, in the very beginning, *to go and complete the rest*.

A task which I am leaving to *you*. Probably, anyway. I make no promises as to where my attention may turn in the future. But y'know, there *are* certain other things I need to do. Even if I develop yet more Art by accident, it may be that I will not have the time to write any of it up.

Beyond all that I have said of fake answers and traps, there are two things I would like you to keep in mind.

The first—that I drew on multiple sources to create my Art. I read many different authors, many different experiments, used analogies from many different fields. *You* will need to draw on multiple sources to create *your* portion of the Art. You should not be getting all your rationality from one author—though there might be, perhaps, a certain centralized website, where you went to post the links and papers that struck you as really important. And a maturing Art will need to draw from multiple sources. To the best of my knowledge there is *no* true science that draws its strength from only one person. To the best of my knowledge that is *strictly* an idiom of cults. A true science may have its heroes, it may even have its lonely defiant heroes, but *it will have more than one*.

The second—that I created my Art in the course of *trying to do some particular thing* that animated all my efforts. Maybe I'm being too idealistic—maybe thinking too much of the way the world *should* work—but even so, I somewhat suspect that you couldn't develop the Art *just* by sitting around thinking to yourself, "Now how can I fight that akrasia thingy?" You'd develop the rest of the Art in the course of trying to *do something*. Maybe even—if I'm not overgeneralizing from my own history—some task difficult enough to strain and break your old understanding and force you to reinvent a few things. But

maybe I'm wrong, and the next leg of the work will be done by direct, specific investigation of "rationality," without any need of a specific application considered more important.

A past attempt of mine to describe this principle, in terms of maintaining a secret identity or day job in which one doesn't teach rationality, was roundly rejected by my audience. Maybe "leave the house" would be more appropriate? It *sounds* to me like a really good, healthy idea. Still—perhaps I am deceived. We shall see where the next pieces of the Art do, in fact, come from.

I have striven for a long time now to convey, pass on, share a piece of the strange thing I touched, which seems to me so precious. And I'm not sure that I ever said the central rhythm into words. Maybe you can find it by listening to the notes. I can say these words but not the rule that generates them, or the rule behind the rule; one can only hope that by *using* the ideas, perhaps, similar machinery might be born inside you. Remember that *all* human efforts at learning arcana slide by default into passwords, hymns, and floating assertions.

I have striven for a long time now to convey my Art. Mostly without success, before this present effort. Earlier I made efforts only in passing, and got, perhaps, as much success as I deserved. Like throwing pebbles in a pond, that generate a few ripples, and then fade away ... This time I put some back into it, and heaved a large rock. Time will tell if it was large enough—if I really *disturbed* anyone deeply enough that the waves of the impact will continue under their own motion. Time will tell if I have created anything that moves under its own power.

I want people to go forth, but also to return. Or maybe even to go forth and stay simultaneously, because this is the Internet and we can get away with that sort of thing; I've learned some interesting things on *Less Wrong*, lately, and if continuing motivation over years is any sort of problem, talking to others (or even *seeing* that others are also trying) does often help.

But at any rate, if I have affected you at all, then I hope you will go forth and confront challenges, and achieve somewhere beyond your armchair, and create new Art; and then, remem-

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bering whence you came, radio back to tell others what you learned.

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